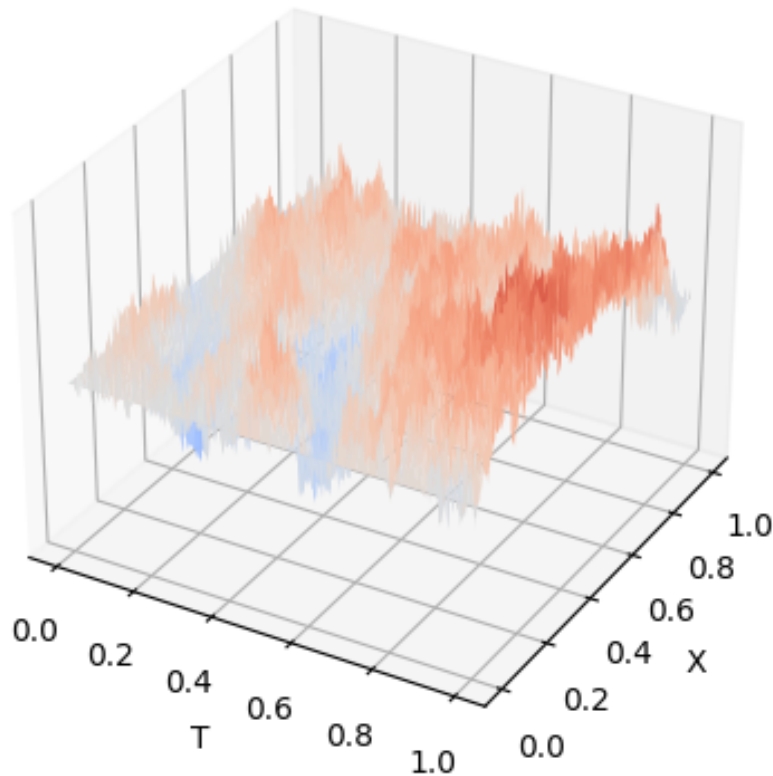




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Simulation of semilinear parabolic SPDEs



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¹Image from titlepage is the middle surface plot from Figure 1.

1. Introduction

This thesis discusses to the numerical analysis of semilinear parabolic stochastic partial differential equations (SPDEs). SPDEs occur naturally in many research fields such as neurobiology (for example in models for spiking neurons), population theory, economics (for example in modeling the structure of interest rates) or physics whenever a particle system is involved (see for example [LR17] or [Tuc13]) and therefore form a very active research area. They typically allow for more precise models than stochastic ordinary differential equations (SODEs) due to the inclusion of the spatial dimension [see Tuc13] and than partial differential equations (PDEs) due to the inclusion of missing or not known data using a stochastic term. Yet, they also combine the difficulties from SODEs and PDEs. More precisely, they suffer from the irregularity of the stochastic term and from the lack of solvability of partial differential equations.

The field of research of SPDEs can be briefly divided into two areas - on the one hand the study of the existence and uniqueness of a solution and on the other hand the numerical analysis of such equations. The former deals with the basics and suffers in particular from the problem that there is no general theory for the solution of such equations and thus one has to deal more or less with each problem separately. Luckily, for specific classifications of SPDEs like for parabolic semilinear SPDEs, the existence and uniqueness of solutions is quite well understood and there are great overviews like [DZ14]. The latter, on the other hand, assumes an existing solution and focuses on the development of numerical approximations and their asymptotic properties. In particular, convergence properties such as the convergence rate are important here. Hence, the main difficulties here lie in the structure of the solution itself. In general, the solution of an SPDE is a continuous stochastic process with values in an infinite dimensional space. In addition, the noise of such a solution is highly irregular. So, it is not easy to approximate such a solution and usually requires high computational effort. Due to the increasing computing power of computers in recent years, this has become more and more possible, so that there are now many papers in this area. Examples are [JK09a] [Dav95], [Kru14], [Wan14] [LP17] and [Bré17].

This work places itself more in the second category and deals with the numerical approximation of semilinear parabolic SPDEs. In particular, let $L^2(0, 1)$ be the space of square integrable functions from $[0, 1]$ to $\mathbb{R}$ and consider the one-dimensional semilinear stochastic heat equation on $L^2(0, 1)$ with additive space-time white noise, Dirichlet boundary condition and zero initial value condition given by

$$\begin{cases} dX(t) = \left[\vartheta \frac{d^2}{dx^2} X(t) + F(X(t)) \right] dt + dW(t), & x \in [0, 1], \quad t \in [0, T] \\ X(0) = 0 \\ X(t)(0) = X(t)(1) = 0, \end{cases} \quad (1)$$

where $\vartheta \in \mathbb{R}$, $T \in (0, \infty)$ and $F : L^2(0, 1) \rightarrow L^2(0, 1)$ is a twice Fréchet differentiable Nemytskii operator induced by a possibly nonlinear function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $F(\nu)(x) = f(\nu(x))$ for every $\nu \in L^2(0, 1)$ and $x \in [0, 1]$. With sufficient assumptions

on F , we can find a mild solution X of (1). X is a continuous stochastic process with values in a proper subspace of $L^2(0, 1)$. Consequently, we need to discretize the space and time to numerically handle X . The space will be discretized by applying the spectral Galerkin method. This leads to a finite N -dimensional approximation X_N of X for $N \in \mathbb{N}$. Additionally, let $M \in \mathbb{N}$ and $\{t_k\}_{k=0, \dots, M}$ with $0 = t_0 < t_1 < \dots < t_M = T$ be a equidistant partition of $[0, T]$ with grid width $t_{k+1} - t_k = T/M =: h$. Then, the exponential Euler method - first introduced in [JK09a] - will be developed to discretize the temporal dimension and generate a fully discrete approximation $X_{N,M}$ of X . We will see that $X_{N,M}$ converges strongly against X with an overall strong convergence rate of $1/3$ with respect to the computational effort M^3 . Moreover, we can express $X_{N,M}$ for t_k by

$$X_{N,M}(t_k) = \sum_{l=1}^N X_{N,M}^l(t_k) \phi_l \quad (2)$$

with

$$\begin{aligned} X_{N,M}^l(t_k) = & e^{-\lambda_l h} X_{N,M}^l(t_{k-1}) + \left(\frac{1 - e^{-\lambda_l h}}{\lambda_l} \right) F_N^l(X_{N,M}(t_{k-1})) \\ & + \left(\frac{1 - e^{-2\lambda_l h}}{2\lambda_l} \right)^{1/2} R_{k-1}^l \end{aligned} \quad (3)$$

with $\lambda_l = \vartheta \pi^2 l^2$ whereby $(-\lambda_l)_{l \geq 1}$ are the eigenvalues of the linear operator $\vartheta \frac{d^2}{dx^2}$ with corresponding eigenfunctions $(\phi_l)_{l \geq 1}$ given by $\phi_l(x) = \sqrt{2} \sin(\pi l x)$ and independent real-valued standard gaussian random variables $(R_{k-1}^l)_{l=1, \dots, N, k=1, \dots, M}$. Let $(U_l(t))_{t \in [0, T]}$ be an one-dimensional Ornstein-Uhlenbeck process $(U_l(t))_{t \in [0, T]}$ satisfying $du_l(t) = -\lambda_l u_l(t) dt + d\beta_l(t)$, where $\beta_l(t)$ $l \geq 1$, $t \in [0, T]$ are independent and identically distributed (iid) real-valued Brownian motions. Knowing that $u_l(t_k)$, $k = 1, \dots, M$ has the AR(1)-structure

$$u_l(t_k) = e^{-\lambda_l h} u_l(t_{k-1}) + \left(\frac{1 - e^{-2\lambda_l h}}{2\lambda_l} \right)^{1/2} R_{k-1}^l$$

we can express (3) in terms of $u_l(t_k)$ by

$$\begin{aligned} X_{N,M}^l(t_k) = & u_l(t_k) + e^{-\lambda_l h} (X_{N,M}(t_{k-1}) - u_l(t_{k-1})) \\ & + \left(\frac{1 - e^{-\lambda_l h}}{\lambda_l} \right) F_N^l(X_{N,M}(t_{k-1})). \end{aligned} \quad (4)$$

Using this expression, we will generalize the idea of [Hil20] and introduce the semi-linear replacement method $\tilde{X}_{J,L,M}$ on a discrete grid $0 = y_0 < y_1 < \dots < y_J = 1$ of $[0, 1]$ with equidistant points $(y_j)_{j=0, \dots, J}$ such that $y_{j+1} - y_j = 1/J =: \delta$ for every $j = 0, \dots, J-1$, $J \in \mathbb{N}$. We will see that $(\phi_l)_{l=1, \dots, J-1}$ is an orthonormal basis with respect to the empirical inner product $\langle \nu, \nu \rangle_J := \frac{1}{J} \sum_{j=0}^J \nu(y_j) \nu(y_j)$. Using this and periodic properties of we can express $X_{N,M}(t_k)$ by

$$X_{N,M}(t_k)(y_j) = \sum_{l=1}^{J-1} \sum_{m \in \mathcal{I}_l} X_{N,M}^m(t_k) \phi_l(y_j)$$

with $\mathcal{I}_l = \{l + 2mJ : m \in \mathbb{N}\} \cap \{2J - l + 2mJ : m \in \mathbb{N}\}$. For large l (e.g. $l \geq LN$ for a threshold $L \in \mathbb{N}$), $(u_l(t_k))_{l \geq LN}$ behave like iid centered gaussian random variables with variances $\frac{1}{2\lambda_l}$ [see Hil20, Chapter 3]. Hence, by assuming a sufficiently smooth F and using (4), $(X_{N,M}^l(t_k))_{l \geq LN}$ also behaves like iid centered gaussian random variables with variances $\frac{1}{2\lambda_l}$. Therefore, the idea is to replace $\sum_{m \in \mathcal{I}_l, m \geq LN} X_{N,M}^l(t_k)$ for every $l = 1, \dots, J-1$ by a centered gaussian random variable $\tilde{R}_k^{l,L} \sim \mathcal{N}(0, \sum_{m \in \mathcal{I}_l, m \geq LN} \frac{1}{2\lambda_m})$. This leads to the approximation $\tilde{X}_{J,L,M}$ expressed by

$$\tilde{X}_{J,L,M}(t_k)(y_j) = \sum_{l=1}^{J-1} \tilde{U}_l^L(t_k) \phi_l(y_j), \quad (5)$$

with

$$\tilde{U}_l^L(t_k) = \sum_{m \in \mathcal{I}_l, m < LN} X_{N,M}^l(t_k) + \tilde{R}_k^{l,L}.$$

Due to the representations (2) and (5) both approximations are simple to simulate. Hence, we will describe methods to evaluate the quality of those approximations using simulations. More precisely, we will first develop an setting to simulate the strong error at time T , even if there is no explicit representation of the solution to (1) available. In this case we will use a reference solution $X_{N_{ref}, M_{ref}}$ with N_{ref} much larger than N and M_{ref} much larger than M . This will result in the simulation of

$$\|X_{N_{ref}, M_{ref}}(T) - X_{N,M}(T)\|_{L^2(\Omega)} \approx \frac{1}{D} \sum_{d=1}^D \sum_{j=1}^J \delta \left(X_{N_{ref}, M_{ref}}^d(T)(y_j) - X_{N,M}^d(T)(y_j) \right)^2$$

with D iid copies of $X_{N_{ref}, M_{ref}}$ and $X_{N,M}(T)$. Then, we will consider central limit theorems known from [HT21] for both spatial and temporal realized quadratic variations of semilinear SPDEs and use them to compare the performance of the exponential Euler method and the semilinear replacement method.

This thesis is organized as follows. Chapter 2 deals with mathematical foundations necessary for the theory of SPDEs. Chapter 3 then introduces SPDEs as an abstract Hilbert space-valued construct. Chapter 4 gives a brief overview of convergence types and spatial and temporal discretization of SPDEs, while in Chapter 5 introduces first the exponential Euler method and then the semilinear replacement methods. Chapter 6 describes the theory of numerical quality criteria for evaluating approximations, and Chapter 7 contains first simulations of the methods introduced in the previous chapter. Finally, chapter 8 recaps everything. While the theoretical part is mainly based on [DZ14], [LPS14] and [LR17], the two main references approximation methods are [JK09a] and [Hil20].

2. Mathematical foundations

The purpose of this chapter is to introduce the mathematical foundations needed for the theory of SPDEs. For this we will first look at functional analytics basics and then introduce Brownian motions on Hilbert spaces and its application to stochastic integrals. The used literature will be introduced at each section.

The following notations are used throughout this paper on a regular basis. Let $(\Omega, \mathcal{F}, \mu)$ be a measurable space with filtration $\mathcal{F}_t$, $t \geq 0$. Further, let H be a separable Hilbert space with inner product $\langle \cdot, \cdot \rangle_H$ and norm $\|\cdot\|_H$ and let H' denote its dual space. We denote the norm on $\mathbb{R}$ by $|\cdot| := \|\cdot\|_{\mathbb{R}}$. The following basic notations from functional analysis are mainly based on [Wer04] and [LR17]. In particular, let A be a linear operator. Then $\mathcal{D}(A)$ denotes its domain and $\text{tr}(A)$ its trace (see Definition A.15). A is of trace-class, if $\text{tr}(A) < \infty$ (See Definition A.16). The operator norm of A is denoted by $\|A\|_{\mathcal{O}_p}$ (see Theorem A.12). Let X be an arbitrary Banach space, we denote by $L^2(X) := L^2(X; \mathbb{R})$ the space of square integrable functions from X to $\mathbb{R}$, if $X = [a, b]$ is an interval for $a < b$, $a, b \in \mathbb{R}$ then we will write $L(0, 1) := L([0, 1])$. More general, we denote by $L^2(\Omega) := L^2((\Omega, \mathcal{F}, \mathbb{P}); \mathbb{R})$ the space of square integrable functions from Ω to $\mathbb{R}$ with respect to the measure $\mathbb{P}$ [e.g. see Wer04, p.13ff.]. Let D be a subspace of $\mathbb{R}$ and Y be a Banach space. Then, $C(D; Y)$ denotes the space of all continuous functions from D to Y and $C^k(D; Y)$ the space of all continuous functions from D to Y with k continuous derivatives [e.g. see LPS14, p.3]. If $Y = \mathbb{R}$ then we write $C(D) := C(D; \mathbb{R})$. Let $B \subset X$ then we denote by $\mathbf{1}_B(x)$ for $x \in X$ the indicator function which is 1 if $x \in B$ and 0 if $x \notin B$. Further, we denote by I the indicator operator or indicator matrix depending on the context. Further, as usual, we will abbreviate "independent and identically distributed" by *iid*.

2.1. Sobolev spaces and weak derivatives

The aim of this section is to give a brief overview of the theory of weak derivatives and Sobolev spaces. The following is based on [Dav95], [AF03], [Tai20], [LPS14] and [Wer04]. First, note that the *differential operator* D^α is given by

$$D^\alpha f(x) = \frac{\partial^{\alpha_1}}{\partial^{\alpha_1} x_1} \cdots \frac{\partial^{\alpha_n}}{\partial^{\alpha_n} x_n} f(x_1, \dots, x_n),$$

where $\alpha = \{\alpha_1, \dots, \alpha_n\}$ is a multi-index containing non-negative integers α_i , $i \in \{1, \dots, n\}$ [see LPS14, p.3]. We call $|\alpha| = \alpha_1 + \dots + \alpha_n$ the degree of α .

Further, we will call an infinite continuously differentiable function φ with compact support *testfunction* and the space of all testfunctions will be denoted by $C_0^\infty(\mathbb{R}^d)$ if the support of φ is a subset of $\mathbb{R}^d$ or more specific $C_0^\infty(\mathcal{O})$, if its support is an non-empty open subset of $\mathbb{R}^d$ [e.g. see LPS14, Definition 1.13]. Note that the space C_0^∞ is dense in L^2 such that any function of L^2 can be approximated arbitrarily good by a sequence of testfunctions. With these tools, we can define the weak derivative.

Definition 2.1 (Weak derivative ([Wer04] p.204))

Let $\mathcal{O} \subset \mathbb{R}^n$ be open, $\alpha = \{\alpha_1, \dots, \alpha_n\}$ a multi-index as previously defined, and $f \in L^2(\mathcal{O})$. $g \in L^2(\mathcal{O})$ is called *weak α -derivative* of f if

$$\langle g, \varphi \rangle_{L^2(\mathcal{O})} = (-1)^{|\alpha|} \langle f, D^\alpha \varphi \rangle_{L^2(\mathcal{O})}, \quad \forall \varphi \in C_0^\infty(\mathcal{O})$$

is satisfied.

If g exists, it is unique [see Wer04, p.204] and thus we denote g with $D^\alpha f$. Further, weak derivatives can be viewed as a generalization of the classical derivative, since the derivative of any differentiable function f satisfies the above equation. This follows directly through integration by parts [e.g. see the discussion in LPS14, p.12f.].

Remark 2.2 (Delta function)

There are functions, that do not have a weak derivative. The following expressions of a classic example of such a function are based on [LPS14, Example 1.43]. Consider $f \in L^2(\mathcal{O})$ given by

$$f(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0, \end{cases}$$

for all $x \in \mathcal{O}$. Then, due to the discontinuity at 0, there is obviously no derivative in the classical sense. Now, consider $\langle f, D\varphi \rangle_{L^2(\mathcal{O})}$ for $\varphi \in C_0^\infty(\mathcal{O})$. Then

$$\langle f, D\varphi \rangle_{L^2(\mathcal{O})} = \int_{\mathcal{O}} f(x) D\varphi(x) dx = \int_{x \in \mathcal{O}, x > 0} D\varphi(x) dx = \varphi(0),$$

holds, using the fundamental theorem of calculus and that $\varphi \in C_0^\infty(\mathcal{O})$. For f to have a weakly derivative, a function $\delta \in L^2(\mathcal{O})$ must exist so that

$$\langle \delta, \varphi \rangle_{L^2(\mathcal{O})} = -\varphi(0)$$

is satisfied for all $\varphi \in C_0^\infty(\mathcal{O})$. In particular, δ needs to satisfy $\int_{-\epsilon}^{\epsilon} \delta(x) dx = 1$ for any $\epsilon > 0$. Using the latter condition, it is clear that δ cannot be a function in $L^2(\mathcal{O})$. Nevertheless, such a construct is from great interest in a variety of applications as for example in the discussion of the covariance of space-time white noise in this paper. To proper define such a construct we would need the theory of Distributions. We will not discuss this topic further but refer to [Wer04, §VIII.5] for an introduction in distribution theory.

Let $\mathcal{O} \subset \mathbb{R}^n$ be an open set and let $W^m(\mathcal{O}) \subset L^2(\mathcal{O})$ be the subspace of all functions $f \in L^2(\mathcal{O})$ for which $D^\alpha f \in L^2(\mathcal{O})$ exists for all $|\alpha| \leq m$. For such a space we can define the norm

$$\|f\|_m := \|f\|_{W^m(\mathcal{O})} = \sum_{|\alpha| \leq m} \|D^\alpha f\|_{L^2(\mathcal{O})}.$$

This approach leads to Sobolev spaces.

Definition 2.3 (Sobolev spaces on $L^2(\mathcal{O})$ [AF03](p.59f))

For any positive integer m we consider three vector spaces on which $\|\cdot\|_m$ is a norm:

- a) $H^m(\mathcal{O}) := \overline{\{f \in C^m(\mathcal{O}) : \|f\|_m < \infty\}}$ with respect to $\|\cdot\|_m$,
- b) $W^m(\mathcal{O}) := \{f \in L^2(\mathcal{O}) : D^\alpha f \in L^2(\mathcal{O}) \quad \forall |\alpha| < m\}$,
- c) $H_0^m(\mathcal{O}) := \overline{C_0^\infty(\mathcal{O})}$ with respect to $\|\cdot\|_m$.

These spaces are called *Sobolev spaces*.

This definition can be generalized to $L^p(\mathcal{O})$ -spaces for $1 \leq p \leq \infty$ [e.g. see AF03, p.59f.]. It can be shown [see AF03, Theorem 3.6] that $W^m(\mathcal{O})$ is a separable Hilbert space with inner product

$$\langle f, g \rangle_m := \langle f, g \rangle_{W^m(\mathcal{O})} = \sum_{|\alpha| \leq m} \langle D^\alpha f, D^\alpha g \rangle_{L^2(\mathcal{O})}.$$

Moreover, using [Tai20, Theorem 4.6], it can be shown that if $\mathcal{O}$ is a bounded Lipschitz domain then $H^m(\mathcal{O}) = W^m(\mathcal{O})$ holds. Thus, from now on we only consider bounded open subspaces $\mathcal{O} \subset \mathbb{R}^n$ we denote the classical Sobolev space by $H^m(\mathcal{O})$. Sobolev spaces can also be defined on whole $\mathbb{R}^n$. In this case, if $\mathcal{O}$ has smooth boundaries $\partial\mathcal{O}$, then we can define $H^m(\mathcal{O})$ by $v|_{\mathcal{O}} = u \in H^m(\mathcal{O})$ for $v \in H^m(\mathbb{R}^n)$ [see Tai20, Definition 4.5]. When $m = 1$ and $\mathcal{O} = (a, b)$ for $a, b \in \mathbb{R}$, then we can define $H_0^1(\mathcal{O})$ also by

$$H_0^1(a, b) := \{f \in H^1(a, b) : f(x) = 0, \quad \forall x \in \{a, b\}\}$$

[see LPS14, p.14]. This is in particular interesting for Dirichlet boundary conditions as we will see later. The next remark provides more useful properties of Sobolev spaces.

Remark 2.4

- (i) The following relations hold: $H_0^m(\mathcal{O}) \subset H^m(\mathcal{O}) \subset H^{\tilde{m}}(\mathcal{O})$ for all $m > \tilde{m}$ and $H^0(\mathcal{O}) = L^2(\mathcal{O})$. Hence, $H^m(\mathcal{O}) \subset L^2(\mathcal{O})$ for all $m > 0$.
- (ii) $C_0^\infty(\mathbb{R})$ is dense in $L^2(\mathbb{R})$. Moreover, if $\mathcal{O}$ satisfies the so-called segment condition (which is a smoothness conditions for the boundary of $\mathcal{O}$ [see AF03, Condition 3.21]) then the space $C_0^\infty(\mathbb{R})|_{\mathcal{O}} := \{f|_{\mathcal{O}} : f \in C_0^\infty(\mathbb{R})\}$ is a dense subset of $H^m(\mathcal{O})$ [see AF03, Theorem 3.22]. Consequently, $H_0^m(\mathcal{O})$ and $H^m(\mathcal{O})$ are dense in $L^2(\mathcal{O})$.
- (iii) The Sobolev embedding theorem states the connection between $H^m(\mathcal{O})$ and $C^k(\overline{\mathcal{O}})$. The relation depends on the dimension of $\mathcal{O}$. We only consider 1-dimensional intervals. Hence, if $f \in H^m(\mathcal{O})$ and if $m > k + 1/2$, $k \in \mathbb{N}$, then there exists a continuous extension of f on $\overline{\mathcal{O}}$ such that this extension is an element of $C^k(\overline{\mathcal{O}})$ [see LPS14, Theorem 1.51].

Example 2.5 (Laplace operator)

Let Δ denote the Laplace operator on $L^2(0, 1)$. Δ is defined as the second order derivative of $u \in L^2(0, 1)$ (e.g. $\Delta u = \frac{d^2 u}{dx^2}$). Thus, the domain of Δ is given by the space all two times differentiable functions $C^2(0, 1) \subset L^2(0, 1)$. However, this is often too strict for applications. In that case, one can also understand Δ in terms of weak derivatives. Then, it is defined on the space of all functions $u \in L^2(0, 1)$ with $D^2 u \in L^2(0, 1)$, which is the Sobolev space of order 2 as we seen before. Hence, we have $\mathcal{D}(\Delta) = H^2(0, 1)$.

The Laplace operator is an important component of the so-called heat equation, which forms an important class of PDEs and, in its stochastic form, of SPDEs. Hence, due to their connection to the Laplace operator, Sobolev spaces are an important concept in the theory of PDEs and therefore also SPDEs. Consider the following example which is based on [LPS14, Example 1.81].

Example 2.6 (Heat equation)

Let $H = L^2(0, 1)$. Consider the following PDE with Dirichlet boundary conditions:

$$\begin{cases} du(t) = Au(t), \\ u(0) = u(1) = 0, \end{cases} \quad (6)$$

where A is the Laplace operator Δ known from the previous example. Problem 6 is called *heat equation*. Since considering only solutions $u \in C^2(0, 1)$ is too restrictive, we also allow weak derivatives. Example 2.5 then tells us that a solution u to Problem (6) must be an element of $H^2(0, 1)$. Taking also the Dirichlet boundary condition into account, u must also be an element of $H_0^1(0, 1)$. Note that elements from $H_0^1(0, 1)$ have bounded support inside of $(0, 1)$ and hence, $u(0) = u(1) = 0$. Combining these assumptions leads to $H^2(0, 1) \cap H_0^1(0, 1)$ as the solution space. Hence, the domain of A in the setting of the heat equation with Dirichlet boundary condition equals $\mathcal{D}(A) = H^2(0, 1) \cap H_0^1(0, 1)$ and thus also $\mathcal{D}(-A) = H^2(0, 1) \cap H_0^1(0, 1)$.

The domain introduced in the previous example also applies to the semilinear stochastic heat equation, which we will learn about in a later chapter. Besides Definition 2.3 Sobolev spaces can also be defined using the Fourier transform [e.g. see Dav95, p.67f]. This allows us to define these spaces also for fractional exponents $r \in \mathbb{R}$.

Remark 2.7 (Fractional Sobolev spaces)

For every $f \in C_0^\infty(\mathbb{R})$ let

$$\hat{f}(y) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}} e^{-ixy} f(x) dx$$

be the Fourier transform of f . Define the norm $\|\cdot\|_r$ for $r \in \mathbb{R}$, by

$$\|f\|_r^2 = \int_{\mathbb{R}} (1 + |y|^2)^r |\hat{f}(y)|^2 dy < \infty, \quad r \in \mathbb{R}.$$

Then, Let $H_0^r(\mathbb{R})$ be the completion of $C_0^\infty(\mathbb{R})$ with respect to the norm $\|\cdot\|_r$. $H_0^r(\mathbb{R})$ is called (fractional) Sobolev space [see LR17, Example 3.1.9].

One can also define fractional sobolev spaces $H^r(\mathcal{O})$ for a suitable open subset $\mathcal{O} \subset \mathbb{R}$ [e.g. see DPV12, Chapter 2] or [Tai20, §4.1.8]. While $H_0^r(\mathbb{R}) = H^r(\mathbb{R})$ since $C_0^\infty(\mathbb{R})$ is a dense subset of $H_0^r(\mathbb{R})$ this does not hold for $H_0^r(\mathcal{O})$ and $H^r(\mathcal{O})$ [see the discussion of DPV12, Theorem 2.4].

2.2. Exponential operator, fractional powers and interpolation spaces

In the theory of SPDEs, linear operators and especially their eigenvalues and eigenfunctions have an important role. In this section we will therefore deal with such operators and describe them in more detail. Note that an operator $A : H \rightarrow H$ is called *linear* if $A(ax + by) = aA(x) + bA(y)$ holds for all $a, b \in \mathbb{R}$ and $x, y \in H$ (see Definition A.10).

Assumption 1

$A : \mathcal{D}(A) \subset H \rightarrow H$ is a linear operator with eigenfunctions $\{\phi_j\}_{j \geq 1}$ creating an orthonormal basis of H and their corresponding negative real-valued eigenvalues $\{-\lambda_j\}_{j \geq 1}$ such that $0 < \lambda_1 \leq \lambda_2 \leq \dots$ and $\lim_{j \rightarrow \infty} \lambda_j = \infty$. Moreover, let A be given by

$$A\nu = \sum_{j=1}^{\infty} -\lambda_j \langle \phi_j, \nu \rangle \phi_j, \quad (7)$$

for all $\nu \in \mathcal{D}(A)$ with $\mathcal{D}(A) = \{\nu \in H : \sum_{j=1}^{\infty} |\lambda_j|^2 |\langle \phi_j, \nu \rangle|^2 < \infty\}$.

If an operator A satisfies the previous assumption, we say that A is of negative kind. Note that if A satisfies Assumption 1 such that in particular Equation 7 holds, then A is also self-adjoint [see LPS14, Lemma 1.86].

Example 2.8

We continue Example 2.5 and 2.6 and consider the heat equation on $L^2(0, 1)$ with Dirichlet boundary conditions. Recall that $A = \Delta$ is the Laplace operator. Note that then

$$\frac{d^2}{dx^2} \sin(\pi j x) = -(\pi j)^2 \sin(\pi j x).$$

Thus, $\sin(\pi j)$ are eigenfunctions of A with eigenvalues $-(\pi j)^2$. Obviously, $\lambda_j = (\pi j)^2$ satisfy the eigenvalue condition given in Assumption 1. When we normalize the above eigenfunctions (e.g. $\phi_j(x) = \sqrt{2} \sin(\pi j x)$), then it is well known that $\{\phi_j\}_{j \geq 1}$ is an orthonormal basis of $L^2(0, 1)$ [e.g. see LPS14, Example 1.81].

Next we like to introduce an operator-valued version of the exponential function. For the rest of this work, the so-called exponential operator plays an important role. The following definition can be found in [LPS14, Definition 3.20].

Definition 2.9 (exponential operator)

Let A be an operator satisfying Assumption 1. Then, for $t \geq 0$, the *exponential* of At is defined by

$$e^{At} \nu := \sum_{j=1}^{\infty} e^{-\lambda_j t} \langle \nu, \phi_j \rangle \phi_j, \quad (8)$$

for all $\nu \in H$.

A simple but useful fact is given by the following lemma.

Lemma 2.10

Suppose Assumption 1 is satisfied for a linear operator A . Then for the exponential operator e^{At} , $t \geq 0$ the following statement holds true for every $v \in H$:

$$\|e^{At}\nu\|_H \leq e^{-\lambda_1 t} \|\nu\|_H \leq \|\nu\|_H. \quad (9)$$

Proof. Let $\nu \in H$ be arbitrary, but fixed. Using Definition 2.9, the fact that $e^{-2\lambda_j t} \leq 1$ for all $t \geq 0$ and Parseval's identity we get

$$\begin{aligned} \|e^{At}\nu\|_H^2 &= \left\| \sum_{j=1}^{\infty} e^{-\lambda_j t} \langle \nu, \phi_j \rangle_H \phi_j \right\|_H^2 \\ &= \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} e^{-\lambda_j t} e^{-\lambda_k t} \langle \nu, \phi_j \rangle_H \langle \nu, \phi_k \rangle_H \langle \phi_j, \phi_k \rangle_H \\ &= \sum_{j=1}^{\infty} e^{-2\lambda_j t} \langle \nu, \phi_j \rangle_H^2 \\ &\leq e^{-2\lambda_1 t} \sum_{j=1}^{\infty} \langle \nu, \phi_j \rangle_H^2 = e^{-2\lambda_1 t} \|\nu\|_H^2 \\ &\leq \sum_{j=1}^{\infty} \langle \nu, \phi_j \rangle_H^2 = \|\nu\|_H^2. \end{aligned}$$

This shows (9). □

As a consequence, we get $\|e^{At}\|_{Op} \leq 1$ for all $v \in H$ as long as A satisfies Assumption 1. We need this fact in later calculations.

Remark 2.11

The structure of e^{At} and A given by (7) and (8) allows to change the order of those operators. Hence, for every $\nu \in \mathcal{D}(A)$

$$A(e^{At}(\nu)) = e^{At}(A(\nu)) \quad (10)$$

holds.

Next, we will define fractional powers and interpolation spaces. These concepts play an important role in the theory of stochastic PDEs, as they describe the smoothness of the solution process. The following definition is based on [LPS14, p.25ff.] and [DZ14, Appendix A.4.3].

Definition 2.12 (Fractional power and interpolation space)

Let $A : \mathcal{D}(A) \subset H \rightarrow H$ be a linear operator which has an orthonormal basis of eigenfunctions $\{\phi_j\}_{j \geq 1}$ and a series of corresponding real-valued, positive eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \dots$ such that for all $\nu \in \mathcal{D}(A)$

$$A\nu = \sum_{j=1}^{\infty} \lambda_j \langle \nu, \phi_j \rangle \phi_j$$

holds. Then we can define fractional powers A^γ , $\gamma \in \mathbb{R}$ by

$$A^\gamma \nu = \sum_{j=1}^{\infty} \lambda_j^\gamma \langle \nu, \phi_j \rangle \phi_j, \quad \forall \nu \in \mathcal{D}(A^\gamma). \quad (11)$$

Moreover, $\mathcal{D}(A^\gamma)$ is a separable Hilbert space with inner product $\langle u, \nu \rangle_\gamma := \langle A^\gamma u, A^\gamma \nu \rangle$ and norm $\|\nu\|_\gamma := \|A^\gamma \nu\|$ and is called *Interpolation space* of A .

Note that when A satisfies Assumption 1, then $(-A)^\gamma$ can be defined by Definition 2.12 with the corresponding interpolation space $\mathcal{D}((-A)^\gamma)$. Some of the most important properties of interpolations spaces are stated in the next remark without its proofs. For these we refer to [LPS14, Lemma 1.89].

Remark 2.13

Let A satisfy the assumptions from Definition 2.12. Then:

- (i) For every $\nu \in \mathcal{D}(A^\alpha)$ and $\alpha, \gamma \in \mathbb{R}$ with $\alpha > \gamma$

$$\|\nu\|_\gamma \leq \lambda_1^{2(\gamma-\alpha)} \|\nu\|_\alpha < \infty$$

holds. Hence, $\mathcal{D}(A^\alpha) \subset \mathcal{D}(A^\gamma)$.

- (ii) Let $\gamma > 0$. Then, it follows that $\mathcal{D}(A^\gamma) \subset \mathcal{D}(A^0) = H \subset \mathcal{D}(A^{-\gamma})$.

- (iii) For any $u \in \mathcal{D}(A)$ and $\nu \in \mathcal{D}(A^{1/2})$,

$$\langle u, \nu \rangle_{1/2} = \langle A^{1/2} u, A^{1/2} \nu \rangle_H = \langle Au, \nu \rangle_H$$

holds.

Obviously, Equation (10) also holds for A^γ as well. Thus, for $\nu \in \mathcal{D}(A^\gamma)$ we know that $A^\gamma e^{At} \nu = e^{At} A^\gamma \nu$ [see LPS14, Lemma 3.22]. Often these interpolation spaces coincide with Sobolev spaces as we can see in the next example.

Example 2.14 (Fractional power of laplace operator)

We continue Example 2.8. Let A be the Laplace operator on $L^2(0, 1)$ in the context of the heat equation with Dirichlet boundary conditions such that $\mathcal{D}(-A) = H^2(0, 1) \cap H_0^1(0, 1)$. We saw that $(-A)$ has eigenfunctions $\phi_j = \sqrt{2} \sin(\pi j)$ and eigenvalues $\lambda_j = (\pi j)^2$. Hence, its fractional power norm for $u \in \mathcal{D}((-A)^\gamma)$ is given by

$$\|u\|_\gamma^2 = \|(-A)^\gamma u\|_H^2 = \sum_{j=1}^{\infty} \lambda_j^{2\gamma} \langle u, \phi_j \rangle^2 = \sum_{j=1}^{\infty} (\pi j)^{4\gamma} \langle u, \phi_j \rangle^2. \quad (12)$$

[see LPS14, Example 1.90]. Then, $\mathcal{D}((-A)^\gamma)$ is equivalent to $H_0^{2\gamma}(0, 1)$ [see LPS14, Example 10.9], where $H_0^{2\gamma}(0, 1)$ is a fractional Sobolev space.

2.3. Basics of semigroup theory

Semigroups play an important role in the theory of the existence and uniqueness of solutions to semilinear parabolic SPDEs. In this section we briefly introduce the concept of semigroups and discuss conditions under which a linear operator generates a so-called C_0 -semigroup. This section is mainly based on [Hai09] and [EN01]. We start with the definition of a semigroup and additionally with its most common classes: The C_0 -semigroup and the analytic semigroup. Note that, while the following can also be defined for Banach spaces, it is enough in our case to consider only Hilbert spaces. The definition is based on [Hai09, Definition 4.1].

Definition 2.15 (Semigroup)

A *semigroup* $S(t)$ on a Hilbert space H is a family of bounded linear operators $\{S(t)\}_{t \geq 0}$ with the properties

- 1) $S(t) \circ S(s) = S(t + s)$ for any $s, t \geq 0$,
- 2) $S(0) = \mathbb{1}$.

Moreover, a semigroup is called

- *strongly continuous*, if the map $(x, t) \Rightarrow S(t)x$ is strongly continuous,
- *analytic*, if there exists $\theta > 0$ such that the operator-valued map $t \mapsto S(t)$ has an analytic extension to $\{\lambda \in \mathbb{C} : |\arg \lambda| < \theta\}$, satisfies the semigroup property there, and is such that $t \mapsto S(e^{i\phi}t)$ is a strong continuous semigroup for every angle ϕ with $|\phi| < \theta$.

While an analytic semigroup is also very important and common in the theory of SPDEs (for example for proving the existence of a solution when the nonlinearity is not globally Lipschitz continuous [see DZ14, §7.2]), in this work we will focus mainly on the strongly continuous semigroup, also called C_0 -semigroup. The properties (1) and (2) from the previous definition are known as *semigroup properties*.

Remark 2.16 (Contraction semigroup)

As stated in [EN01, Chapter 1, Proposition 5.5] for every C_0 -semigroup, there exist constants $\omega \in \mathbb{R}$ and $M \geq 1$ such that $\|S(t)\|_{\mathcal{O}_p} \leq Me^{\omega t}$ holds for every $t > 0$. If this is satisfied for $\omega = 0$ and $M = 1$, then the C_0 -semigroup is also called *contraction semigroup* [see EN01, Chapter 1, Definition 5.6].

Now that we know how a semigroup and in particular a C_0 -semigroup is defined, we will investigate how to generate one. For this we first introduce the infinitesimal generator (or just generator). The following definition is based on [Hai09, p.29].

Definition 2.17 (Generator of a C_0 -semigroup)

The *generator* L of a C_0 -semigroup is given by

$$Lx = \lim_{t \rightarrow 0} \frac{(S(t)x - x)}{t}$$

on the set $\mathcal{D}(L)$ of all elements $x \in H$ such that the limit exists (in the sense of strong convergence in H).

If L is the generator of a C_0 -semigroup, than

$$x_t = S(t)x_0$$

is the solution to the linear equation

$$\partial_t x = Lx, \quad x(0) = x_0 \in \mathcal{B}$$

[e.g. see Hai09, Proposition 4.7]. It can be shown that such a generator of a C_0 -semigroup is unique in the sense that no two different C_0 -semigroups can have the same generator [e.g. see Hai09, Corollary 4.8].

Remark 2.18

Following [LPS14, Lemma 3.21], it can be shown that if a linear operator A satisfies Assumption 1, then $S(t) := e^{At}$ forms a semigroup with A as its infinitesimal generator.

In the Following we will state characteristics and properties under which a linear operator generates such a strongly continuous semigroup. While the classical theorem by Hille and Yosida [e.g. see DZ14, Theorem A.3] is a good way to theoretically proof whether an operator is the generator of a C_0 -semigroup, it turns out to be rather impractical in applications, since it requires knowledge of the so-called *resolvent* of the operator. A more applicable theorem is given by Lumer and Phillips. The most important property we need for this is the dissipativity of an operator. Its definition is based on [EN01, p.82].

Definition 2.19 (Dissipative operator)

A linear operator $(A, \mathcal{D}(A))$ on a Hilbert space H is called *dissipative* if

$$\|(\lambda - A)x\|_H \geq \lambda \|x\|_H$$

for all $\lambda > 0$ and $x \in \mathcal{D}(A)$.

Using the so-called *duality set* $\mathcal{J}(x) := \{x' \in H' : \langle x, x' \rangle_H = \|x'\|_B^2 = \|x\|_B^2\}$, which by the Hahn-Bach theorem [see EN01, p.87] is nonempty for every $x \in H$, we get a different characterization of dissipative operators which can also be find in [EN01, p.88].

Proposition 2.20

An operator $(A, \mathcal{D}(A))$ is dissipative if and only if for every $x \in \mathcal{D}$ there exists $x' \in \mathcal{J}(x)$ such that

$$\operatorname{Re}\langle Ax, j(x) \rangle_H \leq 0.$$

Recall, that the canonical identification of a Hilbertspace H with its dual H' is given by Riesz representation theorem (see Theorem A.7) and thus $\mathcal{J}(f) = \{f\}$ for $f \in H$. Hence, in case that A is a linear Operator on a Hilbertspace H we need to proof if $\operatorname{Re}\langle Af, f \rangle_H \leq 0$ for all $f \in \mathcal{D}(A)$ to ensure that A is dissipative.

While the Lumer-Phillips theorem can be given in a more general way [e.g. see EN01, Chapter 2, Theorem 3.15] we only need the following variant for the special case when A is selfadjoint, which also can be found in [EN01, p.84].

Theorem 2.21 (Lumer-Phillips theorem)

Let A be a densely defined operator on a Hilbert space H . If both A and its adjoint A^* are dissipative, then the closure $\bar{A}$ of A generates a contraction semigroup on H .

Remark 2.22

Let A be a linear operator of negative kind (e.g. A satisfies Assumption 1). Then A is selfadjoint. Moreover, since all eigenvalues are real-valued and negative and H is a separable Hilbertspace, we get

$$\operatorname{Re}\langle Af, f \rangle_H = \langle Af, f \rangle_H \stackrel{(7)}{=} \left\langle \sum_{k=1}^{\infty} -\lambda_k \langle \phi_k, f \rangle_H \phi_k, f \right\rangle_H = - \sum_{k=1}^{\infty} \lambda_k \langle \phi_k, f \rangle_H^2 \leq 0.$$

Hence, A is dissipative and thus, using Theorem 2.21, its closure $\bar{A}$ generates a C_0 -semigroup. In particular it is a contraction semigroup, since

$$\|S(t)\|_{O_P} = \|e^{At}\|_{O_P} \leq 1,$$

as seen in Lemma 4.10. Since every selfadjoint operator is closed ($\bar{A} = A$) and $\mathcal{D}(A)$ is dense in H , A generates a C_0 -semigroup.

Example 2.23 (Heat semigroup)

We consider the context of Example 2.8. Therefore, we consider the heat equation with Dirichlet boundary conditions on $H = L^2(0, 1)$ and let $A = \Delta$ be the Laplace operator. Moreover, the Domain of A is given by $\mathcal{D}(A) = H^2(0, 1) \cap H_0^1(0, 1)$. $\mathcal{D}(A)$ is dense in $L^2(0, 1)$ since both $H^2(0, 1)$ as well as $H_0^1(0, 1)$ are dense in $L^2(0, 1)$ (see Remark 2.4). Further, since A satisfies Assumption 1 it is selfadjoint and thus $A = A^*$. Now, since we only consider real-valued functions $\nu \in L^2(0, 1)$ we only have to show that $\langle A\nu, \nu \rangle_{L^2(0,1)} \leq 0$ for all $\nu \in \mathcal{D}(A)$ in order to show that A is dissipative (See discussion after Proposition 2.20). Using integration by parts [e.g. see Wer04, p.224], we get

$$\langle A\nu, \nu \rangle_{L^2([0,1])} = \int_0^1 \partial_{xx}\nu(x)\nu(x)dx = - \int_0^1 \partial_x\nu(x)\partial_x\nu(x)dx = - \|\partial_x\nu\|_{L^2([0,1])}^2 \leq 0.$$

Therefore, A is the generator of a contraction semigroup $S(t) = e^{At} = e^{\Delta t}$.

2.4. Space-time white noise

While white noise is a purely mathematical construct which is not found in reality [see ZK17, p. 2] it is still a common concept of stochastic theory. Especially its property of being the 'derivative' of a Brownian motion is been used frequently. Note that we will see in later chapters that SPDEs are represented using generalized Brownian motions and therefore the dynamics of an SPDE can be described using space-time white noise. Therefore, the basic theory for this concept will be developed in this section. It is mainly based on [DZ14, Chapter 4].

A zero-mean gaussian process is uniquely defined by its covariance mapping. Therefore, one way to define white noise is to construct a centered gaussian process $\dot{w} = \{\dot{w}(t)|t \in [0, T]\}$ with covariance function

$$\operatorname{COV}(\dot{w}(s), \dot{w}(t)) = \delta(t - s), \quad \forall s, t \in [0, T], \quad (13)$$

where δ represents the delta-function known from Remark 2.2. Space-time white noise can be expressed similarly by considering a zero-mean stochastic process $\dot{W} = \{\dot{W}(t, x) | t \in [0, T] x \in [0, L]^d\}$ with covariance operator

$$\text{COV}(\dot{W}(s, x), \dot{W}(t, y)) = \delta(t - s)\delta(y - x). \quad (14)$$

Note that $\delta(y - x) = \sum_i^d \delta(y_i - x_i)$ for $x, y \in \mathbb{R}^d$. It is easy to see that a stochastic process with such a covariance mapping cannot exist [e.g. see Kal80, p.10]. Hence, we need a more general approach. Thus in following we consider $\dot{w}$ on a functional space. This leads to the analysis on infinite dimensional spaces - especially infinite dimensional Hilbert spaces. Due to the missing infinite dimensional analogue of the Lebesgue measure the gaussian measure theory needs to be redefined. We refer to [LMP16] and the first part of [Hai09].

2.4.1. H -valued random elements

To define a stochastic process on a general Hilbertspace, we first need to define what an H -valued random element (or equally random variable) is and how we can express the expectation and covariance in a more general way to hold for H -valued random elements. This section states a brief overview of the most important aspects of the theory of infinite dimensional random variables needed for this work based on [DZ14, Chapter 1], [Hai09, Chapter 3] and [LMP16]. For more details of this topic and the topic of measure theory in infinite dimensions, we refer to these sources.

An H -valued random element X is a measurable mapping $X : \Omega \rightarrow H$ between two measurable spaces $(\Omega, \mathcal{F})$ and $(H, \tilde{\mathcal{F}})$. Let X and Y be two H -valued random elements, then $\alpha X + \beta Y$ is also an H -valued random element for all $\alpha, \beta \in \mathbb{R}$. Moreover, $\|X\|_H$ is a real-valued random element [see DZ14, Proposition 1.2]. As in the case of real-valued random elements, we can define the expected value

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) \mathbb{P}(d\omega), \quad (15)$$

where the right hand side of (15) is the Bochner integral with the main property

$$\left\| \int_{\Omega} X d\mathbb{P} \right\|_H \leq \int_{\Omega} \|X\|_H d\mathbb{P}.$$

(see Section A.2 and [DZ14, p.18ff.]). Following [DZ14, p.21f.] we can also state the L^p function spaces in a more general way. $L^1(\Omega) := L^1(\Omega, \mathcal{F}, \mathbb{P}; H)$ is the space of equivalence-classes of all Bochner integrable random elements with norm $\|X\|_{L^1(\Omega)} = \mathbb{E}[\|X\|_H]$ and $L^2(\Omega) := L^2(\Omega, \mathcal{F}, \mathbb{P}; H)$ is the space of equivalence-classes of square Bochner integrable random variable with norm

$$\|X\|_{L^2(\Omega)} = \mathbb{E}[\|X\|_H^2]^{1/2}.$$

Now, let $X, Y \in L^2(\Omega)$. Then we can define the covariance of X and Y by

$$\text{COV}(X, Y) = \int_{\Omega} (X - \mathbb{E}[X])(Y - \mathbb{E}[Y]) d\mathbb{P}$$

and the variance by

$$\text{VAR}(X) = \text{COV}(X, X) = \int_{\Omega} (X - \mathbb{E}[X])^2 d\mathbb{P}.$$

For simplicity we assume that X and Y are centered, such that $\mathbb{E}[X] = \mathbb{E}[Y] = 0$. The covariance $\text{COV}(X, Y)$ can also be expressed by an operator $\tilde{Q}$ via $\text{COV}(X, Y) = \tilde{Q}(X)(Y)$ [see Hai09]. Now using Riesz representation Theorem A.7, there exists a unique operator $Q : H \rightarrow H$ such that

$$\text{COV}(X, Y) = \tilde{Q}(X)(Y) = \langle QX, Y \rangle_H.$$

Q is positive symmetric and nuclear [see DZ14, p.26]. Note that we implicitly used the fact that H is a separable Hilbert space such that we can express the dual space H' of H with H itself.

2.4.2. Q-Brownian motion

With these basic definitions, we will now introduce Brownian motions as a process of H -valued random elements. The following is based on Chapter 4 of [DZ14].

At first, let us assume that Q is a positive, symmetric, nuclear operator which is of trace-class (i.e. $\text{tr}(Q) < \infty$, see Definition A.16). Then, following [Hai09, Proposition 3.15] there exists a gaussian measure on H with Q being its covariance operator. Recall that we can find a series of eigenvalues $\{q_l\}_{l \geq 0}$ of Q such that $Q\phi_l = q_l\phi_l$ for eigenfunctions ϕ_l , $l \geq 0$. Next, let $\{X_t\}_{t \geq 0}$ be a family of H -valued random elements. Note that $X = \{X_t\}_{t \geq 0}$ defined on Ω is called H -valued stochastic process on $[0, T]$ and $X(\cdot, \omega)$ for $\omega \in \Omega$ is called trajectory of X_t . We now define a Q -Brownian motion. The definition below is based on [DZ14, Definition 4.2]

Definition 2.24 (Q-Brownian motion)

A H -valued stochastic process $W = \{W(t)\}$, $t \geq 0$ is called Q -Brownian motion if

- (i) $W(0) = 0$,
- (ii) W has continuous trajectories,
- (iii) W has independent increments,
- (iv) $W(t) - W(s) \sim \mathcal{N}(0, (t - s)Q)$.

Note that this definition is similar to the one of a real-valued Brownian motion.

Remark 2.25

When W is a Q -Brownian motion, then for every $u \in H$, $\langle W, u \rangle_H = \langle W(t), u \rangle_H$, $t \geq 0$ is a real-valued Brownian motion with

$$\mathbb{E}[\langle W(t), u \rangle_H \langle W(s), \nu \rangle_H] = t \wedge s \langle Qu, \nu \rangle_H. \quad (16)$$

Even though W is a stochastic process with values in an infinite dimensional space, the next theorem shows that we can still generate such a process with a series of one-dimensional real-valued Brownian motions. This property will be useful in later chapters. The theorem is based on [DZ14, Proposition 4.3].

Theorem 2.26

Assume W is a Q -Brownian motion. Then the following statements hold:

i) W is a gaussian Process on H and

$$\mathbb{E}[W(t)] = 0, \quad \text{VAR}(W(t)) = tQ$$

ii) For arbitrary $t \geq 0$, W has the expansion

$$W(t) = \sum_{l=1}^{\infty} \sqrt{q_l} \phi_l \beta_l(t) \tag{17}$$

Where $\{\phi_l\}_{l \geq 1}$ build an orthonormal basis of H and $\beta_l(t) = \frac{1}{\sqrt{q_l}} \langle W(t), \phi_l \rangle_H$ are real-valued Brownian motions mutually independent on $(\Omega, \mathcal{F}, \mathbb{P})$. The series (17) is convergent in $L^2(\Omega, \mathcal{F}, \mathbb{P})$.

We do not state the proof here but refer to the proof of [DZ14, Proposition 4.3]. But we want to point out that for this to hold, it is crucial for Q to be of trace-class. Otherwise the series (17) will not be well-defined in the sense that it will not converge in $L^2(\Omega, \mathcal{F}, \mathbb{P})$, since the series $\sum_{l=1}^{\infty} q_l = \infty$. There are cases, where it is reasonable to choose a covariance operator Q which is not trace-class, for example $Q = I$. Recall that the identity operator is not trace-class when H is infinite dimensional. In this case the process defined by Definition 2.24 is not a well-defined process on H . To cover this important case, we will now extend the previous theory.

2.4.3. Generalized Brownian motion

Now let Q no longer be necessarily of trace-class. In the case $\text{tr}(Q) = \infty$ we cannot rely on Definition 2.24 anymore, since the gaussian law given in Definition 2.24(iv) cannot be expressed by Q . Nevertheless, we can still consider a Brownian motion as a stochastic process $W = (W(t))_{t \geq 0}$ with continuous trajectories, independent increments and a law $\mathcal{L}$ that satisfies

$$\mathcal{L}(W(t) - W(s)) = \mathcal{L}(W(t - s)) \quad \text{and} \quad \mathcal{L}(W(t)) = \mathcal{L}(-W(t)), \quad \forall t \geq s \geq 0.$$

[see DZ14, Definition 4.1]. We will make use of this representation to define the so-called generalized Brownian motion .

Definition 2.27 (Generalized Brownian motion)

Let $a \in H$ and let W be a H -valued Brownian motion. Further, let $a \mapsto W_a(t)$ with $W_a(t) = \langle a, W(t) \rangle_H$, $t \geq 0$ be a linear transformation from H to the space of stochastic processes, such that its values are real-valued Brownian motions. Further, let it be continuous in the sense that for $t \geq 0$, $\{a_n\}_{n \geq 0} \in H$ with $\lim_{n \rightarrow \infty} a_n = a$ it follows that

$$\lim_{n \rightarrow \infty} \mathbb{E}[|W_{a_n}(t) - W_a(t)|^2] = 0.$$

Then the linear transformation is called *generalized Brownian motion*.

This definition indicates that there exists a self-adjoint, positive definite Hilbert-Schmidt operator Q such that

$$\mathbb{E}[W_a(t)W_b(s)] = s \wedge t \langle Qa, b \rangle_H, \quad s, t \geq 0, \quad a, b \in H$$

[see DZ14, p.84]. For the next expressions we define the space $H_0 = Q^{1/2}(H)$ with the induced norm.

Remark 2.28

- i) If $Q = I$ is the identity operator on H , then the generalized Brownian motion is called *cylindrical Brownian motion*. Its covariance structure is given by

$$\text{COV}(W_a(t), W_b(s)) = s \wedge t \langle a, b \rangle_H, \quad s, t \geq 0, \quad a, b \in H.$$

- ii) We can define $W_a(t)$ via

$$W_a(t) = \sum_{l=1}^{\infty} \langle Q^{1/2} \phi_l, a \rangle_H \beta_l(t). \quad (18)$$

The series (18) indicates the importance of the $Q^{1/2}(H)$. Now, let $\tilde{H}$ be a larger separable Hilbert space such that $Q^{1/2}(H)$ is dense and continuously embedded in $\tilde{H}$, such that the embedding $\iota : Q^{1/2}(H) \rightarrow \tilde{H}$ is a Hilbert-Schmidt operator (see Definition A.17). In this case, the operator $Q_1 = \iota^*$ is trace-class (see Remark A.18). Then the following theorem given by [DZ14, Proposition 4.7] holds.

Theorem 2.29

Let $Q^{1/2}(H)$ and $\tilde{H}$ be as defined above with Hilbert-Schmidt embedding ι . Then the formular

$$W_t = \sum_{j=1}^{\infty} Q^{1/2} \phi_j \beta_j(t) \quad (19)$$

defines an $\tilde{H}$ -valued Brownian motion. Moreover, if $Q_1 = \iota^*$ is the covariance operator of W , then the spaces $Q_1^{1/2}(\tilde{H})$ and $Q^{1/2}(H)$ are the same.

Note that $Q^{1/2}(H)$ is called *reproducing kernel*. When considering $H = L^2(D)$ for $D \in \mathbb{R}^d$, $Q^{1/2}$ is actually an integral operator that determines the spatial correlation of W [see DZ14, §4.1.3]. This theorem shows that even for covariance operators Q which are not trace-class (like the identity operator on an infinite dimensional space), one can define a ι^* -Brownian motion on a bigger space whose reproducing kernel still relies on Q . Since $Q_1^{1/2}(H) = Q^{1/2}(H)$ (where Q_1 is defined in Theorem 2.29), with a little abuse of notation we can suppress the fact that a cylindrical Brownian motion is defined only on a larger space. Thus, in the remaining work, we will not distinguish between trace-class operators and non-trace-class operators Q unless we explicitly state otherwise.

Remark 2.30 (Space-time white noise)

Space-time white noise is defined as the 'derivative' of cylindrical Brownian motion. Since $Q = I$, there is no spatial correlation. To see this consider that for $H = L^2(0, 1)$ $Q = I$ is an integral operator which produces $g(\varphi, \nu) = \varphi \wedge \nu$ as its kernel [see DZ14, Example 4.9]. Thus, its derivative has the delta function δ as its kernel representing the spatial correlation. In particular, this leads to rough paths. On the other hand, if Q is trace-class, then there is more spatial correlation leading to so-called colored noise. In that case, the paths are smoother.

2.5. The stochastic integral

In the theory of real-valued processes, the stochastic integral can be constructed by using simple functions and their property to build a dense subset of the space of all adapted functions in L^2 [see Tra21]. A similar approach can be used to construct the H -valued stochastic integral driven by a Q -Wiener Process as a continuous, square integrable H -valued martingale. The goal of this chapter is to create the environment in which such a stochastic integral is well-defined. While the theory is based on a Q -Wiener Process, where Q is a trace-class operator (e.g. $\text{tr}(Q) < \infty$), it can be extended to the case of generalized Wiener Processes by the fact that it defines a Q -Wiener process on a larger space $\tilde{H} \supset H$ with Hilbert-Schmidt embedding, as seen before. Hence, in the following, we do not distinguish between these variants. The following is based on [DZ14, §4.2].

Let us first define the space $H_0 = Q^{1/2}(H)$. Further, Let $L_2^0 = L_2(H_0, H)$ denote the space of all Hilbert-Schmidt operators from H_0 to H . Similarly, let $L_2(H) = L_2(H, H)$ be the space of all Hilbert-Schmidt operators from H to H . L_2^0 itself is a separable Hilbert space, equipped with the norm

$$\|\Phi\|_{L_2^0} = \|\Phi Q^{1/2}\|_{L_2(H)} = \text{tr} \left((\Phi Q^{1/2})(\Phi Q^{1/2})^* \right).$$

Further, let $\Phi = \{\Phi(t)\}_{t \in [0, T]}$ be a L_2^0 -valued process and define the norm

$$\|\Phi\|_t := \mathbb{E} \left[\int_0^t \|\Phi(s)\|_{L_2^0}^2 ds \right]^{1/2}.$$

We first consider elementary (or simple) processes ϕ (e.g. $\phi(t) = \phi_m, \forall t \in (t_m, t_{m+1}]$). Then, the stochastic integral can be defined by

$$\int_0^t \phi(s) dW(s) = \sum_{m=0}^{k-1} \phi_m (W(t_{m+1} \wedge t) - W(t_m \wedge t)). \quad (20)$$

It can be shown [see DZ14, Proposition 4.20] that if ϕ is an elementary L_2^0 -valued process and $\|\phi\|_t \leq \infty$ for all $t \in [0, T]$ that then the stochastic integral defined in (20) is a continuous square integrable H -valued martingale on $[0, T]$ and

$$\mathbb{E} \left[\left\| \int_0^t \phi(s) dW(s) \right\|_H^2 \right] = \|\phi\|_t^2 = \mathbb{E} \left[\int_0^t \|\phi(s)\|_{L_2^0}^2 ds \right]. \quad (21)$$

Now, similar to the real-valued case, one can show that elementary processes are L_2^0 -predictable (e.g. they are adapted and stochastically continuous [see DZ14, p.72]) and that the space of L_2^0 -predictable elementary processes ϕ with $\|\phi\|_t < \infty$ is dense in the space of all L_2^0 -predictable elementary processes ϕ with $\|\phi\|_t < \infty$. Moreover, for all L_2^0 -predictable elementary processes ϕ with $\|\phi\|_t < \infty$ there exist a sequence $\{\phi_n\}$ of L_2^0 -predictable elementary processes ϕ_n such that $\|\phi_n - \phi\|_t \rightarrow 0$ for $n \rightarrow \infty$. Therefore, we can extend the definition of the stochastic integral to all L_2^0 -predictable processes ϕ with $\|\phi\|_t < \infty$. In addition, (21) also holds for such processes.

2.5.1. Stochastic convolution

Knowing what a stochastic integral in infinite dimensions is, we will now define the stochastic convolution, which is an important concept for the remaining work. The following is mainly based on [DZ14, §5.1.2].

Definition 2.31 (Stochastic convolution)

Let W be a Q -cylindrical Brownian motion and $S(\cdot)$ be a C_0 -semigroup generated by a linear operator A . the process

$$W_A(t) = \int_0^t S(t-s)dW(s), \quad t \geq 0, \quad (22)$$

is called *stochastic convolution*.

By definition $S(t)$ is a bounded linear operator from H to H and Since the space of bounded linear operators from H to H is a subspace of $L_2^0 = L_2^0(H_0, H)$ [see DZ14, p.96], we know that $S(t) \in L_2^0$ for all $t \in [0, T]$. Now, if we assume A satisfies Assumption 1, then it is a generator of a C_0 -semigroup (see Remark 2.22) with representation $S(t) = e^{At}$. Further assume

$$\sum_{l=1}^{\infty} \frac{q_l}{\lambda_l} < \infty$$

is satisfied for λ_l as defined in Assumption 1 and eigenvalues q_l of Q . Then we have

$$\begin{aligned} \int_0^T \|S(t)\|_{L_2^0}^2 dt &= \int_0^T \text{tr}(S(t)QS^*(t)) dt = \int_0^T \sum_{l=1}^{\infty} e^{-2\lambda_l t} q_l dt \\ &\stackrel{(1)}{=} \sum_{l=1}^{\infty} q_l \int_0^T e^{-2\lambda_l t} dt = \sum_{l=1}^{\infty} q_l \left[-\frac{e^{-2\lambda_l t}}{2\lambda_l} \right]_0^T \\ &= \sum_{l=1}^{\infty} q_l \left(\frac{1}{2\lambda_l} - \frac{e^{-2\lambda_l T}}{2\lambda_l} \right) \leq \sum_{l=1}^{\infty} \frac{q_l}{\lambda_l} < \infty, \end{aligned}$$

where (1) follows as a special case of fubinis theorem using the counting measure and the lebesgue measure. Thus, from [DZ14, Theorem 5.2] we know that W_A is gaussian, continuous in mean square and has a predictable version with values in H . Moreover, by (21) we know that

$$\mathbb{E} \left[\left\| \int_0^t S(t-s)dW(s) \right\|_H^2 \right] = \int_0^t \|S(t-s)\|_{L_2^0}^2 ds = \int_0^t \text{tr}(S(t-s)QS(t-s)^*) ds. \quad (23)$$

Note that, by considering $\mathbb{1}_{r \geq s} S(t-r) \in L_2^0$ with

$$\int_0^t \|\mathbb{1}_{r \geq s} S(t-r)\|_{L_2^0} ds = \int_0^t \mathbb{1}_{r \geq s} \|S(t-r)\|_{L_2^0} ds$$

, we get a useful extension of (23) given by

$$\begin{aligned} \mathbb{E} \left[\left\| \int_s^t S(t-r) dW(r) \right\|_H^2 \right] &= \mathbb{E} \left[\left\| \int_0^t \mathbb{1}_{r \geq s} S(t-r) dW(r) \right\|_H^2 \right] \\ &= \int_0^t \|\mathbb{1}_{r \geq s} S(t-r)\|_{L_2^0}^2 dr = \int_s^t \|S(t-r)\|_{L_2^0}^2 dr. \end{aligned} \quad (24)$$

Now since A satisfies Assumption 1 we know that A is selfadjoint and thus we can show that W_A actually is more regular, in the sense that its values are in $\mathcal{D}((-A)^\gamma)$ for some $\gamma \in (0, 1)$. For this let $H = L^2(0, 1)$. Following [DZ14, §5.5.1], it can be shown that if A is an elliptic operator of order 2 (like the Laplace operator) then there exists a constant c_γ dependent on γ such that

$$\|(-A)^\gamma S(t)x\|_{L^2(0,1)} \leq c_\gamma t^\gamma \|x\|_{L^2(0,1)}$$

is satisfied. Moreover, consider the next typical assumption concerning SPDEs [e.g. see JK09a, Assumption 5].

Assumption 2

Let A satisfying Assumption 1. Then there exists a sequence $q_l \geq 0$, $l \geq 1$, of positive real numbers and a real number $\gamma \in (0, 1)$ such that

$$\sum_{l=1}^{\infty} \frac{q_l}{\lambda_l^{1-2\gamma}} < \infty \quad (25)$$

is satisfied. Further, for all eigenfunctions ϕ_l

$$\|\phi_l(x)\| \leq c$$

holds for $l \in \mathbb{N}$, $x \in [0, 1]$.

So, let A satisfying Assumption 2, then it can be shown that W_A is a continuous process with values in $\mathcal{D}((-A)^\gamma)$ [see DZ14, Theorem 5.25].

Remark 2.32

In other literature, [e.g. AL15], (25) is often written as

$$\left\| A^{((\beta-1)/2)} \right\|_{L_2^0} < \infty,$$

for $\beta > 0$, where L_2^0 and L_2 are spaces of Hilbert-Schmidt operators defined in Section 2.5. When choosing $\beta = 2\gamma$, it is easy to see that this representation corresponds with the one from the previous assumption.

$$\left\| A^{((\beta-1)/2)} \right\|_{L_2^0} = \left\| A^{((\beta-1)/2)} Q^{1/2} \right\|_{L_2} = \text{tr}(A^{(\beta-1)} Q) = \sum_{l=1}^{\infty} \lambda_l^{\beta-1} q_l = \sum_{l=1}^{\infty} (\lambda_l)^{2\gamma-1} q_l.$$

Example 2.33

Let us extend Example 2.14 and 2.23. Thus, let $A = \Delta$ be the Laplace operator in the context of the heat equation with Dirichlet boundary conditions on $L^2(0, 1)$. A is a generator of a C_0 -semigroup $S(t) = e^{At}$ and has eigenvalues $-\lambda_l$ with $\lambda_l = (\pi l)^2$ and eigenfunctions $\phi_l(x) = \sqrt{2} \sin(\pi l x)$ for $l \in \mathbb{N}$, $x \in [0, 1]$. Further, let W be a cylindrical Brownian motion such that $Q = I$ and thus $q_l = 1$ for all $l \in \mathbb{N}$. Then

$$\sum_{l=1}^{\infty} \frac{q_l}{\lambda_l^{1-2\gamma}} = \sum_{l=1}^{\infty} \frac{1}{(\pi l)^{2(1-2\gamma)}} < \infty \iff \gamma \in (0, 1/4)$$

and $\sqrt{2} \sin(\pi l x) \leq \sqrt{2}$ for all $x \in [0, 1]$, $l \in \mathbb{N}$. Hence, W_{Δ} with $W_{\Delta}(t) = \int_0^t e^{\Delta(t-s)} dW(s)$ is a continuous process with values in $\mathcal{D}((-A)^{\gamma})$ for $\gamma \in (0, 1/4)$.

3. Semilinear parabolic SPDEs

The goal of this chapter is to understand semilinear parabolic SPDEs and their main properties so that we can create an environment in which the existence and uniqueness of solutions to such differential equations can be proved. The following is mainly based on [LR17, Chapter 2 and 4], [LPS14, Chapter 10] and [DZ14, Chapter 7].

Let us consider the following abstract functional equation

$$\begin{cases} dX(t) = [AX(t) + F(t, X(t))] dt + G(t, X(t))dW(t), & t \in [0, T] \\ X(0) = x_0 \end{cases} \quad (26)$$

with linear operator $A : \mathcal{D}(A) \subset H \rightarrow H$, possibly nonlinear functions $F : [0, T] \times H \rightarrow H$ and $G : H \rightarrow L_2^0(H)$ and Q -Brownian motion W . Q does not need to be trace-class, such that W can be a cylindrical Brownian motion with associated space-time white noise. Note that throughout this work, we will call F *nonlinearity* even if it might also be a linear function. When A satisfies Assumption 1 such that it is the infinitesimal generator of a C_0 -semigroup, then (26) is also called *stochastic semilinear evolution equation* [see LPS14, p.431]. Equation (26) is said to have *multiplicative noise* if G depends on the solution process X and *additive noise* if G is at most dependent on t [e.g. see LPS14, p.431]. In this work, we will consider only problems with additive noise. More precisely, G is a constant and for simplicity $G \equiv 1$. Moreover, only mappings $F : H \rightarrow H$ which depend only on X but not on t will be considered. Hence, Problem (26) will be of the form

$$\begin{cases} dX(t) = [AX(t) + F(X(t))] dt + dW(t), & t \in [0, T] \\ X(0) = x_0. \end{cases} \quad (27)$$

Example 3.1 (Semilinear stochastic heat equation)

We extend Example 2.33. Let $A = \Delta$ be the Laplace operator, let $H = L^2(0, 1)$, let F be as described above and let W be a cylindrical Brownian motion. Then the *semilinear stochastic heat equation* with Dirichlet boundary conditions and with additive noise is given by

$$\begin{cases} dX(t) = [\Delta X(t) + F(X(t))] dt + dW(t), & t \in [0, T] \\ X(t)(0) = X(t)(1) = 0 \\ X(0) = x_0. \end{cases} \quad (28)$$

Note that extending the heat equation to the semilinear stochastic case does not affect the properties of A described in previous examples.

Remark 3.2

Note that we call the upper equations *semilinear*, since we are linear in the highest-order derivative and its coefficient is independent of X or any of its derivatives [see LR17, p.39].

As discussed in [LR17, §4.1] partial differential equations and thus also its stochastic versions do not have a general solution theory but instead require individual analysis. PDEs and hence SPDEs can be classified in

- *parabolic* ($\frac{\partial u}{\partial t} + Au = A_1 u + f$),
- *hyperbolic* ($\frac{\partial^2 u}{\partial t^2} + Au = A_1 u + B \frac{\partial u}{\partial t} + f$),
- *elliptic* ($Au = f$)

equations, where A is a so-called elliptic operator (See [LR17, Definition 4.1.2], $f \in L^2((0, T); H)$, and $A_1 : H \rightarrow H$, $B : H \rightarrow H$ are bounded linear operators. Note that it can be shown that for example the Laplace operator on $L^2(0, 1)$ with Dirichlet boundary condition is an elliptic operator. The names of these classes are based on the conic section [see LR17, p.146f.]. Each of those classes needs its own analysis of the solution. In this work we will focus on parabolic SPDEs.

3.1. Solutions of semilinear parabolic SPDEs

In this section, we will first describe the different approaches of solution terms and then discuss the assumptions and results of existence and uniqueness of mild solutions to parabolic semilinear SPDEs of the form (27). This section is mainly based on [DZ14].

3.1.1. Types of solutions

For the following, we will consider problems of the form (27). Note that a *predictable* process is a measurable mapping from $([0, T] \times \Omega, \mathcal{F}_t)$ to $(H, \mathcal{H})$, where $\{\mathcal{F}_t\}_{t \in [0, T]}$ is the filtration of the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ [see DZ14, p.71f.].

Definition 3.3 (Strong solution ([DZ14] p. 122))

An H -valued predictable process $X(t), t \in [0, T]$ is called *strong solution* to (27) if $X(t)$ takes values in $D(A)$, a.e.,

$$\int_0^T |AX(s)| ds < \infty, \quad \mathbb{P} - a.s.$$

and for $t \in [0, T]$

$$X(t) = x_0 + \int_0^t (AX(s) + F(X(s))) ds + W(t), \quad \mathbb{P} - a.s.$$

As stated in [DZ14, p. 122] this definition only works for H -valued Brownian motions and thus only allows Q with $\text{tr}(Q) < \infty$. Hence, it is not suitable for space-time white noise (e.g. when W is a cylindrical Brownian motion). Moreover, in many cases A is the Laplace operator or a variation of it such that solution X must be two times weakly differentiable. Since this is rarely the case, we need a broader Solution concept, which allows also solutions that do not necessarily need to be two times differentiable.

Similar to the introduction of weak derivatives the idea here is to shift the required regularity of X towards a different element, which is sufficiently smooth. In this case we need $z \in \mathcal{D}(A^*) \subset H$, where A^* is the adjoint of A . We can then consider $\langle X(t), z \rangle_H$ instead of $X(t)$ and insert the definition of the strong solution to get

$$\langle X(t), z \rangle_H = \langle x_0, z \rangle_H + \int_0^t (\langle AX(s), z \rangle_H + \langle F(X(s)), z \rangle_H) ds + \langle W(t), z \rangle_H.$$

Note that

$$\langle AX(s), z \rangle_H = \langle X(s), A^* z \rangle_H$$

such that this construction allows X to be a process in H . Additionally, the noise $\langle W(t), z \rangle_H$ is a generalized Brownian motion and thus also well-defined for $Q = I$ as we saw in Section 2.4.3. We formulize this idea in the next definition.

Definition 3.4 (Weak solution [DZ14] p. 122)

An H -valued process $X(t), t \in [0, T]$ is said to be a *weak solution* of (27) if the trajectories of $X(\cdot)$ are $\mathbb{P}$ -a.s. Bochner integrable and if for all $z \in D(A^*)$ and all $t \in [0, T]$ we have

$$\langle X(t), z \rangle_H = \langle x_0, z \rangle_H + \int_0^t (\langle X(s), A^* z \rangle_H + \langle F(X(s)), z \rangle_H) ds + \langle W(t), z \rangle_H, \quad (29)$$

$\mathbb{P} - a.s.$

Note that in this work we will consider only selfadjoint linear operator, such that $A = A^*$. The next kind of solution is called *mild-solution*. It is based on a semigroup approach (see Section 2.3) and uses the stochastic convolution (see Section 2.5.1).

Definition 3.5 (Mild solution ([DZ14] p. 187))

Let $S(\cdot)$ be a C_0 -Semigroup in H . A predictable H -valued process $X(t), t \in [0, T]$ is said to be a *mild solution* of (27) if

$$\mathbb{P} \left(\int_0^T |X(s)|^2 ds < \infty \right) = 1$$

and, for arbitrary $t \in [0, T]$, we have

$$X(t) = S(t)x_0 + \int_0^t S(t-s)F(X(s))ds + \int_0^t S(t-s)dW(s), \quad \mathbb{P} - a.s. \quad (30)$$

Remark 3.6

A strong solution is always also a weak solution and a weak solution is always also a mild solution. In addition, in our context the two concepts of weak and mild solution are interchangeable in the sense that a weak solution is also a mild solution and vice versa [see DZ14, Theorem 6.5].

In the context of SPDEs, it is common to use the mild solution as the basis for the question of the existence and uniqueness of a solution. Thus, the results from the semigroup Section 2.3 and the stochastic convolution Section 2.5.1 will be used in the following.

3.1.2. Existence and uniqueness of a mild solution

Having learned which variants of solutions can be found and which are reasonable for SPDEs to consider, we will now state the necessary preconditions that our SPDE must have in order to have a unique solution. We will concentrate on the existence and uniqueness of the mild solution given in Definition 3.5. While the section could be kept more general, we focus here on the special case of semilinear parabolic SPDEs with additive noise i.e. equation of the kind (27) with (Q) not necessarily being of trace-class. We will concentrate on the case, where the nonlinear map F is globally Lipschitz continuous. We do not give any proofs in this section but rather refer to the corresponding sections in [DZ14, Chapter 7].

Recall that if A satisfies Assumption 1 and thus is of negative kind, it generates a C_0 -semigroup $S(t), t \in [0, T]$ on H (See Remark 2.22). Moreover, we know that $S(t) \in L_2^0$ for every $t \in [0, T]$ (see discussion of Section 2.5.1). For the existence of a solution, we also need the existence of a square integrable mapping which bounds $\|S(t)\|_{L_2^0}$ from above for any $t \in [0, T]$.

Assumption 3

For any $t > 0$ and $x \in H$ there exists a locally square integrable mapping

$$K : [0, T] \rightarrow [0, +\infty), \quad t \rightarrow K(t),$$

such that

$$\|S(t)\|_{L_2^0} \leq K(t), \quad t > 0. \quad (31)$$

Example 3.7

This example is based on [DZ14, Example 7.6]. We continue Example 3.1. In particular, let $A = \Delta$ be the Laplace operator with eigenvalues $-\lambda_l = -\pi^2 l^2$ for $l \geq 1$ generating the C_0 -semigroup $S(\cdot)$ given by $S(t) = e^{At}$. Then

$$\begin{aligned} \|S(t)\|_{L_2^0}^2 &= \text{tr}(S^2(t)) = \sum_{l=1}^{\infty} \langle S^2(t)\phi_l, \phi_l \rangle_{L^2(0,1)} = \sum_{l=1}^{\infty} \|S(t)\phi_l\|_{L^2(0,1)}^2 \\ &= \sum_{l=1}^{\infty} \sum_{h=1}^{\infty} e^{-2\pi^2 h^2 t} \langle \phi_l, \phi_h \rangle_{L^2(0,1)}^2 = \sum_{h=1}^{\infty} e^{-2\pi^2 h^2 t} =: K^2(t). \end{aligned}$$

Thus, K is obviously locally square integrable and thus Assumption 3 is satisfied.

Next, we will state assumptions on $F : H \rightarrow H$ that ensure that F is, in particular, a globally Lipschitz continuous mapping.

Assumption 4

The mapping $F : H \rightarrow H, \nu \mapsto F(\nu)$ is measurable from $(H, \mathcal{B}(H))$ to $(H, \mathcal{B}(H))$ and there exists a constant $C > 0$ such that for all $\nu, v \in H$ we have

$$\|F(\nu) - F(v)\|_H \leq \|\nu - v\|_H \quad \text{and} \quad \|F(\nu)\|_H \leq C(1 + \|\nu\|_H). \quad (32)$$

Following [DZ14, §7.1.1] and especially [DZ14, Theorem 7.5], we will now state the theorem about existence and uniqueness of a solution $(X_t)_{t \geq 0}$ satisfying (30). For this let us first define the solution space $\mathcal{H}_p$ for $p \geq 2$ as the Banach space of all H -valued predictable processes Y defined on the time interval $[0, T]$ such that

$$\left(\sup_{t \in [0, T]} \mathbb{E} [\|Y(t)\|_H^p] \right)^{1/p} < \infty$$

[see DZ14, p.189].

Theorem 3.8 (Existence of unique mild solution)

Let Assumptions 1, 3 and 4 be satisfied and let $p \geq 2$. Then for an arbitrary $\mathcal{F}_0$ -measurable initial condition x_0 such that $\mathbb{E}[\|x_0\|^p] \leq \infty$ there exists a unique mild solution X of (27) in $\mathcal{H}_p$ and there exists a constant C_T , independent of x_0 such that

$$\sup_{t \in [0, T]} \mathbb{E}[\|X(t)\|^p] \leq C_T(1 + \mathbb{E}[\|\xi\|^p]).$$

Finally, if there exists $\alpha \in (0, 1/2)$ such that

$$\int_0^1 s^{-2\alpha} K^2(s) ds < \infty, \tag{33}$$

where K is the function from Assumption 3, then the solution $X(\cdot)$ is continuous $\mathbb{P}$ -a.s.

We refer to [DZ14, p.197ff.] for the proof. When we consider $H = L^2(0, 1)$, a common class of nonlinearities is given by the so-called *Nemytskii Operators*, which allow to transfer results of a real-valued function onto an abstract $L^2(0, 1)$ -valued mapping.

Definition 3.9 (Nemytskii operator)

Let $f : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$, $(x, y) \mapsto f(x, y)$ be a function such that

- i) $f(\cdot, y) : [0, 1] \rightarrow \mathbb{R}$ is measurable for all $y \in \mathbb{R}$,
- ii) $f(y, \cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is continuous for almost all $x \in [0, 1]$.

Then, the operator $F : L^2(0, 1) \rightarrow L^2(0, 1)$, $\nu \mapsto F(\nu)$, given by

$$F(\nu)(x) = f(x, \nu(x)),$$

for every $\nu \in L^2(0, 1)$ is called *Nemytskii operator*.

Note, that the previous definition is based on a more general approach stated in [Rou05, §1.3]. Note that we will only consider one-argument Nemytskii operators, such that $F : L^2(0, 1) \rightarrow L^2(0, 1)$ is given by $F(\nu)(x) = f(\nu(x))$. The next lemma, which is based on [LPS14, Lemma 3.30], states that it is enough to consider a globally Lipschitz continuous real-valued function in order for the Nemytskii operator to be globally Lipschitz continuous.

Lemma 3.10

Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is globally Lipschitz continuous. Then the Nemytskii operator $F : L^2(0, 1) \rightarrow L^2(0, 1)$, given by $F(\nu)(x) = f(\nu(x))$, $\nu \in L^2(0, 1)$, $x \in \mathbb{R}$ is well-defined and globally Lipschitz continuous.

The proof can be found in [LPS14, p.105].

Corollary 3.11

Let $f \in C_0^\infty(\mathbb{R})$, then the Nemytskii operator $F : L^2(0, 1) \rightarrow L^2(0, 1)$ induced by f , such that $F(\nu)(x) = f(\nu(x))$ for all $\nu \in H$, $x \in (0, 1)$ is globally Lipschitz continuous.

Proof. Let $f \in C_0^\infty(\mathbb{R})$, then f is continuous. Moreover, f' is also continuous and has compact support and thus is bounded. Thus, there exists $M > 0$ such that $f'(z) \leq M$ for all $z \in \mathbb{R}$. Hence, using the mean-value Theorem A.24 it follows that for any $x, y \in \mathbb{R}$ there exists a $\xi \in [x, y]$ such that

$$|f(x) - f(y)| \leq f'(\xi) |x - y| \leq M |x - y|,$$

where M is independent of x and y . Therefore, f is globally Lipschitz continuous and applying Lemma 3.10 F is also globally Lipschitz continuous. $\square$

Example 3.12

Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} e^{1/(x^2-1)}, & x \in (-1, 1) \\ 0, & \text{otherwise,} \end{cases} \quad (34)$$

[e.g. see Wer04, p.203]. Then, $f \in C_0^\infty(\mathbb{R})$ and thus following the previous corollary the corresponding Nemytskii operator $F : L^2(0, 1) \rightarrow L^2(0, 1)$ given by $F(\nu)(x) = f(\nu(x))$ is globally Lipschitz continuous.

Example 3.13 (One-dimensional semilinear stochastic heat equation)

We continue Example 3.1. In particular, let $H = L^2(0, 1)$ and let $A = \Delta = \frac{d^2}{dx^2}$ be the Laplace operator with Dirichlet boundary condition, let W be a cylindrical Brownian Motion, let $f \in C_0^\infty(\mathbb{R})$ and $F : L^2(0, 1) \rightarrow L^2(0, 1)$ the Nemytskii operator induced by f and let $x_0 = 0$. Then (28) becomes

$$\begin{cases} \frac{d}{dt}X(t) = \left[\frac{d^2}{dx^2}X(t) + f(X(t)) \right] dt + dW(t), & x \in [0, 1], \quad t \in [0, T] \\ X(t)(0) = X(t)(1) = 0 \\ X(0) = 0. \end{cases} \quad (35)$$

(35) is a one-dimensional semilinear stochastic heat equation. Combining Example 2.8, Example 3.7 and Corollary 3.11 we see that Assumptions 1, 3 and 4 are satisfied. Hence, all conditions for Theorem 3.8 are satisfied and thus there exists a solution X to (35).

In many cases it is too restrictive to consider a globally Lipschitz continuous function F defined on the whole space H . Therefore, we now briefly summarize the case where F is locally Lipschitz continuous on a subspace $V \subset H$. We're following [DZ14, §7.2.1].

Remark 3.14

Let V be a Banach space, continuously and densely embedded in H , and let the initial condition $\xi \in V$ be satisfied. Let A generate a C_0 -semigroup and let A_V be the part of A in V . Further, let the following Assumptions be satisfied.

- 1) Let either A_V generate a C_0 -semigroup $S(\cdot)$ on V or A generate an analytic semigroup $S_V(\cdot)$ on V [see DZ14, Hypothesis 7.3].
- 2) Let $V \subset \mathcal{D}(F)$ and let $F_V : V \rightarrow V$ be locally Lipschitz continuous and bounded on bounded subsets of V [see DZ14, Hypothesis 7.4(i)].
- 3) There exists an increasing function $a : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that

$$[A_{V,n}x + F(x + y), x^*]_{V \times V^*} \leq a(\|y\|_V)(1 + \|x\|_V)$$

for all $x, y \in V$, $n \in \mathbb{N}$ and $x^* \in \partial \|x\|_V$, with $\partial \|x\|_V$ being the subdifferential of the V -norm at point $x \in V$ and $[\cdot, \cdot]_{V \times V^*}$ being the duality form [see DZ14, Hypothesis 7.4(ii)].

Then according to [DZ14, Theorem 7.7], there exists a mild solution to equation (27). The solution is an element of $C([0, +\infty); V)$, if A_E generates a C_0 -semigroup on V and an element of $C([0, +\infty); V) \cap L_{loc}^\infty([0, +\infty); V)$ if A generates an analytic semigroup on V .

Example 3.15

As in Example 3.13 we consider the one-dimensional semilinear stochastic heat equation with additive space-time white noise and Dirichlet boundary conditions given by (35) on $L^2(0, 1)$ with A being the Laplace operator $A = \frac{d^2}{dx^2}$. Let $V = C([0, 1])$. Following [DZ14, Example 7.8] A_V is the generator of an analytic Semigroup $S_V(\cdot)$ on V . Moreover, when F is the Nemytskii operator induced by $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $F(\nu)(x) = f(\nu(x))$ for all $\nu \in L^2(0, 1)$ and $x \in X$, where f is a polynomial of odd degree and negative leading coefficient, then, following [DZ14, Example 7.8] and [DZ14, Example 7.10], the other assumptions stated in the previous remark are satisfied and there exist a mild solution X of (35) with values in $C([0, +\infty); C([0, 1])) \cap L_{loc}^\infty([0, +\infty); C([0, 1]))$.

4. Numerical approximations of semilinear parabolic SPDEs

We saw that under proper Assumptions there exists a unique solution to Problem (27). Often, however, this solution cannot be represented explicitly, which is especially a problem if one wants to simulate the solution process. Therefore, this chapter provides a brief introduction to numerical methods which allow to simulate such solution processes. Note that the solution of (27) is a stochastic process with values in an infinite dimensional Hilbert space $\mathcal{D}((-A)^\gamma)$. Thus, in the first step, the dimensions have to be reduced so that we consider only a finite dimensional problem. Then we can discretize the time dimension to obtain a discrete version of the continuous finite dimensional problem. Once both steps have been performed, we call the resulting approximation of the problem a fully discretized approximation. But before we look at these steps in more detail, we will first introduce the terms of convergence that describe the quality of the approximation.

4.1. Strong and weak convergence

Following are the two most used ways of measuring the approximation error - the strong (or mean-square) and the weak convergence. Depending on what is of interest, one may either use the strong convergence, in case one is interested in pathwise properties of the solution, or the weak convergence, in case one is more interested in distributional properties of the solution. Research considering strong convergence can be found for example in [JK09a], [JKW11] and [JK09b] and research considering weak convergence can be found for example in [AL15], [Bré17] and [Wan14].

Let $N \in \mathbb{N}$ denote the spatial discretization parameter, such that our space discretization gets better for larger N . For calculation purposes, define an $\eta \in \mathbb{R}_+$ such that $\eta \rightarrow 0$ for $N \rightarrow \infty$. We will further define N and η in the discussion of Galerkin methods. Further, let $M \in \mathbb{N}$ denote the temporal discretization, let $0 = t_0 < t_1 < \dots < t_M = T$ be a equidistant partition of $[0, T]$, $0 < T < \infty$ and let $h := t_{k+1} - t_k = T/M$ for all $k = 0, \dots, M-1$ denote the grid width. Obviously, the grid gets finer for larger M . Then, denote by $X_{N,M}$ the (fully) discrete approximation of the solution X of Equation (27). Note that in the following $N, M \rightarrow \infty$ means $\min\{N, M\} \rightarrow \infty$. Recall that for $X \in H$ $\|X\|_{L^2(\Omega)}$ is given by

$$\|X\|_{L^2(\Omega)} = \mathbb{E}_\Omega \left[\|X\|_H^2 \right]^{1/2}. \quad (36)$$

We can then define the strong convergence as follows.

Definition 4.1 (Strong convergence)

Let $N, M \in \mathbb{N}$, h and η defined as before. Let $X_{N,M}$ be the fully discrete approximation of the solution X of the semilinear SPDE (27). Then $X_{N,M}$ *converges strongly* (or *in mean-value*) to X if

$$\sup_{t_k} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \rightarrow 0, \quad \text{for } N, M \rightarrow \infty. \quad (37)$$

It converges strongly with order p_1 in time and p_2 in space if there exists a constant $C > 0$ independent of N and M such that

$$\sup_{t_k} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \leq C(h^{p_1} + \eta^{p_2}).$$

Moreover, for given t_k , $k = 0, \dots, M$ $\|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)}$ is called *strong error at time t_k* .

Remark 4.2

When $H = L^2(0, 1)$, then the error defined in Equation (37) can be expressed by

$$\|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} = \mathbb{E} \left[\int_0^1 |X(t_k, x) - X_{N,M}(t_k, x)|^2 dx \right]^{1/2}. \quad (38)$$

Note that the strong error depends on the difference of each path, as seen in (36), and therefore pathwise properties are important. This can be loosened so that distributional properties are more important. To do this we will now define the weak convergence.

Definition 4.3 (Weak convergence)

Let $N, M \in \mathbb{N}$, h and η defined as before. Let $X_{N,M}$ be the fully discrete approximation of the solution X of the semilinear SPDE (27). Then, $X_{N,M}$ *converges weakly* to X at time T if

$$|\mathbb{E}[\varphi(X(T))] - \mathbb{E}[\varphi(X_{N,M}(T))]| \rightarrow 0, \quad \text{for } N, M \rightarrow \infty,$$

for all sufficiently smooth functions $\varphi : H \rightarrow \mathbb{R}$. It converges weakly at time T with order $\tilde{p}_1$ in time and order $\tilde{p}_2$ in space if there exists a constant $C > 0$ independent of N and M such that

$$|\mathbb{E}[\varphi(X(T))] - \mathbb{E}[\varphi(X_{N,M}(T))]| \leq C(h^{\tilde{p}_1} + \eta^{\tilde{p}_2}) \quad (39)$$

for all sufficiently smooth functions $\varphi : H \rightarrow \mathbb{R}$. The left hand side of (39) is called *weak error*.

Next to individual convergence rates for time and space, we can also consider an overall convergence rate, such that the strong convergence with overall convergence rate p is given by

$$\sup_{t_k} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \leq C(K_{h,\eta})^p.$$

and the weak convergence with overall convergence rate $\tilde{p}$ is given by

$$|\mathbb{E}[\varphi(X(T))] - \mathbb{E}[\varphi(X_{N,M}(T))]| \leq C(K_{h,\eta})^{\tilde{p}},$$

with $K_{h,\eta} > 0$ being dependent on h and η .

Remark 4.4

Often it is of interest to get an optimal convergence rate with respect to computational costs. For example consider the strong convergence of a numerical scheme

with computational costs equal to its arithmetical operations $K = M \cdot N$ and let $h = 1/M$ and $\eta = 1/N$ such that the strong convergence rate is of the form

$$C\left(\frac{1}{M^{p_1}} + \frac{1}{N^{p_2}}\right).$$

Then, following [KP92, p.213], the optimal total convergence rate with respect to the computational costs is given by $\frac{p_1 p_2}{p_1 + p_2}$ with $M = K^{\frac{p_2}{p_1 + p_2}}$ $N = K^{\frac{p_1}{p_1 + p_2}}$ and we get the upper bound

$$CK^{-\frac{p_1 p_2}{p_1 + p_2}}.$$

For example, let $p_1 = 1$ and $p_2 = 1/2$, then the optimal strong convergence rate with respect to the computational costs is given by $1/3$.

Following the discussion on Page 3 in [Kru14], we will show the connection between weak and strong convergence rates. Let $\varphi : H \rightarrow \mathbb{R}$ be an arbitrary sufficiently smooth functional. In particular, let φ be Fréchet differentiable, see Definition A.23, and let its derivative be of linear growth (i.e. there exists a constant C such that $\|\varphi'(x)\|_{Op} \leq C(1 + \|x\|_H)$). Then, consider the following calculations:

$$\begin{aligned} & |\mathbb{E}[\varphi(X(T))] - \mathbb{E}[\varphi(X_{N,M}(T))]| \\ &= |\mathbb{E}[\varphi(X(T)) - \varphi(X_{N,M}(T))]| \\ &= |\mathbb{E}[\varphi(X_{N,M}(T) + (X(T) - X_{N,M}(T))) - \varphi(X_{N,M}(T))]| \\ &\leq \mathbb{E}[\|\varphi(X_{N,M}(T) + (X(T) - X_{N,M}(T))) - \varphi(X_{N,M}(T))\|] \\ &\stackrel{(1)}{\leq} \mathbb{E}\left[\sup_{\xi \in I} C(1 + \|\xi\|_H) \|X(T) - X_{N,M}(T)\|_H\right] \\ &\stackrel{(2)}{\leq} \mathbb{E}\left[(C(1 + \|X(T)\|_H + \|X_{N,M}(T)\|_H))^2\right]^{1/2} \mathbb{E}\left[\|X(T) - X_{N,M}(T)\|_H^2\right]^{1/2} \\ &\stackrel{(3)}{\leq} 2C \left(1 + \|X(T)\|_{L^2(\Omega)} + \|X_{N,M}(T)\|_{L^2(\Omega)}\right) \|X(T) - X_{N,M}(T)\|_{L^2(\Omega)}, \end{aligned}$$

where we used the mean-value Theorem A.24 with

$$I = \{X_{N,M}(T) + s(X(T) - X_{N,M}(T)) | s \in [0, 1]\}$$

in (1). (2) follows by the fact that there exists $s \in [0, 1]$ such that

$$\begin{aligned} \|\xi\|_H &= \|X_{N,M}(T) + s(X(T) - X_{N,M}(T))\|_H \\ &\leq s\|X(T)\|_H + (1-s)\|X_{N,M}(T)\|_H \leq \|X(T)\|_H + \|X_{N,M}(T)\|_H. \end{aligned}$$

(3) then follows by using the inequality $(x + y)^2 \leq 2x^2 + 2y^2$ and $(x + y)^{1/2} \leq x^{1/2} + y^{1/2}$.

Thus, if $X_{N,M}$ converges strongly to X with strong convergence rate p , then the previous calculations indicate that

$$|\mathbb{E}[\varphi(X(T))] - \mathbb{E}[\varphi(X_{N,M}(T))]| \leq \tilde{C} \|X(T) - X_{N,M}(T)\|_{L^2(\Omega)} \leq \tilde{C} \hat{C} (K_{h,\eta})^p,$$

with $2C(1 + \|X(T)\|_{L^2(\Omega)} + \|X_{N,M}(T)\|_{L^2(\Omega)}) \leq \tilde{C}$ and $\hat{C}$ being a constant given by the strong convergence. Consequently, we can always find an overall weak convergence order $\tilde{p}$ which is at least as fast as the overall strong convergence order p .

Remark 4.5

Note that testfunctions are also called smooth functions. There is no general definition of its space $\mathcal{T}$ but rather it is chosen as smooth as needed [see Kru14]. We already defined testfunctions as elements of $C_0^\infty(H; \mathbb{R})$ in Chapter 2. Another commonly used space is $C_b^2(H; \mathbb{R})$, the space of all bounded continuous functions $\varphi : H \rightarrow \mathbb{R}$ which are first and second order bounded Fréchet differentiable [e.g. see Bré17]. The choice of the space of testfunctions strongly influence the rate of weak convergence as shown in [Bré17]. If $\mathcal{T}$ is sufficiently smooth, a general rule of thumb is that the weak convergence is twice as fast as the strong convergence, see for example [Wan14].

The above definitions are based on [ZK17, p.80f.] and describe full discretization convergence. Similarly one could define convergence only for time discretization or only for space discretization. Further, Remark 4.4 indicates that convergence results of fully discrete schemes are combined convergence results of temporal and spatial discretization. The next section is devoted to the discretization of the spatial dimension.

4.2. Spatial discretization

A spatial discretization is usually done by the so-called *Galerkin*-method, based on the weak solution. Following [Kru14, §3.2] and [LPS14, §3.5 and §10.6] we will introduce the main idea behind the Galerkin method: To reduce the dimension by projecting the solution space $\mathcal{D}((-A)^\gamma)$ onto a finite dimensional subspace $\tilde{H} \subset \mathcal{D}((-A)^\gamma)$ and find a weak solution $\tilde{X}$ with values in $\tilde{H}$ such that for all $\tilde{\nu} \in \tilde{H}$ the equation

$$\tilde{X}(t) = \tilde{S}(t)\tilde{X}(0) + \int_0^t \tilde{S}(t-s)\tilde{F}(\tilde{X}(s))ds + \int_0^t \tilde{S}(t-s)d\tilde{W}(s)$$

is satisfied, where $\tilde{S}(\cdot)$ is a C_0 -semigroup generated by the projection $\tilde{A}$ of A . Note that even though we call it spatial discretization, it is much more a truncation of the dimensions of an infinite dimensional space.

In most cases a solution X to Equation (27) does not take values in $\mathcal{D}(-A)$. Thus, we introduced the weak solution to allow also a less regular H -valued solution satisfying Equation 29 for every $z \in \mathcal{D}(-A)$. Within this equation, we consider the inner product $\langle X(s), (-A)\nu \rangle_H$ for $s \in [0, t]$. Using Remark 2.13(iii) we see that $\langle \psi, (-A)\nu \rangle_H = \langle (-A)^{1/2}\psi, (-A)^{1/2}\nu \rangle_H = \langle \psi, \nu \rangle_{1/2}$ for $\psi \in \mathcal{D}((-A)^{1/2})$, $\nu \in D(-A)$ which leads to

$$\langle X(t), \nu \rangle_H = \langle x_0, \nu \rangle_H + \int_0^t (\langle X(s), \nu \rangle_{1/2} + \langle F(X)(s), \nu \rangle_H) ds + \langle W(s), \nu \rangle_H.$$

Doing it like this requires the solution process X to be in $\mathcal{D}((-A)^{1/2})$, which in general is still too restrictive, especially when considering space-time white noise. Hence, we will consider the more general equation

$$\langle X(t), \nu \rangle_H = \langle x_0, \nu \rangle_H + \int_0^t (\langle X(s), \nu \rangle_\gamma + \langle F(X)(s), \nu \rangle_H) ds + \langle W(s), \nu \rangle_H,$$

where X only needs to be a $\mathcal{D}((-A)^\gamma)$ -valued process. Recall that in our case the weak solution and the mild solution are interchangeable (see Remark 3.6), so we will consider the mild solution here. In the following, let Assumptions 1, 3 and 4 be satisfied such that (27) has a $D((-A)^\gamma)$ -valued solution X .

First, we will briefly introduce the Galekin method in a general way and then present the spectral Galerkin method as a special case in more details. While the existence of a finite dimensional solution of the spectral Galerkin method will be shown here, the strong convergence result will be postponed to the next Chapter. We start by defining the so-called *Ritz-Projection* as a special case of a projection operator (see Definition A.19).

Definition 4.6

Let $\tilde{P} : \mathcal{D}((-A)^\gamma) \rightarrow \tilde{H}$ be a projection operator, which satisfies

$$\langle \tilde{P}\nu, \psi \rangle_\gamma = \langle \nu, \psi \rangle_\gamma, \quad \forall \nu \in \mathcal{D}((-A)^\gamma), \psi \in \tilde{H}.$$

Then $\tilde{P}$ is called *Ritz-Projection* or *Ritz-Projector*.

Further, define the matrix operator $\tilde{A} = \tilde{P}A|_{\tilde{H}} : \tilde{H} \rightarrow \tilde{H}$, the nonlinearity $\tilde{F} = \tilde{P}F|_{\tilde{H}} : \tilde{H} \rightarrow \tilde{H}$, the Brownian motion $\tilde{W} := \tilde{P}W$ and the initial value $\tilde{x}_0 := \tilde{P}x_0$. Then we can define the finite dimensional version of (27) by

$$\begin{cases} d\tilde{X}(t) = [\tilde{A}\tilde{X}(t) + \tilde{F}(\tilde{X}(t))] dt + d\tilde{W}(t), & t \in [0, T] \\ \tilde{X}(0) = \tilde{x}_0. \end{cases} \quad (40)$$

Remark 4.7

- (i) In practice, we will define a sequence of finite dimensional subspaces $(\tilde{H}_\eta)_{\eta \geq 0}$ of $\mathcal{D}((-A)^\gamma)$ such that our finite dimensional approximation $\tilde{X}_\eta$ corresponding to $\tilde{H}_\eta$ gets closer to the real solution X for $\eta \rightarrow 0$.
- (ii) In the setting of Remark 4.7(i) it can be shown that under additional assumptions to $\tilde{P}$ [see Kru14, Assumptions 3.3 and 3.5] the solution of (40) converges strongly towards the infinite dimensional solution for $\eta \rightarrow 0$. See [Kru14, Theorem 3.10] for more details.

There are two main concepts on how to create these subspaces - the *Galerkin finite element method (Galerkin FEM)* which is widely applicable since it does not rely on eigenvalues and eigenfunctions of A and the *spectral Galerkin method* which is often been used when eigenvalues and eigenfunctions of A are known like in the case when A is the Laplace operator on $H = L^2(0, 1)$ (see Example 2.8). Note that in both cases the additional assumptions of Remark 4.7(ii) are naturally satisfied (see [Kru14, §3.2] for a more details). We will use the spectral Galerkin method in the rest of the paper. Thus, the Galerkin FEM will only be briefly introduce in the next example. It is based on [Kru14, Example 3.6], [LPS14, §3.6, §10.8] and [AL15].

Example 4.8 (Galerkin Finite Element method)

For simplicity, let us only consider $H = L^2(0, 1)$ (it can be expanded on higher dimensions $H = L^2(D)$ with $D \in \mathbb{R}^d$, $d = 1, 2, 3$, [see Kru14, Example 3.6]). Further, let $\eta = 1/N$ denote the grid width of the partition $0 = y_0 < y_1 < \dots < y_N = 1$ of $[0, 1]$. Then $\tilde{H} = H_\eta$ is defined as the space of all continuous, piecewise linear functions $\ell_\eta : [0, 1] \rightarrow \mathbb{R}$ with $\ell_\eta(0) = \ell_\eta(1) = 0$, such that ℓ is linear on every piece $[y_{j+1} - y_j]$, $j = 0, \dots, N-1$. Due to the importance of the mesh width η we denote the Ritz operator $\tilde{P} = P_\eta$, the projected linear operator $\tilde{A} = A_\eta : H_\eta \rightarrow H_\eta$ and the nonlinearity $\tilde{F} = F_\eta$. Then we seek to find the finite element approximation $\tilde{X} = X_\eta$ of

$$dX_\eta = [A_\eta X_\eta + F_\eta(X_\eta(t))] dt + dW(t).$$

As stated in for example [AL15] A_η is the generator of an analytical semigroup S_η such that there exists a mild solution X_η which satisfies

$$X_\eta(t) = S_\eta(t)X_\eta(0) + \int_0^t S_\eta(t-s)F_\eta(X_\eta(s))ds + \int_0^t S_\eta(t-s)dW(s).$$

Furthermore, under sufficient assumptions this solution converges weakly to X with order 2β for $\beta \in (0, 1/2)$ when considering the space of testfunctions $C_b^2(H; \mathbb{R})$ [see AL15, Theorem 1.1].

4.2.1. Spectral Galerkin method

Recall that we assume A to satisfy Assumption 1. Hence, we know that its eigenfunctions build a orthonormal basis of H . When the eigenfunctions $(\phi_j)_{j \geq 0}$ of A are known, then a natural way to define finite dimensional Subspaces $H_N \subset H$ is by taking the span of the first N eigenfunctions of A

$$H_N := \text{span}\{\phi_1, \dots, \phi_N\}.$$

The following is based on [Kru14, Example 3.7] and [LPS14, Chapter 3 and 10] but generalized to $\mathcal{D}((-A)^\gamma)$.

By definition (see Section 2.2), A^γ has the same eigenfunctions $(\phi_l)_{l \geq 0}$ as A with corresponding negative eigenvalues given by $\{-\lambda_l^\gamma\}_{l \geq 0}$. Hence, the finite dimensional spaces H_N are the same regardless of considering A , $-A$, A^γ or $-A^\gamma$ for $\gamma \in (0, 1)$. Using the assumption that the eigenvalues of $(-A)$ are real-valued, positive and ordered (e.g. $0 < \lambda_1 \leq \lambda_2 \leq \dots$) we can choose $\eta = \lambda_N^{-\gamma}$ such that $\eta \rightarrow 0$ for $N \rightarrow \infty$. To emphasize the importance of N we denote the finite dimensional projections of X , A , F , W and x_0 by $X_N = \tilde{X}$, $A_N := \tilde{A} : H_N \rightarrow H_N$, $F_N := \tilde{F} : H_N \rightarrow H_N$, $W_N = \tilde{W}$ and $x_0^N := \tilde{x}_0$. Note that $(-A_N)$ is a diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_N$. In particular, this setup leads to an easy to handle Ritz-projection. Let $\tilde{P} : \mathcal{D}((-A)^\gamma) \rightarrow H_N$ be the Ritz-projector as defined in Definition 4.6. For any $\nu \in \mathcal{D}((-A)^\gamma)$ we can represent $\tilde{P}$ by using the basis $\{\phi_l\}_{l=1, \dots, N}$ of H_N as

$$\tilde{P}(\nu) = \sum_{l=1}^N \langle \tilde{P}(\nu), \phi_l \rangle_H \phi_l.$$

and further, note that for every $\nu \in \mathcal{D}((-A)^\gamma)$

$$\begin{aligned}
 \langle \tilde{P}(\nu), \phi_l \rangle_H &= \langle \tilde{P}(\nu), (-A)^{2\gamma} (-A)^{-2\gamma} \phi_l \rangle_H = \langle \tilde{P}(\nu), (-A)^{2\gamma} \frac{1}{\lambda_l^{2\gamma}} \phi_l \rangle_H \\
 &= \frac{1}{\lambda_l^{2\gamma}} \langle (-A)^\gamma \tilde{P}(\nu), (-A)^\gamma \phi_l \rangle_H = \frac{1}{\lambda_l^{2\gamma}} \langle \tilde{P}(\nu), \phi_l \rangle_\gamma \\
 &\stackrel{(1)}{=} \frac{1}{\lambda_l^{2\gamma}} \langle \nu, \phi_l \rangle_\gamma = \langle \nu, (-A)^{2\gamma} \frac{1}{\lambda_l^{2\gamma}} \phi_l \rangle_H \\
 &= \langle \nu, (-A)^{2\gamma} (-A)^{-2\gamma} \phi_l \rangle_H = \langle \nu, \phi_l \rangle_H,
 \end{aligned}$$

where we used that ϕ_l is an eigenfunction of $(-A)^{-2\gamma}$ with eigenvalue $\lambda^{-2\gamma}$ and the definition of the Ritz-Projection. Consequently,

$$\tilde{P}(\nu) = \sum_{l=1}^N \langle \nu, \phi_l \rangle_H \phi_l.$$

Let $H_N^\perp := \text{span}\{\phi_{N+1}, \phi_{N+2}, \dots\}$. Then $H_N^\perp$ is the orthogonal complement of H_N (see Definition A.20) such that for any $\psi \in H$ there exists

$$p := \sum_{l=1}^N \langle \psi, \phi_l \rangle_H \phi_l \in H_N \quad \text{and} \quad q := \sum_{l=N+1}^{\infty} \langle \psi, \phi_l \rangle_H \phi_l \in H_N^\perp$$

such that $\psi = p + q$. Note, that an orthogonal projector of H is defined as an operator $P_N : H \rightarrow H_N$ such that $P_N(\psi) = p$ (see Definition A.22). Therefore, $\tilde{P}$ matches the orthogonal projector P_N of H restricted to $\mathcal{D}((-A)^\gamma)$ (e.g. $\tilde{P} = P_N|_{\mathcal{D}((-A)^\gamma)}$). Hence, without loss of generality, we will use the orthogonal projector P_N instead of the Ritz projector $\tilde{P}$. Moreover, with similar arguments as before we see that $(I - P_N) : H \rightarrow H_N^\perp$ is given by $(I - \tilde{P})(\nu) = \sum_{l=N+1}^{\infty} \langle \nu, \phi_l \rangle_H \phi_l$.

Note that using Representation (17), $P_N W = W_N$ can be expressed by

$$W_N(t) = \sum_{l=1}^N \sqrt{q_k} \phi_l \beta_l(t),$$

where $\beta_l, l = 1, \dots, N$ are iid one-dimensional real-valued Brownian motions. Altogether, Problem (40) becomes

$$\begin{cases} dX_N(t) = [A_N X_N(t) + F_N(X_N(t))] dt + dW_N(t), & t \in [0, T] \\ X_N(0) = x_0^N. \end{cases} \quad (41)$$

We will now show that Theorem 3.8 still holds such that a solution to (41) exists. In particular, we will show that Assumptions 1, 3 and 4 are also satisfied for the finite dimensional projections.

Lemma 4.9

Let $A : H \rightarrow H$ and $F : H \rightarrow H$ such that Assumptions 1, 3 and 4 are satisfied. Then the finite dimensional projections $A_N = P_N A|_{H_N} : H_N \rightarrow H_N$ and $F_N = P_N F|_{H_N} : H_N \rightarrow H_N$ also satisfy these assumptions.

Proof. Note that by definition A_N has eigenfunctions ϕ_l , with eigenvalues $(-\lambda_l)$, $l = 1 \dots N$ such that $\lambda_i \leq \lambda_l$ for all $i < l$. Moreover, by construction $\{\phi_l\}_{l=1, \dots, N}$ is an orthonormal basis of H_N . Therefore, Assumption 1 is satisfied and A_N is of negative kind. This also ensures, that A_N is the generator of a C_0 -semigroup $S_N(t) = e^{A_N t}$, $t \in [0, 1]$. Further, we have

$$\begin{aligned} \|S_N(t)\|_{L_2^0} &= \text{tr}(S_N(t)Q_N S_N(t)^*) = \sum_{l=1}^N e^{-2\lambda_l t} q_l \\ &\leq N e^{-2\lambda_1 t} \max_{l=1, \dots, N} q_l < \infty. \end{aligned}$$

Hence, Assumption 3 is naturally satisfied. Lastly, note that for $\nu \in H_N$ we have the equality $\|\nu\|_H = \|\nu\|_{H_N}$. By using the linearity of P_N and that $\|P_N\|_{Op} = 1$ (see Theorem A.21), we get

$$\begin{aligned} \|F_N(\nu) - F_N(\psi)\|_H &= \|P_N(F(\nu) - F(\psi))\|_H \\ &\leq \|P_N\|_{Op} \|F(\nu) - F(\psi)\|_H \leq C \|\nu - \psi\|_H \end{aligned}$$

and

$$\|F_N(\nu)\|_H \leq \|P_N\|_{Op} \|F(\nu)\|_H \leq C(1 + \|\nu\|_H).$$

and thus Assumption 4 also is satisfied. $\square$

Since A_N satisfies Assumption 1 it is the generator of a C_0 -semigroup $S_N(t) = e^{A_N t}$, $t \in [0, T]$. The following quiet useful inequality will be used as a smoothening factor in later calculations in Chapter 5.

Lemma 4.10

Suppose Assumption 1 is satisfied for A_N and let $0 < t \leq T$. Then

$$\|A_N e^{A_N t}\|_{Op} \leq \frac{1}{t}. \quad (42)$$

Proof. Let $\nu \in \mathcal{D}(A_N)$. Then

$$\|A_N e^{A_N t}\|_{Op} = \sup_{\|\nu\|_{H_N} \leq 1} \|A_N e^{A_N t} \nu\|_{H_N}.$$

First, note that for all $N \in \mathbb{N}$ and all $k = 1, \dots, N$

$$e^{-\lambda_k t} \leq 1/(\lambda_k t + 1) \leq 1/(\lambda_k t)$$

holds and thus $\lambda_k e^{-\lambda_k t} \leq 1/t$. Hence,

$$\|A_N e^{A_N t} \nu\|_{H_N}^2 = \sum_{k=1}^N (\lambda_k e^{-\lambda_k t})^2 \langle e_k, \nu \rangle_{H_N}^2 \leq \frac{1}{t^2} \sum_{k=1}^N \langle e_k, \nu \rangle_{H_N}^2 = \frac{1}{t^2} \|\nu\|_{H_N}^2.$$

Consequently,

$$\|A_N e^{A_N t}\|_{Op} = \sup_{\|\nu\|_{H_N} \leq 1} \|A_N e^{A_N t} \nu\|_{H_N} \leq \sup_{\|\nu\|_{H_N} \leq 1} \frac{1}{t} \|\nu\|_{H_N} = \frac{1}{t}.$$

$\square$

Another quiet useful lemma is the following. It and its proof is based on [LPS14, Lemma 1.94] and [Kru14, p.57].

Lemma 4.11

Let A be a linear operator which suffices Assumption 1, let $\nu \in \mathcal{D}((-A)^\gamma)$, $\gamma \in (0, 1)$ and Let $P_N : H \rightarrow H_N$ be the previously introduced orthogonal projection operator. Then

$$\|(I - P_N)\nu\|_H \leq \frac{1}{(\lambda_{N+1})^\gamma} \|(-A)^\gamma \nu\|_H$$

holds.

Proof. For any $\nu \in \mathcal{D}((-A)^\gamma)$ we know that

$$(I - P_N)(\nu) = \sum_{l=N+1}^{\infty} \langle \nu, \phi_l \rangle_H \phi_l.$$

Hence, with $\lambda_{N+1} \leq \lambda_j$ for all $l > N + 1$ we get

$$\begin{aligned} \|(I - P_N)(\nu)\|_H^2 &= \sum_{l=N+1}^{\infty} \langle \nu, \phi_l \rangle_H^2 = \sum_{l=N+1}^{\infty} \langle \nu, (-A)^\gamma (-A)^{(-\gamma)} \phi_l \rangle_H^2 \\ &= \sum_{l=N+1}^{\infty} \lambda_j^{-\gamma} \langle \nu, (-A)^\gamma \phi_l \rangle_H^2 \leq \lambda_{N+1}^{-\gamma} \sum_{l=N+1}^{\infty} \langle (-A)^\gamma \nu, \phi_l \rangle_H^2 \\ &= \lambda_{N+1}^{-\gamma} \|(-A)^\gamma \nu\|_{H_N^\perp}^2 \leq \lambda_{N+1}^{-\gamma} \|(-A)^\gamma \nu\|_H^2, \end{aligned}$$

where we used that due to orthogonality $\|\nu\|_H^2 = \|\nu\|_{H_N}^2 + \|\nu\|_{H_N^\perp}^2$. Thus, the assertion holds. $\square$

Now, let $\mathcal{H}_p^N$ for $p \geq 2$ be the Banach space of all H_N -valued predictable processes Y_N defined on the time interval $[0, T]$ such that

$$\left(\sup_{t \in [0, T]} \mathbb{E} [\|Y_N(t)\|_{H_N}^p] \right)^{1/p} < \infty.$$

Then Theorem 3.8 can be applied to the spektral Galerkin approximation, as the following corollary shows.

Corollary 4.12

Let Assumptions 1, 3 and 4 be satisfied and let $p \geq 2$. Then for an arbitrary $\mathcal{F}_0$ -measurable initial condition x_0 such that $\mathbb{E}[\|x_0\|^p] < \infty$ there exists a unique mild solution X_N of (41) in $\mathcal{H}_p^N$ and there exists a constant C_T , independent of x_0^N such that

$$\sup_{t \in [0, T]} \mathbb{E}[\|X_N(t)\|^p] \leq C_T(1 + \mathbb{E}[\|x_0^N\|^p]).$$

Then there exists a unique mild solution X_N to Problem (41) of the form

$$dX_N(t) = [A_N X_N(t) + F_N(X_N(t))] dt + dW_N(t).$$

where $(S_N(t))_{t \in [0, T]} = (e^{A_N t})_{t \in [0, T]}$.

Proof. Due to Lemma 4.9 we know that A_N , F_N and $S_N(t)$ satisfy Assumptions 1, 3 and 4. Moreover, note that

$$\mathbb{E}[\|x_0^N\|_{H_N}^p] = \mathbb{E}[\|P_N x_0\|_{H_N}^p] \leq \mathbb{E}[\|P_N\|_{O_p}^p \|x_0\|_{H_N}^p] = \mathbb{E}[\|x_0\|_{H_N}^p] < \infty.$$

Therefore, the conditions of Theorem 3.8 are satisfied in the setting of Problem (41) and thus the assertion follows directly by Theorem 3.8. $\square$

The strong convergence of the spectral Galerkin method will be shown in Chapter 5.

4.3. Temporal discretization

In addition to the spatial component, the temporal component must also be discretized in order to apply a numerical scheme. Thanks to the above explanations, we consider here an N -dimensional SODE whose solution exists and is given by a continuous stochastic process with N -dimensional values. Therefore, classical methods can be used to discretize SODEs. In this section, one of the most used temporal discretization methods - the *semi-implicit Euler-Maruyama method* - will be briefly introduced. For this let $M \in \mathbb{N}$ and let $(t_k)_{k=1,\dots,M}$ be a partition of $[0, T]$ such that $0 = t_0 < t_1 < \dots < t_M = T$ with $t_{k+1} - t_k = T/M =: h$ is satisfied. The following derivation of the basic idea behind the semi-implicit Euler-Maruyama method is based on [LPS14, §3.1].

4.3.1. The semi-implicit Euler-Maruyama method

First consider the following initial value problem. Find the solution $u : [0, T] \rightarrow \mathbb{R}^d$ such that

$$\begin{cases} \frac{du}{dt} = f(u) \\ u(0) = u_0, \quad u_0 \in \mathbb{R}^d \end{cases} \quad (43)$$

with $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ holds. Using [LPS14, Lemma 3.1], we see that the solution (if existent) has the form $u(t) = u_0 + \int_0^t f(u(s))ds$, $t \in [0, T]$. If we use the grid points $(t_k)_{k=1,\dots,M}$, we can write $u(t_{k+1})$ as

$$u(t_{k+1}) = u_0 + \sum_{j=0}^k \int_{t_j}^{t_{j+1}} f(u(s))ds = u_{t_k} + \int_{t_k}^{t_{k+1}} f(u(s))ds.$$

A simple and convenient way to approximate this is to choose f to be constant on the intervals $[t_k, t_{k+1})$ (or $(t_k, t_{k+1}]$) for $k = 0, \dots, M-1$. If we choose $f(u(s)) = f(u(t_k))$ for $s \in [t_k, t_{k+1})$ we get

$$u(t_{k+1}) \approx u_{t_k} + hf(u(t_k)). \quad (44)$$

If we choose on the other hand $f(u(s)) = f(u(t_{k+1}))$ for $s \in (t_k, t_{k+1}]$ we get

$$u(t_{k+1}) \approx u_{t_k} + hf(u(t_{k+1})). \quad (45)$$

Note that $\int_{t_k}^{t_{k+1}} ds = t_{k+1} - t_k = h$. (44) is called *explicit Euler method* - explicit in the sense that only already known quantities are used for the calculation. (45) is called *implicit Euler method* - implicit in the sense that one must solve a nonlinear system of equations with d equations to get $u(t_{k+1})$ [see LPS14, p.90].

Now consider the slightly more complex initial value problem

$$\begin{cases} \frac{du}{dt} = Au + f(u) \\ u(0) = u_0, \quad u_0 \in \mathbb{R}, \end{cases}$$

where A is a linear operator of negative kind (see Assumption 1). The solution u (if existent) is then given by

$$u(t) = u_0 + \int_0^t Au(s) + f(u(s))ds$$

and thus as before we have

$$u(t_{k+1}) = u(t_k) + \int_{t_k}^{t_{k+1}} Au(s) + f(u(s))ds.$$

Now we can approximate $u(t_{k+1})$ by approximating Au implicitly (i.e. $Au(s) = Au(t_{k+1})$ for all $s \in (t_k, t_{k+1}]$) and $f(u)$ explicitly (i.e. $f(u(s)) = f(u(t_k))$ for all $s \in [t_k, t_{k+1})$). The resulting scheme

$$u(t_{k+1}) \approx u(t_k) + hAu(t_{k+1}) + hf(u(t_k))$$

is called the *semi-implicit Euler method*.

Remark 4.13

Note that the implicit method has higher computational costs than the explicit one, since a possibly nonlinear d -dimensional system of equations must additionally be solved in each step. Nevertheless, the implicit method is often used in practice. This is because it allows a more stable procedure and thus a coarser temporal grid. The semi-implicit method tries to combine both advantages. By considering $f(u)$ explicitly, only a much less computationally expensive linear system has to be solved in each step. Choosing $f(u)$ sufficiently regular, the term Au is the one that is more sensitive to grid adjustments. Hence, one then uses the stability advantage of the implicit method for Au [see LPS14, p.96]. For more information about the stability of explicit and implicit methods we refer to [LPS14, p.95f.] for the deterministic case and to [LPS14, p.335] for the stochastic case.

That said, let X_N be a strong solution of (40). Then X_N has the form

$$X_N(t) = x_0 + \int_0^t A_N X_N(s) + F_N(X_N(s))ds + \int_0^t dW_N(s)$$

and thus, as before

$$X_N(t_{k+1}) = X_N(t_k) + \int_{t_k}^{t_{k+1}} A_N X_N(s) + F_N(X_N(s))ds + \int_{t_k}^{t_{k+1}} dW_N(s).$$

Note that $\int_{t_k}^{t_{k+1}} dW_N(s) = W_N(t_{k+1}) - W_N(t_k) =: \Delta W_N(t_k)$ is an increment of a Brownian motion and thus $\Delta W_N(t_k) \sim \mathcal{N}(0, h)$. Moreover, A_N is a linear operator of negative kind. Hence, by approximating AX_N by the implicit Euler method and $F_N(X)$ by the explicit Euler method we get the following so-called *semi-implicit Euler-Maruyama method*:

$$\hat{X}_{N,M}(t_{k+1}) \approx \hat{X}_{N,M}(t_k) + hA_N\hat{X}_{N,M}(t_{k+1}) + hF_N(\hat{X}_{N,M}(t_k)) + \Delta W_N(t_k). \quad (46)$$

Remark 4.14

Considering the semilinear heat equation with additive space-time white noise and Dirichlet boundary conditions on $H = L^2(0, 1)$ known from Example 3.1 with F satisfying Assumption 4, it can be shown [e.g. see LPS14, §10.6 und §10.7] that the fully discrete approximation $\hat{X}_{N,M}$, given by the spectral Galerkin method in combination with the semi-implicit Euler-Maruyama method converges strongly against the infinite dimensional continuous solution X as $h \rightarrow 0$, $N \rightarrow \infty$ with fixed $h\lambda_N$. Note that in this special case we showed in Example 2.33 that Assumption 2 is satisfied for $\gamma \in (0, 1/4)$. Hence, following [LPS14, Corollary 10.38]), the strong convergence rate in this case is given by

$$\sup_{t_k} \|X - \hat{X}_{N,M}\|_{L^2(\Omega)} \leq \hat{C}(M^{-(\frac{1}{4}+\epsilon)}),$$

for a constant $\hat{C} > 0$ and any arbitrary small ϵ . It can be shown, that if one does not assume the condition of fixed $h\lambda_N$, that one then gets the convergence rates

$$\sup_{t_k} \|X - \hat{X}_{N,M}\|_{L^2(\Omega)} \leq C(N^{-(\frac{1}{2}+\epsilon)} + M^{-(\frac{1}{4}+\epsilon)}),$$

for a constant $C > 0$ and any arbitrary small ϵ . Note that we do not give the proof here, however the spatial convergence rate will be shown in the next chapter. Equation (46) implies that this method needs $N \cdot M$ arithmetical operations and thus its computational costs are $K = N \cdot M$. Hence, we can apply Remark 4.4 and obtain a combined convergence rate of

$$\sup_{t_k} \|X - \hat{X}_{N,M}\|_{L^2(\Omega)} \leq CK^{\frac{\frac{1}{2} \frac{1}{4}}{\frac{1}{2} + \frac{1}{4}}} = CK^{\frac{1}{6}}, \quad (47)$$

with respect to the computational costs.

Moreover, as shown in [DG01], this rate cannot be improved as long as one considers $\Delta W(t_k) = \int_{t_k}^{t_{k+1}} dW(s)$, $k = 0, \dots, M-1$ for the approximation of the noise. However, as the paper of Davie and Gaines suggests, we will see that one could improve the convergence rate by approximating the functional $W_{A_N}(t_k) = \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW(s)$ instead of using $\Delta W(t_k)$. Here $e^{A_N(t_{k+1}-s)}$ is to be used as smoothing factor to reduce the roughness of the space-time white noise. Formulating this and stating an improved convergence rate is the goal of the next chapter.

5. The exponential Euler method

In this chapter we will introduce the exponential Euler method which was first stated in [JK09a]. First, we will discuss the special preliminaries needed for the method and show why there exists a solution under this assumptions. Then we will discretize the space dimension by the spectral Galerkin method known from Section 4.2.1 and afterwards we will introduce the exponential Euler scheme to discretize the time dimension. The goal is to show the strong convergence of the spatial and the temporal discretization and state a convergence rate of the full discretization. Thereafter, we will use this method as our basis and develop a slightly modified version based on the idea of [Hil20]. We will call the new version *semilinear replacement method*. In the following, Problem (27) with W being a Q -Brownian motion with Q not necessary of trace-class will be considered.

5.1. Preliminaries

In the following, we will combine previous assumptions and new ones taken from [JK09a] to first ensure the existence of a $\mathcal{D}((-A)^\gamma)$ -valued solution process X to (27) and second give a convergence rate of the fully discrete approximation $X_{N,M}$ generated by a combination of the spectral Galerkin method and the exponential Euler method. We assume that Assumptions 1, 2, 3 and 4 are satisfied. Moreover, the initial value x_0 satisfy the following assumption.

Assumption 5 (Initial value condition)

The initial value x_0 is a $\mathcal{D}((-A)^\gamma)$ -valued random variable which satisfies

$$\mathbb{E}[\|(-A)^\gamma x_0\|_H^4] < \infty,$$

where $\gamma > 0$ is given in Assumption 2.

Remark 5.1

Note that since

$$\|(-A)^\gamma x_0\|_H^2 = \sum_{k=1}^{\infty} \lambda_k^{2\gamma} \langle x_0, \phi_k \rangle_H^2 \geq \lambda_1^{2\gamma} \|x_0\|_H^2$$

holds, it follows that $\lambda_1^{4\gamma} \|x_0\|_H^4 \leq \|(-A)^\gamma x_0\|_H^4$ is also satisfied. Thus, the above assumption also ensures that

$$\mathbb{E}[\|x_0\|_H^4] < \infty. \quad (48)$$

Consequently, applying Theorem 3.8, there exists a mild solution $X = (X(t))_{t \in [0, T]}$ to (27) given by

$$X(t) = e^{At} x_0 + \int_0^t e^{A(t-s)} F(X(s)) ds + \int_0^t e^{A(t-s)} dW(s), \quad (49)$$

which is a predictable stochastic process with values in $\mathcal{D}((-A)^\gamma)$. Moreover, supposing the previous remark, there exists a constant C_T , such that

$$\sup_{t \in [0, T]} \mathbb{E}[\|X(t)\|_H^4] \leq C_T (1 + \mathbb{E}[\|x_0\|_H^4]) \quad (50)$$

is satisfied. Under Assumption 2 W is a predictable process in $\mathcal{D}((-A^\gamma))$ (see Section 2.5.1) and thus, so is X .

With the knowledge of the existence of a solution, we will now discuss the conditions that the numerical scheme needs to satisfy the upcoming convergence result. In particular, we need enough regularity of the nonlinearity F . The following assumption formulizes these conditions.

Assumption 6 (Nonlinearity F)

The nonlinearity $F : H \rightarrow H$ is two times continuously Fréchet differentiable (see Definition A.23), it and its derivatives satisfy

$$\|F'(\nu) - F'(v)\|_{O_p} \leq L \|\nu - v\|_H, \quad (51)$$

$$\|(-A)^{(-r)} F'(\nu) (-A)^r \psi\|_H \leq L \|\psi\|_H \quad (52)$$

for all $\nu, v \in H$, $\psi \in \mathcal{D}(-A)^r$ and $r = 0, 1/2, 1$ and they satisfy

$$\|A^{-1} F''(\nu)(\psi_1, \psi_2)\|_H \leq L \|(-A)^{-(1/2)} \psi_1\|_H \|(-A)^{-(1/2)} \psi_2\|_H, \quad (53)$$

for all $\nu, \psi_1, \psi_2 \in H$, where $L > 0$ is a positive constant.

Given a mapping F satisfying Assumption 6, then Assumption 4 is implicitly contained and thus also satisfied, as the next corollary shows.

Corollary 5.2

Let $F : H \rightarrow H$ be a non-linear function satisfying Assumption 6. Then F is globally Lipschitz continuous and

$$\|F(\nu)\|_H \leq C(1 + \|\nu\|_H) \quad \forall \nu \in H. \quad (54)$$

Proof. Since F' is continuously Fréchet differentiable, it is continuous and thus locally bounded. Further, due to the generalized mean-value theorem A.24, F is locally Lipschitz continuous. Now, considering (52) for $r = 0$ we get an upper bound L of $\|F'(\nu)\|_{O_p}$, for every $\nu \in H$ (recall that $A^0 = \mathbf{1}$). Therefore, we get a global upper bound for $\|F(\nu) - F(v)\|_H$ with Lipschitz constant L given by Equation (52). Hence, for all $\nu, v \in H$ there exists a $\xi \in H$ such that

$$\|F(\nu) - F(v)\|_H \leq \|F'(\xi)\|_{O_p} \|\nu - v\|_H \leq L \|\nu - v\|_H$$

and thus F is globally Lipschitz continuous.

Let $\nu \in H$ be arbitrary. Since $F(0) \in H$, we know that there exists a $\tilde{C}$ such that $\|F(0)\|_H \leq \tilde{C}$. Since F is globally Lipschitz continuous, we get

$$\|F(\nu)\|_H \leq \|F(\nu) - F(0)\|_H + \|F(0)\|_H \leq L \|\nu\|_H + \tilde{C} \leq C(1 + \|\nu\|_H),$$

where $C = \max\{L, \tilde{C}\}$. □

When $H = L^2(0, 1)$, then there exists a simple way to create a nonlinearity F which satisfies the previous assumption.

Remark 5.3

Let $H = L^2(0, 1)$, let $F : L^2(0, 1) \rightarrow L^2(0, 1)$ be the Nemytskii operator induced by $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $F(\nu)(x) = f(\nu(x))$ for all $\nu \in L^2(0, 1)$, $x \in (0, 1)$ and let $f \in C_0^\infty(\mathbb{R})$. Then one can show that

- $F'(\nu)(\psi)(x) = f'(\nu(x))\psi(x)$
- $F''(\nu)(\psi_1, \psi_2)(x) = f''(\nu(x))\psi_1(x)\psi_2(x)$

for all $\nu, \psi, \psi_1, \psi_2 \in L^2(0, 1)$ and $x \in (0, 1)$ [e.g. see JKW11, p.916] and thus F satisfies Assumption 6.

5.2. The numerical scheme: Spatial and temporal discretization

We know that under the assumptions of this chapter, there exists a unique mild solution $X \in \mathcal{D}((-A)^\gamma)$ of Problem (27), where $\mathcal{D}((-A)^\gamma)$ is an infinite dimensional space. To simulate the solution, we now need to discretize it. We will start by truncating the spatial dimension, to get a N -dimensional solution X_N and show that this solution converges strongly towards the infinite dimensional solution for $N \rightarrow \infty$. Afterwards we will discretize the temporal dimension by considering the process only on a fine grid on $[0, T]$ with gridn with T/M . We will obtain a discrete N -dimensional solution $X_{N,M}$ which converges strongly against X_N for $M \rightarrow \infty$. Combining these two assertions we will state the convergence rate of the fully discrete solution $X_{N,M}$.

5.2.1. Spatial discretization: The spectral Galerkin method

We have already learned in Section 4.2 how to discretize the spatial dimensions using a Galerkin method. The method of choice here is the spectral Galerkin method, which is convenient, since we concentrate on additive noise and only consider examples where the eigenvalues and eigenfunctions of A are known. Therefore, we consider $H_N = \text{span}\{\phi_1, \dots, \phi_N\}$, where $(\phi_l)_{l=1, \dots, N}$ are the first N eigenfunctions of A . let $P_N : H \rightarrow H_N$ be the orthogonal projection and A_N, F_N, W_N and x_0^N all defined as in Section 4.2.1. Further, let $(S_N(t))_{t \in [0, T]}$ with $S_N(t) = e^{A_N t}$ be the C_0 -semigroup generated by A_N . Note that A_N also satisfies Assumption 1 and thus $S_N(\cdot)$ exists. By presupposing Assumptions 1, 3 and 4 to be satisfied, Corollary 4.12 ensures the existence of a finite dimensional solution X_N to (41). Together with Assumption 2 and 5 we know that X_N is an $\mathcal{D}((-A_N)^\gamma)$ -valued process which satisfies

$$X_N(t) = e^{A_N t} x_0^N + \int_0^t e^{A_N(t-s)} F_N(X_N(s)) ds + \int_0^t e^{A_N(t-s)} dW_N(s). \quad (55)$$

The next Lemma shows that Assumptions 5 and 6 are also satisfied for x_0^N and $F_N = P_N F|_{H_N}$.

Lemma 5.4

Let F_N and x_0^N be defined as above. Then Assumptions 5 and 6 are satisfied for all $\nu \in H$ and each $N \in \mathbb{N}$.

Proof. The first thing we take care of is nonlinearity. Let

$$H_N^\perp := \text{span}\{\phi_1, \dots, \phi_N\}^C = \text{span}\{\phi_{N+1}, \dots\}$$

be the orthogonal complement of H_N (see Definition A.20) and $P_N^\perp = (I - P_N)$ the projection as introduced in Theorem A.21. Define $F_N^\perp := P_N^\perp F$. Then it holds that $F_N \perp F_N^\perp$ in the sense that $\langle F_N, F_N^\perp \rangle_H = 0$ and $F(\nu) = F_N(\nu) + F_N^\perp(\nu)$ for all $\nu \in H$. Thus, we have

$$\|F(\nu)\|_H = \|F_N(\nu) + F_N^\perp(\nu)\|_H = \|F_N(\nu)\|_H + \|F_N^\perp(\nu)\|_H.$$

Further, by the linearity of the differential operator D we have

$$F'(\nu) = DF(\nu) = D(F_N(\nu) + F_N^\perp(\nu)) = DF_N(\nu) + DF_N^\perp(\nu). \quad (56)$$

Hence,

$$\begin{aligned} \|F'(\nu) - F'(v)\| &= \|DF_N(\nu) + DF_N^\perp(\nu) - DF_N(v) - DF_N^\perp(v)\| \\ &= \|DF_N(\nu) - DF_N(v) + DF_N^\perp(\nu) - DF_N^\perp(v)\| \\ &= \|DF_N(\nu) - DF_N(v)\| + \|DF_N^\perp(\nu) - DF_N^\perp(v)\| \end{aligned}$$

and thus

$$\|F'_N(\nu) - F'_N(v)\| \leq \|F'(\nu) - F'(v)\| \leq L \|\nu - v\|$$

for all $\nu, v \in H$. With similar arguments we have

$$\|F'_N(\nu)v\| \leq L \|v\|.$$

Further, using that A is a linear operator and similar arguments as before, we get

$$\begin{aligned} \|(-A)^{-r} F'(\nu) (-A)^r v\| &= \|(-A)^{-r} (DF_N(\nu) - DF_N^\perp(\nu)) (-A)^r v\| \\ &= \|(-A)^{-r} DF_N(\nu) (-A)^r v\| + \|(-A)^{-r} DF_N^\perp(\nu) (-A)^r v\| \end{aligned}$$

and thus

$$\|(-A)^{-r} F'_N(\nu) (-A)^r v\| \leq \|(-A)^{-r} F'(\nu) (-A)^r v\| \leq L \|v\|$$

for $r \in \{0, 1/2, 1\}$. Similar to (56) we get $F''(\nu) = D^2 F_N(\nu) + D^2 F_N^\perp(\nu)$ and thus

$$\|A^{-1} F''(\nu)(\psi_1, \psi_2)\| = \|A^{-1} D^2 F_N(\nu)(\psi_1, \psi_2)\| + \|A^{-1} D^2 F_N^\perp(\nu)(\psi_1, \psi_2)\|$$

and therefore

$$\|A^{-1} F''_N(\nu)(\psi_1, \psi_2)\| \leq L \|(-A)^{-1/2} \psi_1\| \|(-A)^{-1/2} \psi_2\|.$$

Now let's consider the initial value Assumption 5. Using the spectral representation of $(-A)^\gamma x_0^N$, Parseval's identity and that $\lambda_k > 0$ for all $k > 0$ we get

$$\|(-A)^\gamma x_0^N\|_H^2 = \sum_{k=1}^{\infty} \lambda_k^\gamma \langle x_0^N, \phi_k \rangle_H^2 = \sum_{k=1}^N \lambda_k^\gamma \langle u_0, \phi_k \rangle_H^2 \leq \sum_{k=1}^{\infty} \lambda_k^\gamma \langle x_0, \phi_k \rangle_H^2 = \|(-A)^\gamma x_0\|_H^2$$

and thus

$$\mathbb{E} \left[\|(-A)^\gamma x_0^N\|_H^4 \right] \leq \mathbb{E} \left[\|(-A)^\gamma x_0\|_H^4 \right] < \infty.$$

Note that for $\nu \in H_N$ we know that $A_N \nu = A \nu$, $(-A_N)^r \nu = (-A)^r \nu$ and $A_N^{-1} \nu = A^{-1} \nu$. Therefore, previous calculations all also hold for A_N . $\square$

Since Assumption 5 is satisfied for x_0^N , Corollary 4.12 implies that for X_N there exists a constant C_T such that

$$\sup_{t \in [0, T]} \mathbb{E} \left[\|X_N(t)\|_H^4 \right] \leq C_T \left(1 + \mathbb{E} \left[\|x_0^N\|_H^4 \right] \right) \quad (57)$$

is satisfied. Next, we state the convergence result for the spatial discretization.

Proposition 5.5

Suppose Assumptions 1, 2, 3, 4, 5 and 6 are satisfied. Then, there exists a constant $C_T^{(S)} > 0$, independent of N , such that for any $t \in [0, T]$

$$\|X(t) - X_N(t)\|_{L^2(\Omega)} \leq \frac{C_T^{(S)}}{\lambda_{N+1}^\gamma}. \quad (58)$$

Proof. Note, that we have $\|X\|_{L^2(\Omega)} = \mathbb{E}[\|X\|_H^2]^{1/2}$ for $X \in H$. Let $N \in \mathbb{N}$ and $0 \leq t \leq T$. Recall, that by Lemma 4.11

$$\|(I - P_N)\nu\|_H \leq \lambda_{N+1}^{-\gamma} \|\nu\|_\gamma$$

for $\nu \in \mathcal{D}((-A)^\gamma)$. Further define

$$P_N X(t) = e^{A_N t} x_0^N + \int_0^t e^{A_N(t-s)} F_N(X(s)) ds + \int_0^t e^{A_N(t-s)} dW_s^N.$$

Then

$$P_N X(t) - X_N(t) = \int_0^t e^{A_N(t-s)} (F_N(X(s)) - F_N(X_N(s))) ds.$$

Since F satisfies Assumption 6 F and thus also F_N are globally Lipschitz continuous (see Corollary 5.2) such that

$$\|F_N(X(s)) - F_N(X_N(s))\|_H \leq L \|X(s) - X_N(s)\|_H$$

holds. Further, recall that $\|e^{A(t-s)}\nu\|_H \leq \|\nu\|_H$ (see Lemma 2.10). Then,

$$\begin{aligned} \|X(t) - X_N(t)\|_H &\leq \|X(t) - P_N X(t)\|_H + \|P_N X(t) - X_N(t)\|_H \\ &\leq \lambda_{N+1}^{-\gamma} \|X(t)\|_\gamma + \int_0^t \|e^{A(t-s)} (F_N(X(s)) - F_N(X_N(s)))\|_H ds \\ &\leq \lambda_{N+1}^{-\gamma} \|X(t)\|_\gamma + L \int_0^t \|X(s) - X_N(s)\|_H ds. \end{aligned}$$

By using the Hölder inequality (see Theorem A.1) we get

$$\begin{aligned} \left(\int_0^t \|X(s) - X_N(s)\|_H ds \right)^2 &\leq \int_0^t (\mathbf{1}_{[0,T]}(s))^2 ds \int_0^t \|X(s) - X_N(s)\|_H^2 ds \\ &\leq T \int_0^t \|X(s) - X_N(s)\|_H^2 ds. \end{aligned}$$

Hence, using the fact that we can interchange the integral and the expectation due to Fubini's theorem, we get

$$\begin{aligned} 2L^2 \mathbb{E} \left[\left(\int_0^t \|X(s) - X_N(s)\|_H ds \right)^2 \right] &\leq 2L^2 \mathbb{E} \left[T \int_0^t \|X(s) - X_N(s)\|_H^2 ds \right] \\ &= 2L^2 T \int_0^t \mathbb{E} [\|X(s) - X_N(s)\|_H^2] ds. \end{aligned}$$

Then, by using that $a^2 + b^2 \geq 2ab$ such that $(a + b)^2 \leq 2a^2 + 2b^2$ holds, we get

$$\begin{aligned} \mathbb{E} [\|X(t) - X_N(t)\|_H^2] &\leq 2\lambda_{N+1}^{-2\gamma} \mathbb{E} [\|X(t)\|_\gamma^2] + 2L^2 \mathbb{E} \left[\left(\int_0^t \|X(s) - X_N(s)\|_H ds \right)^2 \right] \\ &\leq 2\lambda_{N+1}^{-2\gamma} \mathbb{E} [\|X(t)\|_\gamma^2] + L^2 T \int_0^t \mathbb{E} [\|X(s) - X_N(s)\|_H^2] ds. \end{aligned}$$

Next, we note that

$$2\lambda_{N+1}^{-2\gamma} \mathbb{E} [\|X(t)\|_\gamma^2] \leq 2\lambda_{N+1}^{-2\gamma} \sup_{0 \leq t \leq T} \mathbb{E} [\|X(t)\|_\gamma^2] =: B < \infty, \quad (59)$$

since by (57) we know that $\sup_{0 \leq t \leq T} \mathbb{E} [\|X(t)\|_\gamma^4] < \infty$ and thus also

$$\sup_{0 \leq t \leq T} \mathbb{E} [\|X(t)\|_\gamma^2] < \infty.$$

Gronwall's inequality (see Lemma A.5) states that if $u(t) \leq B + G \int_0^t u(s) ds$ is satisfied then $u(t) \leq B e^{Gt}$ holds. Therefore, by setting B as in (59) and $G = 2L^2 T$ and using that $t \leq T$ we get

$$\mathbb{E} [\|X(t) - X_N(t)\|_H^2] \leq 2\lambda_{N+1}^{-2\gamma} \sup_{0 \leq t \leq T} \mathbb{E} [\|X(t)\|_\gamma^2] e^{2L^2 T^2}.$$

Consequently,

$$\mathbb{E} [\|X(t) - X_N(t)\|_H^2]^{1/2} \leq \sqrt{2} \lambda_{N+1}^{-\gamma} \sup_{0 \leq t \leq T} \mathbb{E} [\|X(t)\|_\gamma^2]^{1/2} e^{L^2 T^2} \leq \frac{C_T^{(S)}}{\lambda_{N+1}^\gamma},$$

with $C_T^{(S)} = \sqrt{2} e^{L^2 T^2} \sup_{0 \leq t \leq T} \mathbb{E} [\|X(t)\|_\gamma^2]^{1/2}$. □

5.2.2. Temporal discretization: The exponential Euler scheme

The previous chapter provides a strongly converging approximation X_N with values in a finite dimensional space. However, X_N is still continuous in time and thus

difficult to simulate. The task for this section is to discretize the temporal dimension so that we get a solution process $X_{N,M}$ converging strongly against X with the fastest possible convergence rate. For this, we introduce the so-called exponential Euler scheme, as first introduced in [JK09a].

Let $M \in \mathbb{N}$ and let $\{t_k\}_{k=0,\dots,M}$ $t_0 < t_1 < \dots < t_M = T$ be an equidistant partition of $[0, T]$ with grid width $h := t_{k+1} - t_k = T/M$ for all $k = 0, \dots, M-1$. For any t_{k+1} the mild solution form of $X_N(t_{k+1})$ can be expressed by

$$\begin{aligned}
 X_N(t_{k+1}) &= e^{A_N t_{k+1}} x_0^N + \int_0^{t_{k+1}} e^{A_N(t_{k+1}-s)} F_N(X_N(s)) ds \\
 &\quad + \int_0^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s) \\
 &= e^{A_N(t_{k+1}-t_k)} e^{A_N t_k} x_0^N \\
 &\quad + e^{A_N(t_{k+1}-t_k)} \int_0^{t_k} e^{A_N(t_k-s)} F_N(X_N(s)) ds \\
 &\quad + \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} F_N(X_N(s)) ds \\
 &\quad + e^{A_N(t_{k+1}-t_k)} \int_0^{t_k} e^{A_N(t_k-s)} dW_N(s) \\
 &\quad + \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s) \\
 &= e^{A_N h} X_N(t_k) + \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} F_N(X_{N,M}(s)) ds \\
 &\quad + \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s).
 \end{aligned} \tag{60}$$

In this section we consider X_N only on discrete points t_k , $k = 0, \dots, M$. Hence, it is reasonable to approximate $F(X_N(s))$ by $F(X_N(t_k))$ for $s \in [t_k, t_{k+1})$. Then, it follows that for every $k = 0, \dots, M-1$

$$\begin{aligned}
 \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} F_N(X_N(s)) ds &\approx \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} F_N(X_N(t_k)) ds \\
 &= \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} ds F_N(X_N(t_k)) \\
 &\stackrel{(1)}{=} A_N^{-1} (e^{A_N h} - I_N) F_N(X_N(t_k))
 \end{aligned} \tag{62}$$

holds, where (1) follows since A is a negative diagonal matrix with negative diagonal entries $\{-\lambda_1, \dots, -\lambda_N\}$ and thus

$$\int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} ds = A_N^{-1} \left[e^{A_N(s-t_{k+1})} \right]_{t_k}^{t_{k+1}} = A_N^{-1} (e^{A_N h} - I_N),$$

where I_N is the N -dimensional identity matrix. With this we can now define the fully discrete Process $X_{N,M}$. Let $X_{N,M}(t_0) = x_0^N$. We can express the process in two

forms. First, when combining (60) and (62), by

$$\begin{aligned} X_{N,M}(t_{k+1}) &= e^{A_N t_{k+1}} x_0^N + \sum_{j=0}^k \int_{t_j}^{t_{j+1}} e^{A_N(t_{k+1}-s)} F_N(X_{N,M}(t_j)) ds \\ &\quad + \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s) \end{aligned} \quad (63)$$

and second, when combining (61) and (62), by

$$\begin{aligned} X_{N,M}(t_{k+1}) &= e^{A_N h} X_{N,M}(t_k) + A_N^{-1} (e^{A_N h} - I_N) F_N(X_{N,M}(t_k)) \\ &\quad + \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s). \end{aligned} \quad (64)$$

With this, we will now state the convergence result of the temporal discretization.

Proposition 5.6

Suppose Assumptions 1, 2, 3, 5 and 6 are satisfied. Then there exists a constant $C_T^{(T)}$ such that

$$\sup_{k=0, \dots, M} \|X_{N,M}(t_k) - X_N(t_k)\|_{L^2(\Omega)} \leq C_T^{(T)} \frac{\log(M)}{M}. \quad (65)$$

Proof. First, note that throughout this proof we frequently use the Bochner Inequality A.8, e.g. for a Bochner integrable function $g \in H$ we use

$$\left\| \int g(x) dx \right\|_H \leq \int \|g(x)\|_H dx.$$

Another inequality that will be frequently used throughout this proof is Hölders inequality (see Theorem A.1) in two variants. The first one is given by

$$\begin{aligned} \left(\int_{t_l}^{t_{l+1}} \|g(x)\|_H dx \right)^2 &= \left(\int_{t_l}^{t_{l+1}} \mathbf{1}_{[0,T]}(x) \|g(x)\|_H dx \right)^2 \\ &\leq \underbrace{\int_{t_l}^{t_{l+1}} (\mathbf{1}_{[0,T]}(x))^2 dx}_{=h} \int_{t_l}^{t_{l+1}} \|g(x)\|_H^2 dx \end{aligned} \quad (66)$$

for $g \in H$ and the second one is given by

$$\begin{aligned} \mathbb{E} \left[(\|g(x)\|_H^2 \|p(x)\|_H^2) \right] &\leq \mathbb{E} \left[\|g(x)\|_H^4 \right]^{1/2} \mathbb{E} \left[\|p(x)\|_H^4 \right]^{1/2} \\ &= \left\| \|g(x)\|_H^2 \right\|_{L^2(\Omega)} \left\| \|p(x)\|_H^2 \right\|_{L^2(\Omega)} \end{aligned} \quad (67)$$

for $g, p \in H$.

Let $N, M \in \mathbb{N}$ be arbitrary with $M \geq 2$. Recall that $X_{N,M}(t_k)$ is given by (63) and $X_N(t_k)$ is given by (55). Now define another process $Y := (Y(t_k))_{k=0, \dots, M}$ by

$$\begin{aligned} Y(t_{k+1}) &:= e^{A_N(t_{k+1})} x_0^N + \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} F_N(X_N(t_l)) ds \\ &\quad + \int_0^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s). \end{aligned} \quad (68)$$

Using Y we get

$$\begin{aligned} & \|X_{N,M}(t_{k+1}) - X_N(t_{k+1})\|_{L^2(\Omega)} \\ & \leq \|X_{N,M}(t_{k+1}) - Y(t_{k+1})\|_{L^2(\Omega)} + \|Y(t_{k+1}) - X_N(t_{k+1})\|_{L^2(\Omega)}. \end{aligned}$$

Let us first consider $\|X_{N,M}(t_{k+1}) - Y(t_{k+1})\|_{L^2(\Omega)}$.

Using the Expressions (63) of $X_{N,M}(t_{k+1})$ and (68) of $Y(t_{k+1})$ we get

$$(X_{N,M}(t_{k+1}) - Y(t_{k+1})) = \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} (F_N(X_{N,M}(t_l)) - F_N(X_N(t_l))) ds.$$

Then

$$\begin{aligned} & \|X_{N,M}(t_{k+1}) - Y(t_{k+1})\|_{L^2(\Omega)} \\ & = \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} (F_N(X_{N,M}(t_l)) - F_N(X_N(t_l))) ds \right\|_{L^2(\Omega)} \\ & \stackrel{(1)}{\leq} \sum_{l=0}^k \left\| \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} (F_N(X_{N,M}(t_l)) - F_N(X_N(t_l))) ds \right\|_{L^2(\Omega)} \\ & = \sum_{l=0}^k \mathbb{E} \left[\left\| \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} (F_N(X_{N,M}(t_l)) - F_N(X_N(t_l))) ds \right\|_H^2 \right]^{1/2} \\ & \stackrel{(2)}{\leq} \sum_{l=0}^k \mathbb{E} \left[\left(\underbrace{\int_{t_l}^{t_{l+1}} \|e^{A_N(t_{k+1}-s)}\|_{Op}}_{\substack{2.10 \\ \leq 1}} \underbrace{\|F_N(X_{N,M}(t_l)) - F_N(X_N(t_l))\|_H}_{\substack{5.2 \\ \leq L \|X_{N,M}(t_l) - X_N(t_l)\|_H}} \right)^2 \right]^{1/2} \\ & \stackrel{(3)}{\leq} \sum_{l=0}^k \mathbb{E} \left[\left(\int_{t_l}^{t_{l+1}} 1 ds \times L \|X_{N,M}(t_l) - X_N(t_l)\|_H \right)^2 \right]^{1/2} \\ & \stackrel{(4)}{=} \sum_{l=0}^k h L \mathbb{E} \left[\|X_{N,M}(t_l) - X_N(t_l)\|_H^2 \right]^{1/2} = \sum_{l=0}^k h L \|X_{N,M}(t_l) - X_N(t_l)\|_{L^2(\Omega)} \end{aligned} \tag{69}$$

where (1) follows from the triangle inequality, (2) follows from the Bochner inequality A.8 and $\|Ax\|_H = \|A\|_{Op} \|x\|_H$, (3) follows by using Lemma 2.10 and Corollary 5.2 and (4) follows from $h = t_{l+1} - t_l$ for every $l = 0, \dots, N-1$.

Next consider $\|Y(t_{k+1}) - X_N(t_{k+1})\|_{L^2(\Omega)}$. We first find an upper bound using the Taylor approximation of $X_N(t_{k+1})$ (see Theorem A.3) and then show that each of its components is smaller than $(\tilde{C} \log(M)/M)$ for some $\tilde{C} < \infty$. We then conclude that

$$\|Y(t_{k+1}) - X_N(t_{k+1})\|_{L^2(\Omega)} \leq \frac{\hat{C} \log(M)}{M}.$$

First note, that $F_N(X_N(s))$ can be expressed using its Taylor expansion at $X_N(t_l)$ by

$$\begin{aligned} & F_N(X_N(s)) \\ & = F_N(X_N(t_l)) + R \\ & = F_N(X_N(t_l)) + \left(\int_0^1 F'_N(X_N(t_l) + r(X_N(s) - X_N(t_l))) dr \right) (X_N(s) - X_N(t_l)). \end{aligned}$$

For the following, let

$$I_{l,s} := \int_0^1 F'_N(X_N(t_l) + r(X_N(s) - X_N(t_l)))dr,$$

denote the integral from the previous equation, such that

$$F(X_N(s)) - F(X_N(t_l)) = I_{l,s}(X_N(s) - X_N(t_l)). \quad (70)$$

Further, note that

$$\begin{aligned} e^{A_N(s-t_l)} X_N(t_l) &= e^{A_N t_l + A_N(s-t_l)} x_0^N \\ &\quad + \int_0^{t_l} e^{A_N(t_l-r) + A_N(s-t_l)} F_N(X_N(r)) dr \\ &\quad + \int_0^{t_l} e^{A_N(t_l-r) + A_N(s-t_l)} dW_N(r) \\ &= e^{A_N s} u_0^N \\ &\quad + \int_0^s e^{A_N(s-r)} F_N(X_N(r)) dr + \int_0^s e^{A_N(s-r)} dW_N(r) \\ &\quad - \int_{t_l}^s e^{A_N(s-r)} F_N(X_N(r)) dr - \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \\ &= X_N(s) - \int_{t_l}^s e^{A_N(s-r)} F_N(X_N(r)) dr - \int_{t_l}^s e^{A_N(s-r)} dW_N(r), \end{aligned}$$

such that

$$X_N(s) = e^{A_N(s-t_l)} X_N(t_l) + \int_{t_l}^s e^{A_N(s-r)} F_N(X_N(r)) dr + \int_{t_l}^s e^{A_N(s-r)} dW_N(r). \quad (71)$$

Using (70) and (71) and $I_{l,s} = F'_N(X_N(t_l)) + (I_{l,s} - F'_N(X_N(t_l)))$ we get

$$\begin{aligned} \|Y(t_{k+1}) - X_N(t_{k+1})\|_H &= \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} (F_N(X_N(s)) - F_N(X_N(t_l))) ds \right\|_H \\ &= \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} I_{l,s} (X_N(s) - X_N(t_l)) ds \right\|_H \\ &\leq E_1 + E_2 + E_3 + E_4, \end{aligned}$$

with

$$\begin{aligned} E_1 &= \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} I_{l,s} (e^{A_N(s-t_l)} - I) X_N(t_l) ds \right\|_H, \\ E_2 &= \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} I_{l,s} \int_{t_l}^s e^{A_N(s-r)} F_N(U_{t_l}^N) dr ds \right\|_H, \\ E_3 &= \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} F'_N(X_N(t_l)) \int_{t_l}^s e^{A_N(s-t_l)} dW_N(r) ds \right\|_H, \\ E_4 &= \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} (I_{l,s} - F'_N(X_N(t_l))) \int_{t_l}^s e^{A_N(s-t_l)} dW_N(r) ds \right\|_H. \end{aligned}$$

We now like to bound every E_i in $L^2(\Omega)$ -norm by $C \frac{\log(M)}{M}$. For this we will start with E_1 .

(E_1)

$$\begin{aligned} E_1 &= \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} I_{l,s}(e^{A_N(s-t_l)} - I) X_N(t_l) ds \right\|_H \\ &\leq \sum_{l=0}^k \int_{t_l}^{t_{l+1}} \left\| e^{A_N(t_{k+1}-s)} I_{l,s}(e^{A_N(s-t_l)} - I) X_N(t_l) \right\|_H ds. \end{aligned}$$

In the following we often need that $\|e^{A_N t} \nu\|_H \leq |e^{-\lambda_1 t}| \|\nu\|_H$ (see Lemma 2.10). From this it also follows that $e^{A_N t} \nu \in H_N$ for $\nu \in H_N$. Further, from Assumption 6 we know that $\|F'_N(\nu)v\| \leq L\|v\|$ for $\nu, v \in H$. Now, if we set $\nu := X_N(t_l) + r(X_N(s) - X_N(t_l))$ and $v := (e^{A_N(s-t_l)} - I)X_N(t_l)$ we get

$$\begin{aligned} &\left\| I_{l,s}(e^{A_N(s-t_l)} - I)X_N(t_l) \right\|_H \\ &\stackrel{(1)}{\leq} \int_0^1 \left\| F'_N(X_N(t_l) + r(X_N(s) - X_N(t_l)))(e^{A_N(s-t_l)} - I)X_N(t_l) \right\|_H dr \\ &\leq \int_0^1 L \left\| (e^{A_N(s-t_l)} - I)X_N(t_l) \right\|_H dr \\ &\leq L |e^{-\lambda_1(s-t_l)} - 1| \|X_N(t_l)\|_H, \end{aligned}$$

where (1) follows from the definition of $I_{l,s}$ and the Bochner inequality. This calculation also shows that $I_{l,s}(e^{A_N(s-t_l)} - I)X_N(t_l) \in H_N$. Hence

$$\begin{aligned} \left\| e^{A_N(t_{k+1}-s)} I_{l,s}(e^{A_N(s-t_l)} - I)X_N(t_l) \right\|_H &\leq |e^{-\lambda_1(t_{k+1}-s)}| \left\| I_{l,s}(e^{A_N(s-t_l)} - I)X_N(t_l) \right\|_H \\ &\leq L |e^{-\lambda_1(t_{k+1}-s)}| |e^{-\lambda_1(s-t_l)} - 1| \|X_N(t_l)\|_H \\ &\leq \|X_N(t_l)\|_H. \end{aligned}$$

Due to (57) we know that $\sup_{0 \leq s \leq T} \mathbb{E}[\|X_N(s)\|_H^2]^{1/2}$ is finite and therefore

$$\begin{aligned} &\left\| \int_{t_k}^{t_{k+1}} \left\| e^{A_N(t_{k+1}-s)} I_{k,s}(e^{A_N(s-t_k)} - I)X_N(t_k) \right\|_H ds \right\|_{L^2(\Omega)} \\ &\leq L \left\| \int_{t_k}^{t_{k+1}} \|X_N(t_k)\|_H ds \right\|_{L^2(\Omega)} \\ &\leq Lh \sup_{0 \leq s \leq T} \mathbb{E}[\|X_N(s)\|_H^2]^{1/2} = C^{(1)}h, \end{aligned} \tag{72}$$

holds with $C^{(1)} = L \sup_{0 \leq s \leq T} \mathbb{E}[\|X_N(s)\|_H^2]^{1/2}$. Next, consider the smoothening factor for $s \in [t_l, t_{l+1}]$. By using Lemma 4.10 we get

$$\left\| A_N e^{A_N(t_{k+1}-s)} \right\| \leq \frac{1}{t_{k+1} - s} \leq \frac{1}{t_{k+1} - t_{l+1}} = \frac{1}{t_k - t_l} = \frac{1}{h(k-l)} \tag{73}$$

Note that $e^{A_N t} = A_N A_N^{-1} e^{A_N t} = A_N e^{A_N t} A_N^{-1}$ (as seen in Remark 2.11). Then

$$\begin{aligned}
 & \|E_1\|_{L^2(\Omega)} \\
 & \leq \sum_{l=0}^k \left\| \int_{t_l}^{t_{l+1}} \left\| e^{A_N(t_{k+1}-s)} I_{l,s} (e^{A_N(s-t_l)} - I) X_N(t_l) \right\|_H ds \right\|_{L^2(\Omega)} \\
 & \stackrel{(1)}{\leq} C^{(1)} h + \sum_{l=0}^{k-1} \left\| \int_{t_l}^{t_{l+1}} \left\| e^{A_N(t_{k+1}-s)} I_{l,s} (e^{A_N(s-t_l)} - I) X_N(t_l) \right\|_H ds \right\|_{L^2(\Omega)} \\
 & = C^{(1)} h + \sum_{l=0}^{k-1} \left\| \int_{t_l}^{t_{l+1}} \left\| A_N e^{A_N(t_{k+1}-s)} A_N^{-1} I_{l,s} (e^{A_N(s-t_l)} - I) X_N(t_l) \right\|_H ds \right\|_{L^2(\Omega)} \\
 & = C^{(1)} h + \sum_{l=0}^{k-1} \left\| \int_{t_l}^{t_{l+1}} \left\| A_N e^{A_N(t_{k+1}-s)} \right\|_{Op} \left\| A_N^{-1} I_{l,s} (e^{A_N(s-t_l)} - I) X_N(t_l) \right\|_H ds \right\|_{L^2(\Omega)} \\
 & \stackrel{(2)}{\leq} C^{(1)} h + \sum_{l=0}^{k-1} \left\| \int_{t_l}^{t_{l+1}} \frac{1}{h(k-l)} \left\| A_N^{-1} I_{l,s} (e^{A_N(s-t_l)} - I) X_N(t_l) \right\|_H ds \right\|_{L^2(\Omega)} \\
 & \stackrel{(3)}{\leq} C^{(1)} h + \sum_{l=0}^{k-1} L \frac{1}{(k-l)} \left\| \int_{t_l}^{t_{l+1}} \|X_N(t_l)\|_H ds \right\|_{L^2(\Omega)} \\
 & \stackrel{(4)}{\leq} C^{(1)} h + \sum_{l=0}^{k-1} L \frac{1}{(k-l)} h \sup_{0 \leq s \leq T} \mathbb{E} [\|X_N(s)\|_H^2]^{1/2} \\
 & \leq C^{(1)} h + L h \sup_{0 \leq s \leq T} \mathbb{E} [\|X_N(s)\|_H^2]^{1/2} \sum_{l=0}^{k-1} \frac{1}{(k-l)} \\
 & \stackrel{(5)}{=} C^{(1)} h + C^{(1)} h \sum_{l=1}^k \frac{1}{l} \leq C^{(1)} h + C^{(1)} h \sum_{l=1}^M \frac{1}{l},
 \end{aligned}$$

where (1) follows by using (72), (2) follows by using (73), (3) follows by the fact that $e^{-\lambda_k h} - 1 \leq \frac{1}{\lambda_k h + 1} - 1 \leq \lambda_k h + 1 - 1 = \lambda_k h$ such that

$$\begin{aligned}
 \left\| A_N^{-1} (e^{A_N h} - I) X_N(t_l) \right\|_H &= \left\| \sum_{k=1}^N \lambda_k^{-1} (e^{-\lambda_k h} - 1) \langle e_k, X_N(t_l) \rangle e_k \right\|_H \\
 &\leq h \left\| \sum_{k=1}^N \langle e_k, X_N(t_l) \rangle e_k \right\|_H = h \|X_N(t_l)\|_H,
 \end{aligned}$$

(4) follows as in (72) and (5) follows by reordering the sum.

Now, contemplating Riemann sums, we see that

$$\sum_{l=2}^M \frac{1}{l} \leq \int_1^M \frac{1}{x} dx = \log(M)$$

and hence $\sum_{l=1}^M \frac{1}{l} \leq 1 + \log(M)$. Further using

$$\begin{aligned}
 C^{(1)} h (2 + \log(M)) &\leq \tilde{C}^{(1)} h \log(M) \\
 \iff \\
 C^{(1)} \frac{2 + \log(M)}{\log(M)} &= C^{(1)} \left(\frac{2}{\log(M)} + 1 \right) \leq \tilde{C}^{(1)}
 \end{aligned}$$

and that the logarithm is monotone increasing and $M \geq 2$, such that $\log(2) \leq \log(M)$ and thus choosing $\tilde{C}^{(1)} := C^{(1)}(1 + (2/\log(2)))$ we get

$$\begin{aligned} C^{(1)}h + C^{(1)}h \sum_{l=1}^M \frac{1}{l} &\leq C^{(1)}h(\log(M) + 2) \\ &\leq \tilde{C}^{(1)} \frac{T}{M} \log(M) = C_T^{(E_1)} \frac{\log(M)}{M}, \end{aligned}$$

with

$$\begin{aligned} C_T^{(E_1)} &:= \tilde{C}^{(1)}T = C^{(1)} \left(1 + \frac{2}{\log(2)}\right) T \\ &= LT \sup_{0 \leq s \leq T} \mathbb{E} \left[\|X_N(s)\|_H^2 \right]^{1/2} \left(1 + \frac{2}{\log(2)}\right). \end{aligned} \tag{74}$$

(E_2) In previous calculations, we saw that

- (1) $\|e^{A_N(t_{k+1}-s)} I_{l,s} \nu\|_H \leq L \|\nu\|_H$, for $s \in [t_l, t_{l+1}]$ and for all $\nu \in H$,
- (2) $\|e^{A_N(s-r)} F_N(X_N(r))\|_H \leq \|F_N(X_N(r))\|_H$ for $r \leq s \leq T$.

Further, from Corollary 5.2 we know that

- (3) $\|F(\nu)\| \leq C(1 + \|\nu\|_H)$ for all $\nu \in H$.

Therefore,

$$\begin{aligned} E_2 &= \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} I_{l,s} \int_{t_l}^s e^{A_N(s-r)} F_N(X_N(r)) dr ds \right\|_H \\ &\leq \sum_{l=0}^k \int_{t_l}^{t_{l+1}} \left\| e^{A_N(t_{k+1}-s)} I_{l,s} \int_{t_l}^s e^{A_N(s-r)} F_N(X_N(r)) dr \right\|_H ds \\ &\stackrel{(1)}{\leq} \sum_{l=0}^k \int_{t_l}^{t_{l+1}} L \left\| \int_{t_l}^s e^{A_N(s-r)} F_N(X_N(r)) dr \right\|_H ds \\ &\stackrel{(2)}{\leq} L \sum_{l=0}^k \int_{t_l}^{t_{l+1}} \int_{t_l}^s \|F_N(X_N(r))\|_H dr ds \\ &\stackrel{(3)}{\leq} CL \sum_{l=0}^k \int_{t_l}^{t_{l+1}} \int_{t_l}^s (1 + \|X_N(r)\|_H) dr ds. \end{aligned}$$

Note that $(\sum_{l=0}^k x_k)^2 \leq (k+1) \sum_{l=0}^k x_k^2 \leq M \sum_{l=0}^k x_k^2$. Hence,

$$\begin{aligned}
 \mathbb{E} [|E_2|^2] &\leq \mathbb{E} \left[\left(CL \sum_{l=0}^k \int_{t_l}^{t_{l+1}} \int_{t_l}^s (1 + \|X_N(r)\|_H) dr ds \right)^2 \right] \\
 &\leq C^2 L^2 M \mathbb{E} \left[\sum_{l=0}^k \left(\int_{t_l}^{t_{l+1}} \int_{t_l}^s (1 + \|X_N(r)\|_H) dr ds \right)^2 \right] \\
 &\stackrel{(66)}{\leq} C^2 L^2 M \sum_{l=0}^k \mathbb{E} \left[\int_{t_l}^{t_{l+1}} h \left(\int_{t_l}^s (1 + \|X_N(r)\|_H) dr \right)^2 ds \right] \\
 &\stackrel{(66)}{\leq} C^2 L^2 M \sum_{l=0}^k \mathbb{E} \left[\int_{t_l}^{t_{l+1}} h^2 \int_{t_l}^s (1 + \|X_N(r)\|_H)^2 dr ds \right] \\
 &\leq C^2 L^2 M h^2 \sum_{l=0}^k \int_{t_l}^{t_{l+1}} \int_{t_l}^s (1 + \mathbb{E} [\|X_N(r)\|_H^2]) dr ds \\
 &\leq C^2 L^2 M h^2 \sum_{l=0}^k h^2 \left(1 + \sup_{0 \leq r \leq T} \mathbb{E} [\|X_N(r)\|_H^2] \right) \\
 &\leq C^2 L^2 M^2 h^4 \left(1 + \sup_{0 \leq r \leq T} \mathbb{E} [\|X_N(r)\|_H^2] \right) \\
 &\stackrel{M=T/h}{\leq} C^2 L^2 T^2 \left(1 + \sup_{0 \leq r \leq T} \mathbb{E} [\|X_N(r)\|_H^2] \right) h^2.
 \end{aligned}$$

Hence, setting $C_T^{(E_2)} = \sqrt{C^2 L^2 T^2 (1 + \sup_{0 \leq r \leq T} \mathbb{E} [\|X_N(r)\|_H^2])}$ we get

$$\|E_2\|_{L^2(\Omega)} = (\mathbb{E} [|E_2|^2])^{1/2} \leq C_T^{(E_2)} h.$$

(E_3) First note that using $\|e^{A_N t} v\|_H \leq \|v\|_H$ (see Lemma 4.10) and $\|F'_N(\nu)v\|_H \leq L \|v\|_H$ for all $\nu, v \in H$ (see Assumption 6), we get

$$\begin{aligned}
 &\left\| \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} F'_N(X_N(t_k)) \int_{t_k}^s e^{A_N(s-r)} dW_N(r) ds \right\|_H \\
 &\leq \int_{t_k}^{t_{k+1}} \left\| e^{A_N(t_{k+1}-s)} F'_N(X_N(t_k)) \int_{t_k}^s e^{A_N(s-r)} dW_N(r) \right\|_H ds \\
 &\leq L \int_{t_k}^{t_{k+1}} \left\| \int_{t_k}^s e^{A_N(s-r)} dW_N(r) \right\|_H ds.
 \end{aligned}$$

Hence,

$$\begin{aligned}
 & \mathbb{E} \left[\left\| \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} F'_N(X_N(t_k)) \int_{t_k}^s e^{A_N(s-r)} dW_N(r) ds \right\|_H^2 \right] \\
 & \leq \mathbb{E} \left[L^2 \left(\int_{t_k}^{t_{k+1}} \left\| \int_{t_k}^s e^{A_N(s-r)} dW_N(r) \right\|_H ds \right)^2 \right] \\
 & \stackrel{(66)}{\leq} \mathbb{E} \left[L^2 h \int_{t_k}^{t_{k+1}} \left\| \int_{t_k}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 ds \right] \\
 & \stackrel{Fubini}{=} L^2 h \int_{t_k}^{t_{k+1}} \underbrace{\mathbb{E} \left[\left\| \int_{t_k}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 \right]}_{\stackrel{(1)}{=} \int_{t_k}^s \|e^{A_N(s-r)}\|_{L_2^0}^2 ds} ds \\
 & \stackrel{(2)}{\leq} C^{(3.1)} L^2 h^2,
 \end{aligned} \tag{75}$$

with $C^{(3.1)} = \int_0^T \|e^{A_N(T-r)}\|_{L_2^0}^2 ds \leq \sum_{k=1}^\infty \frac{q_k}{\lambda_k} < \infty$, where $C^{(3.1)}$ and (1) follow from Section 2.5.1. For the following we need two more calculations. First, note that by using the sum representation of $(-A_N)^{1/2}$ and $e^{A_N t}$ and the fact that $e^{-x} \leq 1/x$ for $x > 0$ we get for $s < t_{k+1}$

$$\begin{aligned}
 \|(-A_N)^{1/2} e^{A_N(t_{k+1}-s)} \nu\|_H &= \sum_{l=1}^N (\lambda_l^{1/2} e^{-\lambda_l(t_{k+1}-s)})^2 \langle \nu, \phi_j \rangle_H^2 \\
 &= \sum_{l=1}^N (\lambda_l e^{-2\lambda_l(t_{k+1}-s)})^2 \langle \nu, \phi_j \rangle_H^2 \\
 &\leq \sum_{l=1}^N \frac{\lambda_j}{2\lambda_j(t_{k+1}-s)} \langle \nu, \phi_j \rangle_H^2 \\
 &= \frac{1}{2(t_{k+1}-s)} \|\nu\|_H^2.
 \end{aligned} \tag{76}$$

And second, using the fact that there exists a constant C such that

$$\mathbb{E} \left[\left\| (-A_N)^{-(1/2)} \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^4 \right] \leq C(s - t_l)^2$$

holds, [see JK09a, Equation (2.3)], we get

$$\begin{aligned}
 & \int_{t_l}^{t_{l+1}} \mathbb{E} \left[\left\| (-A_N)^{-(1/2)} \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 \right] ds \leq \int_{t_l}^{t_{l+1}} \sqrt{C}(s - t_l) ds \\
 & = \sqrt{C} \left[\frac{1}{2}(s - t_l)^2 \right]_{t_l}^{t_{l+1}} = \sqrt{C} \frac{1}{2} (t_{l+1} - t_l)^2 = \sqrt{C} \frac{h^2}{2} =: C^{(3.2)} h^2.
 \end{aligned} \tag{77}$$

We can now find an upper bound for $\mathbb{E}[|E_3|^2]$:

$$\begin{aligned}
 \mathbb{E} [|E_3|^2] &= \mathbb{E} \left[\left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} F'_N(X_N(t_l)) \int_{t_l}^s e^{A_N(s-r)} dW_N(r) ds \right\|_H^2 \right] \\
 &\stackrel{(75)}{\leq} L^2 h^2 C^{(3.1)} \\
 &\quad + \mathbb{E} \left[\left\| \sum_{l=0}^{k-1} \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} F'_N(X_N(t_l)) \int_{t_l}^s e^{A_N(s-r)} dW_N(r) ds \right\|_H^2 \right] \\
 &= L^2 h^2 C^{(3)} \\
 &\quad + \sum_{l=0}^{k-1} \mathbb{E} \left[\left\| \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} F'_N(X_N(t_l)) \int_{t_l}^s e^{A_N(s-r)} dW_N(r) ds \right\|_H^2 \right] \\
 &\stackrel{(66)}{\leq} L^2 h^2 C^{(3.1)} \\
 &\quad + h \sum_{l=0}^{k-1} \mathbb{E} \left[\int_{t_l}^{t_{l+1}} \left\| e^{A_N(t_{k+1}-s)} F'_N(X_N(t_l)) \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 ds \right] \\
 &= L^2 h^2 C^{(3.1)} \\
 &\quad + h \sum_{l=0}^{k-1} \mathbb{E} \left[\int_{t_l}^{t_{l+1}} \left\| (-A_N)^{1/2} e^{A_N(t_{k+1}-s)} \right. \right. \\
 &\quad \quad \left. \left. \times (-A_N)^{-1/2} F'_N(X_N(t_l)) \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 ds \right] \\
 &\stackrel{(1)}{\leq} L^2 h^2 C^{(3.1)} \\
 &\quad + L^2 \frac{h}{2h} \sum_{l=0}^{k-1} (k-l)^{-1} \int_{t_l}^{t_{l+1}} \mathbb{E} \left[\left\| (-A_N)^{-1/2} \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 \right] ds \\
 &\stackrel{(77)}{\leq} L^2 h^2 C^{(3.1)} + \frac{1}{2} L^2 C^{(3.2)} h^2 \sum_{l=1}^k l^{-1} \leq L^2 h^2 C^{(3.1)} + \frac{1}{2} L^2 C^{(3.2)} h^2 (\log(M) + 1) \\
 &\leq \tilde{C}^{(3)} h^2 \log(M),
 \end{aligned}$$

with

$$\tilde{C}^{(3)} = \frac{L^2 C^{(3.1)}}{\log(2)} + \left(\frac{1}{2 \log(2)} + \frac{1}{2} \right) L^2 C^{(3.2)}$$

similarly determined as for E_1 . (1) follows from (76), from

$$\left\| (-A_N)^{-1/2} F'_N(\nu) v \right\|_H \leq L \left\| (-A_N)^{-1/2} v \right\|_H \quad \forall \nu \in \mathcal{D}((-A)^{-1/2})$$

as given in Assumption 6 (note that $H \subset \mathcal{D}((-A)^{-1/2})$) and from using Fubini's theorem to interchange the integrals. Therefore we get

$$\begin{aligned}
 \|E_3\|_{L^2(\Omega)} &= \mathbb{E} \left[\|E_3\|_H^2 \right]^{1/2} \\
 &\leq \sqrt{\tilde{C}^{(3)}} h \sqrt{\log(M)} = \sqrt{\tilde{C}^{(3)}} \frac{T}{M} \frac{\log(M)}{\sqrt{\log(M)}} \\
 &\leq \frac{\sqrt{\tilde{C}^{(3)}} T \log(M)}{\sqrt{\log(2)} M} = C_T^{(E_3)} \frac{\log(M)}{M},
 \end{aligned}$$

$$\text{with } C_T^{(E_3)} = \frac{\sqrt{\tilde{C}^{(3)}} T}{\log(2)}.$$

(E_4) First, note that due to Assumption 6 and the fact that for any $\nu \in H$ $F'_N(\nu)$ is a bounded linear operator we get

$$\begin{aligned}
 &\|(I_{l,s} - F'_N(X_N(t_l)))(\nu)\|_H \\
 &= \left\| \int_0^1 (F'_N(X_N(t_l) + q(X_N(s) - X_N(t_l))) - F'_N(X_N(t_l))) \nu dq \right\|_H \\
 &\leq \int_0^1 \| (F'_N(X_N(t_l) + q(X_N(s) - X_N(t_l))) - F'_N(X_N(t_l))) \nu \|_H dq \quad (78) \\
 &= \int_0^1 \| F'_N(X_N(t_l) + q(X_N(s) - X_N(t_l))) - F'_N(X_N(t_l)) \|_{Op} \|\nu\|_H dq \\
 &\leq L \int_0^1 q \|X_N(s) - X_N(t_l)\|_H \|\nu\|_H dq = \frac{L}{2} \|X_N(s) - X_N(t_l)\|_H \|\nu\|_H.
 \end{aligned}$$

Further, note that

$$\|X_N(s) - X_N(t_l)\|_H^2 \leq 2 \|X_N(s)\|_H^2 + 2 \|X_N(t_l)\|_H^2 \leq 4 \sup_{t \in [0, T]} \|X_N(t)\|_H^2$$

such that

$$\begin{aligned}
 \mathbb{E} \left[\left(\|X_N(s) - X_N(t_l)\|_H^2 \right)^2 \right]^{1/2} &\leq 4 \mathbb{E} \left[\sup_{t \in [0, T]} \|X_N(t)\|_H^4 \right]^{1/2} \\
 &\stackrel{(57)}{\leq} 4 \left(1 + \mathbb{E} \left[\|x_0^N\|_H^4 \right] \right)^{1/2}. \quad (79)
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 & \mathbb{E} \left[\left\| \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} (I_{l,s} - F'_N(X_N(t_l))) \int_{t_l}^s e^{A_N(s-r)} dW_N(r) ds \right\|_H^2 \right]^{1/2} \\
 & \leq \mathbb{E} \left[\left(\int_{t_k}^{t_{k+1}} \left\| e^{A_N(t_{k+1}-s)} (I_{l,s} - F'_N(X_N(t_l))) \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 ds \right)^2 \right]^{1/2} \\
 & = \mathbb{E} \left[\left(\int_{t_k}^{t_{k+1}} \|e^{A_N(t_{k+1}-s)}\|_{Op} \left\| (I_{l,s} - F'_N(X_N(t_l))) \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 ds \right)^2 \right]^{1/2} \\
 & \stackrel{(78)}{\leq} \mathbb{E} \left[\left(\int_{t_k}^{t_{k+1}} \frac{L}{2} \|X_N(s) - X_N(t_l)\|_H \left\| \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 ds \right)^2 \right]^{1/2} \\
 & \stackrel{(1)}{\leq} \frac{L}{2} \sqrt{h} \left(\int_{t_k}^{t_{k+1}} \mathbb{E} \left[\left(\|X_N(s) - X_N(t_l)\|_H \left\| \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 \right)^2 \right] ds \right)^{1/2} \\
 & \stackrel{(67)}{\leq} \frac{L}{2} \sqrt{h} \left(\int_{t_k}^{t_{k+1}} \underbrace{\mathbb{E} [\|X_N(s) - X_N(t_l)\|_H^4]^{1/2}}_{\stackrel{(79)}{\leq} 4(1+\mathbb{E}[\|x_0^N\|_H^4])^{1/2}} \underbrace{\mathbb{E} \left[\left\| \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^4 \right]^{1/2}}_{\leq \sup_{t \in [0,T]} \mathbb{E} \left[\left\| \int_0^t e^{A_N(t-r)} dW_N(r) \right\|_H^4 \right]^{1/2}} ds \right)^{1/2} \\
 & \leq C^{(4.1)} h
 \end{aligned}$$

with

$$C^{(4.1)} := L \left(\left(1 + \mathbb{E} [\|x_0^N\|_H^4] \right)^{1/2} \sup_{t \in [0,T]} \mathbb{E} \left[\left\| \int_0^t e^{A_N(t-r)} dW_N(r) \right\|_H^4 \right]^{1/2} \right)^{1/2},$$

where (1) follows by applying Fubini's theorem and the Bochner inequality. Note that due to [JK09a, Equation (2.3)] we know that

$$\begin{aligned}
 & \lambda_1^{4\gamma} \sup_{t \in [0,T]} \mathbb{E} \left[\left\| \int_0^t e^{A_N(t-r)} dW_N(r) \right\|_H^4 \right] \\
 & \leq \sup_{t \in [0,T]} \mathbb{E} \left[\left\| (-A_N)^\gamma \int_0^t e^{A_N(t-r)} dW_N(r) \right\|_H^4 \right] < \infty
 \end{aligned}$$

and thus $\sup_{t \in [0,T]} \mathbb{E} \left[\left\| \int_0^t e^{A_N(t-r)} dW_N(r) \right\|_H^4 \right] < \infty$. Further, note that by using the following second-order Taylor expansion for $F_N(X_N(s))$ (see Theorem A.3)

$$\begin{aligned}
 F_N(X_N(s)) &= F_N(X_N(t_l)) + F'_N(X_N(t_l))(X_N(s) - X_N(t_l)) \\
 &\quad + \int_0^1 (1-r) F''_N(X_N(t_l) + r(X_N(s) - X_N(t_l))) dr (X_N(s) - X_N(t_l))^2,
 \end{aligned}$$

which reordered equals

$$\begin{aligned} & \overbrace{\left(\frac{(F(X_N(s))^N - F(X_N(t_l)))}{(X_N(s) - X_N(t_l))} - F'_N(X_N(t_l)) \right)}^{(70) I_{l,s}} (X_N(s) - X_N(t_l)) \\ &= \int_0^1 (1-r) F''_N(X_N(t_l) + r(X_N(s) - X_N(t_l))) dr (X_N(s) - X_N(t_l))^2, \end{aligned}$$

we get

$$\begin{aligned} & I_{l,s} - F'_N(X_N(t_l)) \\ &= \int_0^1 (1-r) F''_N(X_N(t_l) + r(X_N(s) - X_N(t_l))) dr (X_N(s) - X_N(t_l)). \end{aligned} \quad (80)$$

Thus,

$$\begin{aligned} & \left\| A_N^{-1} (I_{l,s} - F'_N(X_N(t_l))) \left(\int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right) \right\|_H \\ & \stackrel{(1)}{\leq} \int_0^1 \left\| A_N^{-1} F''_N(X_N(t_l) + q(X_N(s) - X_N(t_l))) (X_N(s) - X_N(t_l)) \right. \\ & \quad \times \left. \left(\int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right) (1-q) \right\|_H dr \\ & \stackrel{(2)}{\leq} \frac{1}{2} \left\| (-A_N)^{-1/2} (X_N(s) - X_N(t_l)) \right\|_H \left\| (-A_N)^{-1/2} \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H, \end{aligned} \quad (81)$$

where (1) follows by using (80) and the Bochner integral property, that linear operators can be pulled into the integral and then using the Bochner inequality.

(2) then follows by Assumption 6. Moreover, we have

$$\begin{aligned} & \mathbb{E} \left[\int_{t_l}^{t_{l+1}} \left\| (-A_N)^{-1/2} (X_N(s) - X_N(t_l)) \right\|_H \left\| (-A_N)^{-1/2} \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H ds \right]^2 \\ & \stackrel{(1)}{\leq} h \int_{t_l}^{t_{l+1}} \mathbb{E} \left[\left\| (-A_N)^{-1/2} (X_N(s) - X_N(t_l)) \right\|_H^2 \left\| (-A_N)^{-1/2} \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^2 \right] ds \\ & \stackrel{(67)}{\leq} h \int_{t_l}^{t_{l+1}} \underbrace{\mathbb{E} \left[\left\| (-A_N)^{-1/2} (X_N(s) - X_N(t_l)) \right\|_H^4 \right]^{1/2}}_{\stackrel{(2)}{\leq} C^{(4.2.1)} h} \\ & \quad \times \underbrace{\mathbb{E} \left[\left\| (-A_N)^{-1/2} \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H^4 \right]^{1/2}}_{\stackrel{(3)}{\leq} C^{(4.2.2)} h} ds \\ & \leq C^{(4.2)} h^4, \end{aligned} \quad (82)$$

where in (1) we first use the fact that $\mathbb{E}[\cdot]^2 \leq \mathbb{E}[(\cdot)^2]$, then apply the first variant of Hölders inequality (66) and then use Fubini's theorem to switch the

expectation and the integral. (2) and (3) follow by using the following bounds given by [JK09a, Equation (2.3)]:

$$\begin{aligned}\mathbb{E} \left[\left\| (-A)^{-1/2} (X_N(t) - X_N(s)) \right\|_H^4 \right] &\leq C(t-s)^2 \\ \mathbb{E} \left[\left\| (-A)^{-1/2} \int_s^t e^{A(t-r)} dW(r) \right\|_H^4 \right] &\leq C(t-s)^2.\end{aligned}$$

Alltogether, we get

$$\begin{aligned}& \|E_4\|_{L^2(\Omega)} \\&= \left\| \left\| \sum_{l=0}^k \int_{t_l}^{t_{l+1}} e^{A_N(t_{k+1}-s)} (I_{l,s} - F'_N(X_N(t_l))) \int_{t_s}^s e^{A_N(s-r)} dW_N(r) ds \right\|_H \right\|_{L^2(\Omega)} \\&\stackrel{(1)}{\leq} C^{(4.1)} h + \sum_{l=0}^{k-1} \left\| \int_{t_l}^{t_{l+1}} \left\| e^{A_N(t_{k+1}-s)} (I_{l,s} - F'_N(X_N(t_l))) \int_{t_s}^s e^{A_N(s-r)} dW_N(r) \right\|_H ds \right\|_{L^2(\Omega)} \\&= C^{(4.1)} h + \sum_{l=0}^{k-1} \left\| \int_{t_l}^{t_{l+1}} \|A_N e^{A_N(t_{k+1}-s)}\|_{Op} \right. \\&\quad \times \left. \left\| A_N^{-1} (I_{l,s} - F'_N(X_N(t_l))) \int_{t_s}^s e^{A_N(s-r)} dW_N(r) \right\|_H ds \right\|_{L^2(\Omega)} \\&\stackrel{(2)}{\leq} C^{(4.1)} h + \sum_{l=0}^{k-1} \frac{1}{2h} (k-l)^{-1} \left\| \int_{t_l}^{t_{l+1}} \left\| (-A_N)^{-1/2} (X_N(s) - X_N(t_l)) \right\|_H \right. \\&\quad \times \left. \left\| (-A_N)^{-1/2} \int_{t_l}^s e^{A_N(s-r)} dW_N(r) \right\|_H ds \right\|_{L^2(\Omega)} \\&\stackrel{(82)}{\leq} C^{(4.1)} h + \frac{1}{2h} \sum_{l=0}^{k-1} (k-l)^{-1} \sqrt{C^{(4.2)}} h^2 \leq C^{(4.1)} h + \frac{1}{2} h \sqrt{C^{(4.2)}} (\log(M) + 1) \\&\leq \tilde{C}^{(4)} h \log(M) = C_T^{(E_4)} \frac{\log(M)}{M},\end{aligned}$$

with

$$\tilde{C}^{(4)} := \frac{C^{(4.1)}}{\log(2)} + \frac{\sqrt{C^{(4.2)}}}{2 \log(2)} + \frac{1}{2} \sqrt{C^{(4.2)}}$$

and $C_T^{(E_4)} := T \tilde{C}^{(4)}$. (1) follows from previous calculations and using the Minkovski inequality and the Bochner inequality, (2) follows from (73) and (81).

Thus, all in all we get the estimation

$$\begin{aligned}& \|Y(t_{k+1}) - X_N(t_{k+1})\|_{L^2(\Omega)} \\&\leq \|E_1\|_{L^2(\Omega)} + \|E_2\|_{L^2(\Omega)} + \|E_3\|_{L^2(\Omega)} + \|E_4\|_{L^2(\Omega)} \\&\leq C_T^{(E_1)} \frac{\log(M)}{M} + C_T^{(E_2)} \frac{1}{M} + C_T^{(E_3)} \frac{\log(M)}{M} + C_T^{(4)} \frac{\log(M)}{M} \\&\leq C_T \frac{\log(M)}{M}\end{aligned}$$

with

$$C_T := C_T^{(E_1)} + \frac{C_T^{(E_2)}}{\log(2)} + C_T^{(E_3)} + C_T^{(E_4)}.$$

Together with (69) we get

$$\begin{aligned} & \|X_{N,M}(t_{k+1}) - X_N(t_{k+1})\|_{L^2(\Omega)} \\ & \leq \|X_{N,M}(t_{k+1}) - Y(t_{k+1})\|_{L^2(\Omega)} + \|Y(t_{k+1}) - X_N(t_{k+1})\|_{L^2(\Omega)} \\ & \leq \sum_{l=0}^k hL \|X_{N,M}(t_l) - X_N(t_l)\|_{L^2(\Omega)} + C_T \frac{\log(M)}{M}. \end{aligned}$$

Now, using the discrete version of Gronwalls inequality (see Lemma A.6 and set $c = C_T \frac{\log(M)}{M}$ and $b_j = hL = LT/M$) we get

$$\begin{aligned} \|X_{N,M}(t_{k+1}) - X_N(t_{k+1})\|_{L^2(\Omega)} & \leq C_T \frac{\log(M)}{M} \exp\left(\sum_{j=1}^k \left(\frac{LT}{M}\right)\right) \\ & \leq C_T \frac{\log(M)}{M} \exp\left(M \frac{LT}{M}\right) \\ & \leq C_T^{(T)} \frac{\log(M)}{M} \end{aligned}$$

$$\text{with } C_T^{(T)} := e^{TL} C_T = e^{TL} \left(C_T^{(E_1)} + \frac{C_T^{(E_2)}}{\log(2)} + C_T^{(E_3)} + C_T^{(E_4)} \right).$$

□

5.2.3. Fully discrete Scheme

Combining the last two sections, we will now state the main theorem of this chapter, leading to strong convergence of the fully discrete scheme and its convergence rate. Note, technically, the fully discrete approximation consists of the spectral Galerkin method combined with the exponential Euler method. However, in the following, with a little abuse of notation, we will refer to the fully discrete approximation as the exponential Euler method.

Theorem 5.7 ([JK09a], Theorem 3.1)

Suppose Assumptions 1, 2, 3, 5 and 6 are satisfied. Then, there exists a constant C_T such that

$$\sup_{k=0, \dots, M} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \leq C_T \left(\lambda_N^{-\gamma} + \frac{\log(M)}{M} \right), \quad (83)$$

holds for all $N, M \geq 2$, where X is the solution of SDE (27), $X_{N,M}$ is the process given by (63), $t_k = k \frac{T}{M}$ for $k = 0, \dots, M$ and $\gamma > 0$ is the constant given by Assumption 2.

Proof. Let $N, M \in \mathbb{N}$ be arbitrary with $M \geq 2$. From Proposition 5.5 we know that

$$\|X(t) - X_N(t_k)\|_{L^2(\Omega)} \leq \frac{C_T^{(S)}}{\lambda_N^\gamma}$$

for all $t \in [0, T]$. Thus, we also have

$$\sup_{k=0, \dots, M} \|X(t_k) - X_N(t_k)\|_{L^2(\Omega)} \leq \frac{C_T^{(S)}}{\lambda_N^\gamma}.$$

Further, from Proposition 5.6, we know that

$$\sup_{k=0, \dots, M} \|X_N(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \leq \frac{C_T^{(T)} \log(M)}{M}.$$

Therefore, combining both and choosing $C_T = \max\{C_T^{(S)}, C_T^{(T)}\}$, we get

$$\begin{aligned} & \sup_{k=0, \dots, M} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \\ &= \sup_{k=0, \dots, M} \|X(t_k) - X_N(t_k) + X_N(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \\ &\leq \sup_{k=0, \dots, M} \|X(t_k) - X_N(t_k)\|_{L^2(\Omega)} + \sup_{k=0, \dots, M} \|X_N(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \\ &\leq \frac{C_T^{(S)}}{\lambda_N^\gamma} \frac{C_T^{(T)} \log(M)}{M} \\ &\leq C_T (\lambda_N^{-\gamma} + \frac{\log(M)}{M}). \end{aligned}$$

□

Remark 5.8

Using L'Hospitals rule of limits [e.g. see For13, p.187], it is easy to see that

$$\lim_{M \rightarrow \infty} \frac{\log(M)}{M^\epsilon} = 0$$

for any $\epsilon > 0$. Hence, for a sufficiently large threshold $\tilde{M}$ it holds that

$$\log(M) \leq M^\epsilon$$

for all $M \geq \tilde{M}$. Since convergence rates are primarily asymptotic statements, we will replace $\log(M)$ by M^ϵ for an arbitrary small $\epsilon > 0$ in later discussions.

As mentioned in Remark 4.4 it is often from interest to choose N and M such that the combined convergence rate given by (83) is optimal with respect to the computational costs of the model. In the case of the exponential Euler method, the computational costs equal the $N \cdot M$ arithmetical operations. We now want to analyze this in more detail for a special case. But before we do this, we would like to make a remark on the handling of the nonlinearity.

Remark 5.9

In most numerical schemes, the nonlinearity must be treated in the form

$$F_N^l(X_{N,M}(t_k)) = \langle F_N(X_{N,M}(t_k)), \phi_l \rangle_H.$$

This requires additional approximation steps, and thus additional computational effort. An efficient way of handling this is by using fast fourier transformation. This would lead to an additional computational effort of $\log(M)$. We won't cover this in this work but refer to [JKW11] for more details.

Following [JK09a, Section 4] and [JKW11, Section 1], we will now further study the previous theorem in the special case of the one-dimensional semilinear heat equation with space-time white noise and Dirichlet boundary conditions known from Example 3.13. Let $H = L^2(0, 1)$, let $A = \frac{d^2}{dx^2}$ be the Laplace operator, let W be cylindrical Brownian motion and let $x_0 = 0$. Further let $f \in C_0^\infty(\mathbb{R})$ and $F : L^2(0, 1) \rightarrow L^2(0, 1)$ be the by f induced Nemytskii operator. Then (27) will be

$$\begin{cases} \frac{d}{dt}X(t) = \left[\frac{d^2}{dx^2}X(t) + f(X(t)) \right] dt + dW(t), & x \in [0, 1], \quad t \in [0, T] \\ X(t)(0) = X(t)(1) = 0 \\ X(0) = 0. \end{cases} \quad (84)$$

From Example 3.13, we know that in this setup Assumptions 1, 3 and 4 are satisfied such that this problem has a unique mild solution X . Moreover, Assumption 5 is obviously satisfied for $x_0 = 0$ and from Example 2.14 we know that Assumption 2 is satisfied for $\gamma \in (0, 1/4)$. Lastly, Assumption 6 is also satisfied since $f \in C_0^\infty(\mathbb{R})$, as we saw in Corollary 5.3. Altogether, Theorem 5.7 can be applied here.

Corollary 5.10

In the previously described setup, let X be the solution to (84) and let $X_{N,M}$ be the process given by (61). Then there exists a constant C_T , so that

$$\sup_{k=0, \dots, M} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \leq C_T \left(N^{-1/2+\epsilon} + \frac{\log(M)}{M} \right) \quad (85)$$

holds for all $N, M \geq 2$, and $\epsilon > 0$ arbitrary small, where $t_k = k \frac{T}{M}$ for $k = 0, \dots, M$.

Proof. As described above, since all Assumptions 1 - 6 are satisfied, Theorem 5.7 can be applied with $\gamma = \frac{1}{4} - \epsilon$ for an arbitrary small $\epsilon > 0$. Moreover, from previous examples we know that $\lambda_N = \pi^2 N^2$ and thus

$$\lambda_N^\gamma = \pi^{-\frac{1}{2}+2\epsilon} N^{-\frac{1}{2}+2\epsilon}.$$

And hence

$$\begin{aligned} \sup_{k=0, \dots, M} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} &\stackrel{\text{Theorem 5.7}}{\leq} C_T \left(\lambda_N^{-\gamma} + \frac{\log(M)}{M} \right) \\ &= C_T \left(\pi^{-\frac{1}{2}+2\epsilon} N^{-\frac{1}{2}+2\epsilon} + \frac{\log(M)}{M} \right) \\ &\stackrel{(1)}{\leq} C_T \left(N^{-\frac{1}{2}+2\epsilon} + \frac{\log(M)}{M} \right), \end{aligned}$$

where we used $0 < \pi^{-\frac{1}{2}+2\epsilon} < 1$ in (1). □

Considering Remark 5.8, (85) becomes

$$\sup_{k=0,\dots,M} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} \leq C_T \left(N^{-1/2+\epsilon} + M^{-1+\epsilon} \right)$$

such that for the one-dimensional stochastic heat equation with space-time white noise the exponential Euler method results in a strong convergence rate of $1/2 - \epsilon$ in space and $1 - \epsilon$ in time and thus the time is twice as regular as the space. Combining this result with the computational costs $N \cdot M$, it is reasonable to choose $N = M^2$. Setting $K = MN$ this leads to $N = K/M = K/(N^{1/2}) \iff N = K^{2/3}$ and similar $M = K^{1/3}$. Thus we get

$$\begin{aligned} \sup_{k=0,\dots,M} \|X(t_k) - X_{N,M}(t_k)\|_{L^2(\Omega)} &\leq C_T \left((K^{2/3})^{-1/2+\epsilon} + (K^{1/3})^{-1+\epsilon} \right) \\ &= 2C_T (K^{-1/3+\epsilon}) \end{aligned}$$

such that the optimal convergence rate with respect to the computational costs $N \cdot M$ is approximately $1/3$. This coincides with the statements in Remark 4.4. Note that for comparison, the semi-implicit Euler Maruyama Scheme has a temporal convergence rate $1/4$ (see Remark 4.14) and thus by Remark 4.4 a combined convergence rate of $1/6$ with respect to computational costs.

However, the exponential Euler method also has its drawbacks. First, one needs to know the eigenvalues of A which in practice is rarely the case. Second, the properties of F being twice Frechet-differentiable and globally Lipschitz continuous on H are too restrictive for many applications. For example Nemytskii operators F induced by $f(x) = -x$, $f(x) = -x^3$ or $f(x) = x - x^3$ fail the global Lipschitz condition.

5.3. Simulate this method

We now like to state a way of simulate (64). Define

$$X_{N,M}^l(t_k) = \langle \phi_l, X_{N,M}(t_k) \rangle_H \quad \text{and} \quad F_N^l = \langle \phi_l, F_N \rangle_H$$

for $l = 1, \dots, N$, such that using the representation (64) of $X_{N,M}(t_k)$ we get

$$\begin{aligned} X_{N,M}^l(t_{k+1}) &= \langle \phi_l, X_{N,M}(t_{k+1}) \rangle_H \\ &= \langle \phi_l, e^{A_N h} X_{N,M}(t_k) \rangle_H \\ &\quad + \left\langle \phi_l, A_N^{-1} (e^{A_N h} - I) F_N(X_{N,M}(t_k)) \right\rangle_H \\ &\quad + \left\langle \phi_l, \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s) \right\rangle_H. \end{aligned} \tag{86}$$

Further, recall that we can express $W_N(t)$ by $W_N(t) = \sum_{l=1}^N \sqrt{q_l} \phi_l \beta_l(t)$ and thus the last term of (86) can be written as

$$\begin{aligned} \left\langle \phi_l, \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s) \right\rangle_H &= \left\langle \phi_l, \sum_{n=1}^N \sqrt{q_n} \phi_n \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} d\beta_n(s) \right\rangle_H \\ &= \sqrt{q_l} \int_{t_k}^{t_{k+1}} e^{-\lambda_l(t_{k+1}-s)} d\beta_l(s). \end{aligned}$$

Moreover, by the nature of the stochastic convolution this integral is centered gaussian (see Section 2.5.1). Using the Ito isometry, its variance can be determined as

$$\begin{aligned}
 \text{VAR} \left(\int_{t_k}^{t_{k+1}} e^{-\lambda_l(t_{k+1}-s)} d\beta_l(s) \right) &= \mathbb{E} \left[\left(\int_{t_k}^{t_{k+1}} e^{-\lambda_l(t_{k+1}-s)} d\beta_l(s) \right)^2 \right] \\
 &= \int_{t_k}^{t_{k+1}} e^{(-\lambda_l(t_{k+1}-s))^2} ds \\
 &= \int_{t_k}^{t_{k+1}} e^{2\lambda_l(s-t_{k+1})} ds \\
 &= \frac{1}{2\lambda_l} \left[e^{2\lambda_l(s-t_{k+1})} \right]_{t_k}^{t_{k+1}} \\
 &= \frac{1}{2\lambda_l} (1 - e^{-2\lambda_l h}).
 \end{aligned}$$

Therefore, we can simulate $\langle \phi_l, \sum_{n=1}^N \sqrt{q_n} e_n \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} d\beta_n(s) \rangle$ by

$$\sqrt{q_l} \int_{t_k}^{t_{k+1}} e^{-\lambda_l(t_{k+1}-s)} d\beta_l(s) \stackrel{\mathcal{D}}{=} \sqrt{q_l} \left(\frac{1}{2\lambda_l} (1 - e^{-2\lambda_l h}) \right)^{1/2} R_k^l$$

where $(R_k^l)_{l=1,\dots,N,k=0,\dots,M-1}$ is a set of iid standard gaussian distributed random variables. Using (86) and previous explanations, we can express $X_{N,M}(t_{k+1})$ as a system of N one-dimensional equations by the following scheme:

$$\begin{aligned}
 X_{N,M}^1(t_k + 1) &= e^{-\lambda_1 h} X_{N,M}^1(t_k) + \frac{(1 - e^{-\lambda_1 h})}{\lambda_1} F_N^1(X_{N,M}(t_k)) + \left(\frac{q_1}{2\lambda_1} (1 - e^{-2\lambda_1 h}) \right)^{1/2} R_k^1 \\
 &\vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad (87) \\
 X_{N,M}^N(t_k + 1) &= e^{-\lambda_N h} X_{N,M}^N(t_k) + \frac{(1 - e^{-\lambda_N h})}{\lambda_N} F_N^N(X_{N,M}(t_k)) + \left(\frac{q_N}{2\lambda_N} (1 - e^{-2\lambda_N h}) \right)^{1/2} R_k^N.
 \end{aligned}$$

$X_{N,M}(t_{k+1})$ is then given by

$$X_{N,M}(t_{k+1}) = \sum_{l=1}^N X_{N,M}^l(t_{k+1}) \phi_l.$$

If one wants to simulate paths of a solution process with the help of the exponential Euler method it is common that one represents these on a grid. Let $H = L^2(0, 1)$ and let $(y_j)_{j=0,\dots,J}$ be a partition of $[0, 1]$ with $y_{j+1} - y_j = 1/J =: \delta$. Then the following algorithm generates a path of X on the meshgrid $[0, T] \times [0, 1]$ using the exponential Euler method:

Algorithm 5.11 (Exponential Euler method)

For $k = 0, \dots, M - 1$ do the following:

- (1) Given $X_{N,M}(t_k)$, compute $X_{N,M}^1(t_{k+1}), \dots, X_{N,M}^N(t_{k+1})$ according to (87).
- (2) Compute $X_{N,M}(t_{k+1})(y_j) = \sum_{l=1}^N X_{N,M}^l(t_{k+1}) \phi_l(y_j)$.

Output: $X_{N,M}(t_{k+1})(y_j) = \sum_{l=1}^N X_{N,M}^l(t_{k+1})\phi_l(y_j)$ for $k = 0, \dots, M$, $j = 0, \dots, J$.

Note that in Step 1 $F_N^l(X_{N,M}(t_k))$ also needs to be approximated. We will discuss how this is done in later Section 6.4.

Example 5.12

Let us consider the one-dimensional semilinear heat equation with additive space-time white noise and Dirichlet boundary conditions known from Example 3.13. Let F be a Nemytskii operator induced by $f : \mathbb{R} \rightarrow \mathbb{R}$ and let $x_0 = 0$. For simulation purposes let us consider a weighted Laplace operator $A = \vartheta \frac{d^2}{dx^2}$. (27) then becomes

$$\begin{cases} dX(t) = \left[\vartheta \frac{d^2}{dx^2} X(t) + f(X(t)) \right] dt + dW(t), & x \in [0, 1], \quad t \in [0, T] \\ X_0 = 0 \\ X(t)(0) = X(t)(1) = 0. \end{cases}$$

Note that the eigenvalues of $-A$ are then given by $\lambda_l = \vartheta \pi^2 l^2$ with corresponding eigenfunctions $\phi_l(y) = \sqrt{2} \sin(\pi l y)$ for $l \geq 1$, $y \in [0, 1]$. We will consider three different nonlinear functions $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

(F1) $f(x) \equiv 0$,

(F2) $f(x) = \begin{cases} 10e^{1/(x^2-1)}, & x \in (-1, 1) \\ 0, & \text{otherwise,} \end{cases}$

(F3) $f(x) = -10x^3$.

Note that by using (F1) we consider the linear stochastic heat equation, (F2) is in $C_0^\infty(\mathbb{R})$ and is known from Example 3.12 and (F3) is a polynomial of odd degree with negative leading coefficient. Further, note that Corollary 5.10 does not hold for (F3) but at least pathwise convergence can be shown for (F3) in this setting [see Jen09]. Thus it still may be from interest to simulate paths using (F3).

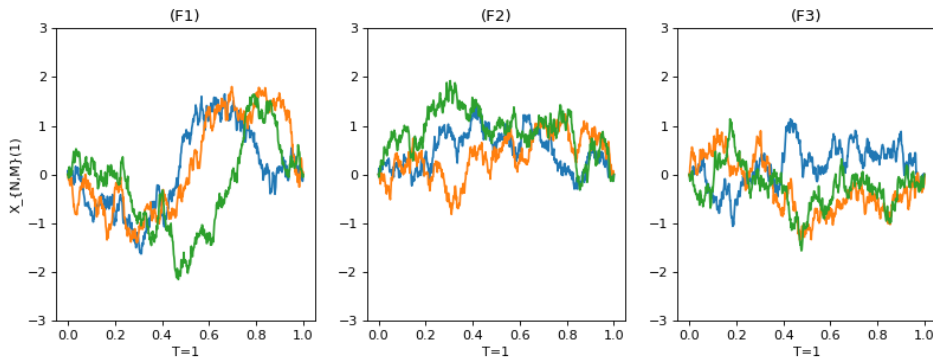


Figure 1: Three Sample paths for different nonlinearities at time $T = 1$ generated by Algorithm 5.11 based in the exponential Euler method with $M = 250$, $N = 250^2$ and $J = 600$.

Following the discussion of Corollary 5.10 we choose $M = 250$ and $N = 250^2$. Let $J = 600$. Figure 1 shows three different paths for each nonlinearity (F1), (F2) and

($F3$) at time $T = 1$ generated by Algorithm 5.11. The plots indicate the influence of the nonlinearity but the influence of space-time white noise makes it difficult to compare. Hence, we like to investigate this further by using the same noise for all three nonlinearities. Figure 2 shows a 3-dimensional surface plot of one sample path

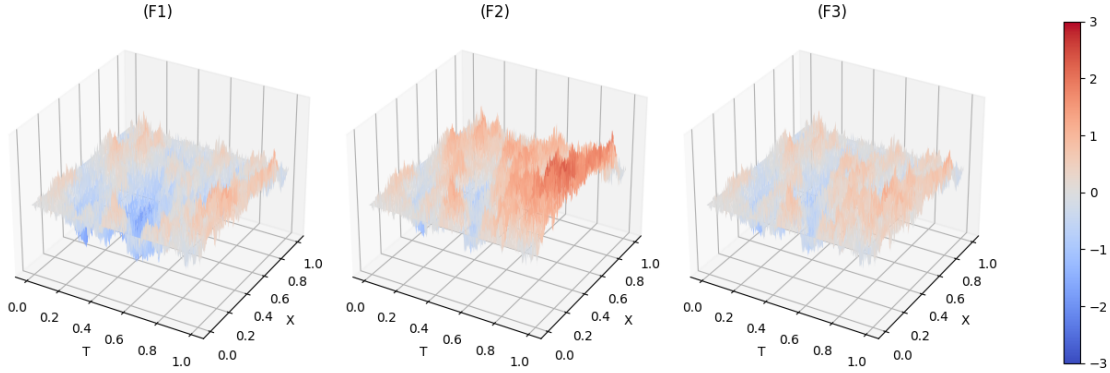


Figure 2: Surface plot of one path for each nonlinearity generated by Algorithm 5.11 based in the exponential Euler method with $M = 250$, $N = 250^2$ and $J = 600$, where the noise is the same for all three nonlinearities.

for each nonlinearity generated by Algorithm 5.11 using the same noise in all three simulations. We clearly see the influence of the nonlinearity. While ($F2$) amplifies the positive values of the approximation, ($F3$) acts like a balancing factor for values that are far away from zero to keep the path around 0. To further analyse the different development of each path, we plot the same sample paths as in Figure 2 in Figure 3 at different times $T = 0$, $T = 0.1$, $T = 0.25$, $T = 0.5$ and $T = 1$. We see that especially for ($F2$) the influence of the nonlinearity increases with increasing time. At time $T = 0.1$ all paths are very similar. While at time $T = 0.25$ ($F1$) and ($F3$) still produce quiet similar paths, but the influence of ($F2$) can already been seen. At later times $T = 0.5$ and especially $T = 1$ the paths of ($F1$) and ($F2$) differ significantly, but on the other hand the ones of ($F1$) and ($F3$) differ only for values that are far away from zero.

5.4. Truncation VS replacement

This section follows the idea of [Hil20] to modify the exponential Euler method known from (87). For this we will first recap the approach of [Hil20] by introducing the connection between the (semi-)linear stochastic heat equation and Ornstein-Uhlenbeck processes and then stating the main idea behind the modified method. After that we will extend this idea into the semilinear context. While we will not state any proofs here, for the linear case, the approach can be justified comparing total variation norms [see Hil20, Theorem 3.3].

Let $H = L^2(0, 1)$. We consider the semilinear stochastic heat equation with Dirichlet boundary conditions and additive space-time white noise as known from Example 3.13. we will denote the Ornstein-Uhlenbeck processes by u . Following the

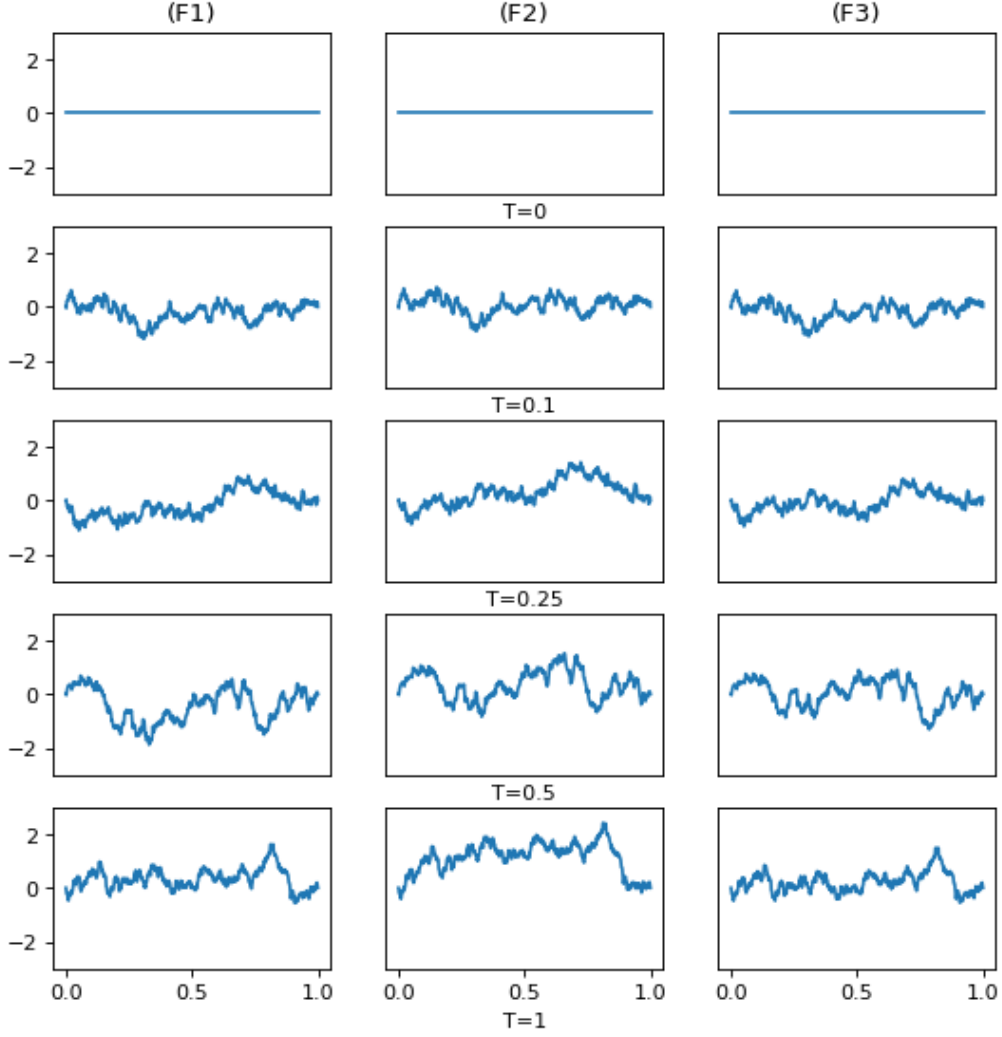


Figure 3: Sample paths for each nonlinearity at different times, generated by Algorithm 5.11 based in the exponential Euler method with $M = 250$, $N = 250^2$ and $J = 600$, where the noise is the same for all three nonlinearities.

idea of [HT21, p.5], we decompose (49) into its linear and its nonlinear part:

$$X(t) = e^{At}x_0 + \underbrace{\int_0^t e^{A(t-s)}dW(s)}_{=:X_t^0} + \underbrace{\int_0^t e^{A(t-s)}F(X(s))ds}_{=:X_t^F}. \quad (88)$$

For this section we set $x_0 = 0$. Thus, when the nonlinear part is zero (e.g. $F \equiv 0$) we can denote the solution by $X = X^0$. It is well known that [e.g. see Hil20] X_t^0 can be represented as a series of independent one-dimensional Ornstein-Uhlenbeck processes $(u_l(t))_{l \geq 1}$ such that

$$X_t^0(y) = \sum_{l=1}^{\infty} u_l(t)\phi_l(y), \quad t \in [0, T], \quad y \in [0, 1]. \quad (89)$$

Equation (89) can be seen by taking the spectral decompositions $e^{At} = \sum_{l=1}^{\infty} e^{-\lambda_l t}\phi_l$

Then,

$$\begin{aligned}
 & X_{N,M}(t_{k+1})(y) \\
 &= \sum_{l=1}^N X_{N,M}^l(t_{k+1})\phi_l(y) \\
 &= \sum_{l=1}^N u_l(t_{k+1})\phi_l(y) \\
 &\quad + \sum_{l=1}^N \left(\frac{(1 - e^{-\lambda_l h})}{\lambda_l} F_N^l(X_{N,M}(t_k)) + e^{-\lambda_l h} (X_{N,M}^l(t_k) - u_l(t_k)) \right) \phi_l(y).
 \end{aligned} \tag{92}$$

In reference to [Hil20] we will call (92) *semilinear truncation method*. Note that, as (92) indicates, it is enough to simulate Ornstein-Uhlenbeck processes $(u_l)_{l=1,\dots,N}$. Everything else can be calculated. This property leads to the next section, where we introduce the semilinear replacement method as a modification of the semilinear truncation method.

5.4.2. The semilinear replacement method

If we evaluate $X(t)$ only at equidistant points $0 = y_0 < \dots < y_J = 1$, with $y_{j+1} - y_j = 1/J$ for $j = 0, \dots, J-1$, $J \in \mathbb{N}$ we can develop a different way of expressing X . For the linear case, this has been done by [Hil20] and is called *replacement method*. For the semilinear case, consider eigenvalues $(\phi_l)_{l \geq 1}$ of $A = \frac{d^2}{dx^2}$ with $\phi_l(y) = \sqrt{2} \sin(\pi l y)$ for $y \in [0, 1]$. Then, for any $l = 1, \dots, J$ the following periodic properties

- (i) $\phi_{l+2mJ}(y_j) = \phi_l(y_j)$,
- (ii) $\phi_{2J-l+2mJ}(y_j) = -\phi_l(y_j)$,
- (iii) $\phi_J(y_j) = 0$,

are satisfied for every y_j , $0 \leq j \leq J$, $m \in \mathbb{N}_0$. Moreover, consider the empirical inner product

$$\langle \nu, v \rangle_J = \frac{1}{J} \sum_{j=0}^J \nu(y_j) v(y_j), \quad \nu, v \in L^2(0, 1)$$

and note that $(\phi_l)_{1 \leq l < J}$ creates an orthonormal basis with respect to the empirical inner product:

$$\langle \phi_l, \phi_k \rangle_J = \frac{2}{J} \sum_{j=0}^J \sin(\pi l y_j) \sin(\pi k y_j) = \delta_{l=k}, \tag{93}$$

where $\delta_{l=k}$ is the delta function which equals 1 if $l = k$ and 0 otherwise (see Remark 2.2). The main idea is to use the above periodic properties and that $(\phi_l)_{1 \leq l < J}$ builds an orthonormal basis to create a finite dimensional sum without directly truncating the spatial dimensions. For this, we define the sets

- $\mathcal{I}_l^+ := \{l + 2mJ : m \in \mathbb{N}_0\}$,

- $\mathcal{I}_l^- := \{2J - l + 2mJ : m \in \mathbb{N}_0\}$,
- $\mathcal{I}^0 := \{mM : m \in \mathbb{N}_0\}$,

for $l = 1, \dots, J-1$ such that we can decompose $\{m : m \in \mathbb{N}_0\}$ by

$$\{m : m \in \mathbb{N}_0\} = \left(\dot{\bigcup}_{l=1, \dots, J-1} (\mathcal{I}_l^+ \dot{\cup} \mathcal{I}_l^-) \right) \dot{\cup} \mathcal{I}^0.$$

Further, we define $\mathcal{I}_l^\neq := \dot{\bigcup}_{n=1, \dots, J-1, n \neq l} (\mathcal{I}_n^+ \dot{\cup} \mathcal{I}_n^-)$.

First, consider the mild solution $X(t)$ given in Example 3.13 and note that $X(t)$ can be expressed by $X(t) = \sum_{l \geq 1} X^l(t) \phi_l$ with $X^l(t) = \langle X(t), \phi_l \rangle_{L^2(0,1)}$. Then define $U_l(t)$ for $l = 1, \dots, J-1$ as

$$\begin{aligned} U_l(t) &:= \langle X(t), \phi_l \rangle_J = \left\langle \sum_{m=1}^{\infty} X^m(t) \phi_m, \phi_l \right\rangle_J = \sum_{m=1}^{\infty} X^m(t) \langle \phi_m, \phi_l \rangle_J \\ &= \sum_{m \in \mathcal{I}_l^+} X^m(t) \langle \phi_m, \phi_l \rangle_J + \sum_{m \in \mathcal{I}_l^-} X^m(t) \langle \phi_m, \phi_l \rangle_J \\ &\quad + \sum_{m \in \mathcal{I}^0} X^m(t) \langle \phi_m, \phi_l \rangle_J + \sum_{m \in \mathcal{I}_l^\neq} X^m(t) \langle \phi_m, \phi_l \rangle_J \\ &= \sum_{m \in \mathcal{I}_l^+} X^m(t) - \sum_{m \in \mathcal{I}_l^-} X^m(t) + 0 + 0, \end{aligned}$$

where we used the above defined periodic properties. Therefore, with respect to the empirical inner product, $X(t)(y_j)$ can be expressed by

$$X(t)(y_j) = \sum_{l=1}^{J-1} \langle X(t), \phi_l \rangle_J \phi_l(y_j) = \sum_{j=1}^{J-1} U_l(t) \phi_l(y_j).$$

Similarly, let $X_{N,M}$ be the approximation given by (64), then we can express $X_{N,M}(t_k)(y_j)$ by

$$X_{N,M}(t_k)(y_j) = \sum_{l=1}^{J-1} \hat{U}_l(t_k) \phi_l(y_j)$$

where $\hat{U}_l(t_k) = \sum_{m \in \mathcal{I}_l^+} X_{N,M}^m(t_k) - \sum_{m \in \mathcal{I}_l^-} X_{N,M}^m(t_k)$, with $X_{N,M}^m(t_k)$ as in Scheme (87). Note that we set $X_{N,M}^m(t_k) = 0$ for $m > N$ for $\hat{U}_l(t_k)$ to be well-defined.

Next, let l be large, such that $\lambda_l \gg 1/h$. Then, using (90), we get

$$u_l(t_{k+1}) = \underbrace{e^{-\lambda_l h}}_{\approx 0} u_l(t_k) + \underbrace{\left(\frac{1 - e^{-2\lambda_l h}}{2\lambda_l} \right)^{1/2}}_{\approx \frac{1}{\sqrt{2\lambda_l}}} R_k^l,$$

where $R_k^l \sim \mathcal{N}(0, 1)$. Thus $(u_l(t_{k+1}))_{k=0, \dots, M-1}$ approximately behave like iid real-valued standard gaussian random variables with variance

$$\text{VAR}(u_l(t_{k+1})) = \frac{1}{2\lambda_l}$$

[see the discussion in Hil20, Chapter 3]. Similar, by using (91) for large l we get

$$X_{N,M}^l(t_{k+1}) = \underbrace{u_l(t_{k+1})}_{\sim \mathcal{N}\left(0, \frac{1}{2\lambda_l}\right)} + \underbrace{\left(\frac{1 - e^{-\lambda_l h}}{\lambda_l}\right) F_N^l(X_{N,M}(t_k))}_{\approx \frac{1}{\lambda_l} F_N^l(X_{N,M}(t_k))} + \underbrace{e^{-\lambda_l h}}_{\approx 0} (X_{N,M}^l(t_k) - u_l(t_k)).$$

For sufficiently smooth F , we can assume that

$$\frac{1}{\lambda_l} F_N^l(X_{N,M}(t_k)) \approx 0$$

and thus $(X_{N,M}^l(t_{k+1}))_{k=0, \dots, M-1}$ also behave like iid real-valued standard gaussian random variables with variance

$$\text{VAR}(X_{N,M}^l(t_{k+1})) = \frac{1}{2\lambda_j}.$$

Example 5.13

Consider the setting of Corollary 5.10. Then, the orthonormal basis is given by $\phi_l(x) = \sqrt{2} \sin(\pi l x)$. Let F be a Nemytskii operator induced by $f \in C_0^\infty(\mathbb{R})$ and let $\text{supp}(f) \subset (0, 1)$ such that $f(0) = f(1) = 0$. Now consider $\langle f, \phi_l \rangle_{L^2(0,1)}$. Using integration by parts twice, we get

$$\begin{aligned} \langle f, \phi_l \rangle_{L^2(0,1)} &= \int_0^1 f(x) \sqrt{2} \sin(\pi l x) dx \\ &= \sqrt{2} \left(\left[f(x) \left(-\frac{1}{\pi l}\right) \cos(\pi l x) \right]_0^1 - \int_0^1 f'(x) \frac{1}{\pi l} \cos(\pi l x) dx \right) \\ &= \sqrt{2} \left(\underbrace{\left[f(x) \left(-\frac{1}{\pi l}\right) \cos(\pi l x) \right]_0^1}_{\stackrel{(1)}{=} 0} - \underbrace{\left[f'(x) \left(-\frac{1}{(\pi l)^2}\right) \sin(\pi l x) \right]_0^1}_{\stackrel{(1)}{=} 0} \right. \\ &\quad \left. + \int_0^1 f''(x) \frac{1}{(\pi l)^2} \sin(\pi l x) dx \right) \\ &= \frac{1}{(\pi l)^2} \langle f'', \phi_l \rangle_{L^2(0,1)}, \end{aligned}$$

where (1) follows by knowing $f(0) = f(1) = f'(0) = f'(1) = 0$. Hence, iteratively we get

$$\left| \langle f, \phi_l \rangle_{L^2(0,1)} \right| = \left| \frac{1}{(\pi l)^{2q}} \langle f^{(q)}, \phi_l \rangle_{L^2(0,1)} \right| \leq \underbrace{\frac{1}{(\pi l)^{2q}} \|f^{(q)}\|_{L^2(0,1)}}_{\stackrel{(2)}{\leq C}} \underbrace{\|\phi_l\|_{L^2(0,1)}}_{=1} \leq \frac{C}{(\pi l)^{2q}}$$

with $f^{(q)}$ being the q th. derivative of f and where (2) follows since $f^{(2q)} \in C_0^\infty(0, 1)$ for every $q \in \mathbb{N}$ and thus its $L^2(0, 1)$ -Norm is bounded. In particular, this shows that $\langle f, \phi_l \rangle_{L^2(0,1)}$ is small for large l . Since f shares its properties with F by the condition $F(\nu)(x) = f(\nu(x))$, also $F_N^l(\nu) = \langle F_N(\nu), \phi_l \rangle_{L^2(0,1)}$ is very small for large l .

Now, Define $\mathcal{I}_l := \mathcal{I}_l^+ \dot{\cup} \mathcal{I}_l^-$. Since gaussian random variables are stable under addition, we can choose a sufficient large L and approximate $\sum_{m \in \mathcal{I}_l, m \geq LJ} X_{N,M}^m(t_k)$ by a real-valued centered gaussian random variable $\tilde{R}_k^{l,L}$ given by

$$\tilde{R}_k^{l,L} \sim \mathcal{N} \left(0, \sum_{m \in \mathcal{I}_l, m \geq LJ} \frac{1}{2\lambda_m} \right),$$

so that we can estimate $\tilde{U}_l(t_k)$ by

$$\tilde{U}_l^L(t_k) = \sum_{m \in \mathcal{I}_l, m < LJ} X_{N,M}^m(t_k) + \tilde{R}_k^{l,L}. \quad (94)$$

Note that $\{\tilde{R}_k^{l,L}\}_{l=1,\dots,J-1, k=1,\dots,M}$, is a set of iid random variables. Let s_l^2 denote the variance of $\tilde{R}_k^{l,L}$:

$$s_l^2 := \sum_{m \in \mathcal{I}_l, m \geq LJ} \frac{1}{2\lambda_m}.$$

Then the *semilinear replacement method* is defined as

$$\tilde{X}_{J,L,M}(t_k)(y_j) := \sum_{l=1}^{J-1} \tilde{U}_l^L(t_k) \phi_l(y_j), \quad (95)$$

for every $k = 0, \dots, M$, $j = 0, \dots, J$. Note that instead of $\tilde{X}_{N,M}$ we denote the semilinear replacement method by $\tilde{X}_{J,L,M}$ to emphasize the importance of J and L . Further, note that unlike the exponential Euler method, the solution $\tilde{X}_{J,L,M}$ is only defined on the discrete grid $(y_j)_{j=0,\dots,J}$. Therefore, $F_N^j(\tilde{X}_{J,L,M})$ must be computed with respect to the empirical inner product, e.g.

$$F_N^l(\tilde{X}_{N,M}(t_k)) = \langle F_N(\tilde{X}_{N,M}(t_k)), \phi_l \rangle_J = \frac{1}{J} \sum_{j=0}^J F_N(x_{t_k})(y_j) \phi_l(y_l).$$

The following remark allows us to exactly calculate s_l^2 for every $l = 1, \dots, J-1$.

Remark 5.14

Following [Hil20, Lemma 3.1], it can be shown that s_l^2 can be expressed in closed form. For this define the symmetric function $\rho : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ with

$$\rho(x, y) = \frac{1}{2}(x(1-y)), \quad x \leq y.$$

Further $\Sigma \in \mathbb{R}^{J+1} \times \mathbb{R}^{J+1}$ with $\Sigma_{i,j} = \rho(y_i, y_j)$ for $i, j = 0, \dots, J$ and define $b_l = \sqrt{2}(\sin(\pi l y_0), \dots, \sin(\pi l y_J))^T \in \mathbb{R}^{J+1}$, then

$$s_l^2 = \frac{1}{J^2} b_l^T \Sigma b_l - \sum_{m \in \mathcal{I}_l, m < LJ} \frac{1}{2\lambda_l}.$$

The following algorithm then states how to calculate $\tilde{X}_{N,M}(t_{k+1})$.

Algorithm 5.15 (Semilinear replacement method)

Choose $L \in \mathbb{N}$ and set $N = LJ$. Set $\tilde{X}_{J,L,M}(0) = 0$.

For $0 \leq k \leq M - 1$ do the following:

(1) Given $(\tilde{X}_{J,L,M}(t_k)(y_j))_{j=0,\dots,J}(t_{k+1})$, compute $X_{N,M}^1(t_{k+1}), \dots, X_{N,M}^J(t_{k+1})$ according to (87).

(2) For $1 \leq l < J$ do the following:

(2.1) Compute s_l^2 according to Remark 5.14 and generate $\tilde{R}_{k+1}^{l,L} \sim \mathcal{N}(0, s_l^2)$.

(2.2) Compute

$$\tilde{U}_l^L(t_{k+1}) = \sum_{m \in \mathcal{I}_l, m < LJ} X_{N,M}^m(t_{k+1}) + \tilde{R}_{k+1}^{l,L}.$$

(3) Compute $\tilde{X}_{J,L,M}(t_{k+1})(y_j)$ for $0 \leq j \leq J$ by

$$\tilde{X}_{J,L,M}(t_{k+1})(y_j) = \sum_{l=1}^{J-1} \tilde{U}_l^L(t_{k+1}) \phi_l(y_j).$$

Output: $\tilde{X}_{J,L,M}(0)(y_j) = 0$ $\tilde{X}_{J,L,M}(t_{k+1})(y_j) = \sum_{l=1}^{J-1} \tilde{U}_l^L(t_{k+1}) \phi_l(y_j)$ for $1 \leq k \leq M - 1$, $0 \leq j \leq J$.

Example 5.16

Consider the situation of Example 5.12. In particular, consider a one-dimensional semilinear stochastic heat equation with additive space-time white noise and Dirichlet boundary conditions on $L^2(0, 1)$ and let F be a Nemystkii operator induced by a real-valued function f given by either (F1) ($f \equiv 0$), (F2) ($f \in C_0^\infty(\mathbb{R})$) or (F3) (f is a polynomial of odd degree with negative leading coefficient). Let $M = 250$, $J = 100$ and Let $L = 6$. Figure 4 shows three sample paths for each nonlinearity

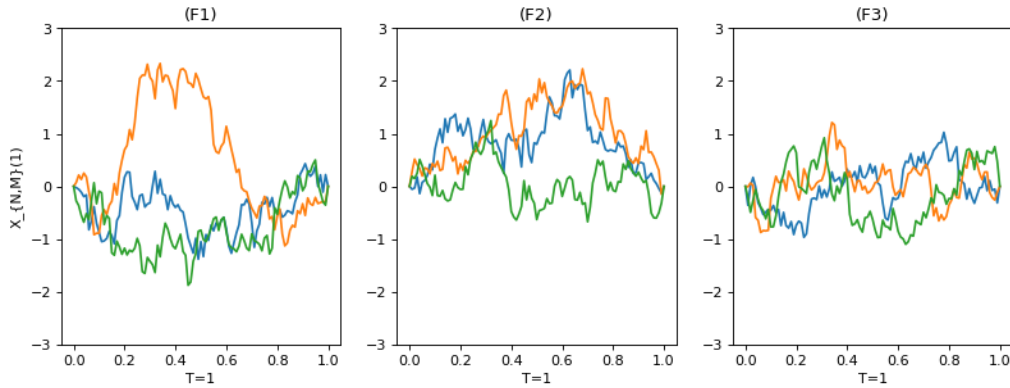


Figure 4: Three Sample paths for different nonlinearities at time $T = 1$ generated by Algorithm 5.15 based in the semilinear replacement method with $M = 250$, $J = 100$ and $L = 6$.

generated by Algorithm 5.15. Similar to Example 5.12, one can see the impact of the nonlinear functions. While (F2) again causes on average larger values than in the linear case with (F1), (F3) acts as a balancing factor around zero.

Note that the semilinear replacement method is only well-defined for $(y_j)_{j=0,\dots,J}$, which is why the strong error of the approximation can also only be considered with respect to the empirical norm given by

$$\|\nu\|_J := \frac{1}{\sqrt{J}} \left(\sum_{j=0}^J (\nu(y_j))^2 \right)^{1/2}$$

for any $\nu \in L^2(0, 1)$.

Remark 5.17

Let X be the solution to Problem (27) and let $\tilde{X}_{J,L,M}$ be an fully discrete approximation of that solution. Then the strong Error with respect to the empirical norm is given by

$$\sup_{t_k} \mathbb{E} \left[\left\| X(t_k) - \tilde{X}_{J,L,M}(t_k) \right\|_J^2 \right]^{1/2}.$$

For the special case of the linear replacement method with $L = 1$ and $F \equiv 0$ a first possible approach might be to use the connection to the exponential Euler method. Note that $J = N$ in this special case. For this let $X_{J,M}$ be the process generated by the exponential Euler method. Then we get the following:

$$\begin{aligned} \tilde{X}_{J,1,M}(t_{k+1})(y_j) &= \sum_{l=1}^{J-1} \left(\sum_{m \in \mathcal{I}_l, m < J} X_{J,M}^m(t_{k+1}) + \tilde{R}_{k+1}^{l,1} \right) \phi_l(y_j) \\ &= \sum_{l=1}^{J-1} \left(X_{J,M}^l(t_{k+1}) + \tilde{R}_{k+1}^{l,1} \right) \phi_l(y_j) \\ &= \sum_{l=1}^{J-1} X_{J,M}^l(t_{k+1}) \phi_l(y_j) + \sum_{l=1}^{J-1} \tilde{R}_{k+1}^{l,1} \phi_l(y_j) \\ &= X_{J,M}(t_{k+1})(y_j) + \sum_{l=1}^{J-1} \tilde{R}_{k+1}^{l,1} \phi_l(y_j). \end{aligned} \tag{96}$$

Then using the connection to the exponential Euler method and its strong convergence result, we get

$$\mathbb{E} \left[\left\| X(t_k) - \tilde{X}_{J,L,M}(t_k) \right\|_J^2 \right] \leq \mathbb{E} \left[\left\| X(t_k) - X_{N,M}(t_k) \right\|_J^2 \right] + \mathbb{E} \left[\left\| X_{N,M}(t_k) - \tilde{X}_{J,L,M}(t_k) \right\|_J^2 \right]$$

First, consider the second term. Using (96) we get

$$\begin{aligned} \mathbb{E} \left[\left\| X_{N,M}(t_k) - \tilde{X}_{J,L,M}(t_k) \right\|_J^2 \right] &= \mathbb{E} \left[\left\| \sum_{l=1}^{J-1} \tilde{R}_k^{l,1} \phi_l \right\|_J^2 \right] \\ &= \mathbb{E} \left[\frac{1}{J} \sum_{j=0}^J \sum_{l=1}^{J-1} (\tilde{R}_k^{l,1} \phi_l(y_j))^2 \right] \\ &= \sum_{l=1}^{J-1} \mathbb{E} \left[(\tilde{R}_k^{l,1})^2 \right] \|\phi_l\|_J^2 = \sum_{l=1}^{J-1} \sum_{m \in \mathcal{I}_l, m \geq J} \frac{1}{2\lambda_m^2} \\ &\leq \sum_{m \geq J} \frac{1}{2\lambda_m^2}. \end{aligned}$$

And thus $\mathbb{E} \left[\left\| X_{N,M}(t_k) - \tilde{X}_{J,L,M}(t_k) \right\|_J^2 \right] \rightarrow 0$ for $J \rightarrow \infty$. Then, for the remaining term we get

$$\begin{aligned} & \mathbb{E} \left[\left\| X(t_k) - X_{N,M}(t_k) \right\|_J^2 \right] \\ & \leq \mathbb{E} \left[\left\| X(t_k) - X_{N,M}(t_k) \right\|_{L^2(0,1)}^2 \right] \\ & \quad + \mathbb{E} \left[\left\| X(t_k) - X_{N,M}(t_k) \right\|_J^2 \right] - \left(\mathbb{E} \left[\left\| X(t_k) - X_{N,M}(t_k) \right\|_{L^2(0,1)}^2 \right] \right) \\ & \stackrel{(1)}{\leq} C_T (\lambda_{N+1}^{-\gamma} + \frac{\log(M)}{M}) + R, \end{aligned}$$

with $R := \mathbb{E} \left[\left\| X(t_k) - X_{N,M}(t_k) \right\|_J^2 \right] - \left(\mathbb{E} \left[\left\| X(t_k) - X_{N,M}(t_k) \right\|_{L^2(0,1)}^2 \right] \right)$. (1) follows by using the convergence result from Theorem 5.7. Now one must show that the difference of the empirical norm to the $L^2(0,1)$ -norm converges to 0. This and the general semilinear case will be postponed to future research. Note, however, that with a general nonlinear term an error term is added in each iteration compared to the exponential Euler method, so it becomes more difficult to show the convergence of the semilinear replacement method using the exponential Euler method. In particular, this additional error term is no longer gaussian distributed due to the influence of nonlinearity, which makes the whole analysis even more difficult.

6. Numerical error analysis

In the previous chapters we described how to numerically approximate a semilinear parabolic SPDE. In particular, we introduced the fully discrete exponential Euler scheme and its modification, the semilinear replacement method. We also stated a strong convergence rate of the exponential Euler method. In this chapter we will look at ways to illustrate convergence rates and to compare numerical schemes using simulations. Before we will introduce three concepts - the strong error approximation, the weak error approximation and the quadratic variation approximation - we will outline three problems that may appear.

1) *No explicit solution.*

When considering SPDEs, we may be able to show that under certain conditions a solution exists, but in most cases this solution can not be stated explicitly. For example, as we saw in the last chapter, the solution of a linear stochastic heat equation can be expressed explicitly as an infinite sum of Ornstein-Uhlenbeck processes [see Hil20]. But as soon as we add a nonlinear term F and thus consider the semilinear stochastic heat equation known from Example 3.13, we cannot state an explicit solution anymore. In this case it is not possible to directly simulate sample paths of the solution process. This is problematic, since the strong error (and also the weak error) is reliant on the solution process. In such cases we will use a so-called reference solution. Due to the known strong (or weak) convergence, we assume that by choosing N and M sufficiently large, the approximation $X_{N,M}$ is sufficiently close to X such that we can use this approximation as our reference instead of the real solution process. This will be discussed in more detail in Section 6.1.

2) *Computational costs.*

Simulation methods often need a high computational effort to accomplish good results. This effort includes computational operations (like arithmetical operations and trigonometrical operations) as well as evaluations of the nonlinearity F and simulations of independent gaussian random variables [see JKW11, p.909]. In the discussion of Corollary 5.10 we mentioned that the exponential Euler method has $N \cdot M$ arithmetical operations such that its computational costs are proportional to $N \cdot M$ and that with respect to these computational costs it is optimal to choose $N = M^2$. Note that the smaller the strong (or weak) convergence rate, the larger N and M must be to achieve good results. Further note that as we saw in Theorem 5.7, the convergence rate is directly dependent on γ and Assumption 2 indicates that γ is smaller the more irregular the noise is. In particular, when using the highly irregular space-time white noise, we need to choose N and M to be large to get a reliable simulation. Additionally, every below introduced concept rely on either an expected value, which will be approximated by Monte-Carlo methods, or on the $L^2(0, 1)$ -norm, which will be approximated by the trapezium rule, or on both. In all cases, we need to generate a large number of independent copies of the underlying approximation, which multiplies the computational effort of one simulations by the number of copies [e.g. see the discussions in LP17].

3) *Comparing simulated paths.*

By Definition 4.1 the strong error is the mean of the error given by the pathwise comparison between the solution process and the approximation at a given time $t \in [0, T]$. Hence, to accurately simulate this error, we must ensure that the underlying Brownian motion of both paths is the same whenever we compare two paths at any time. Otherwise, the error can become very large. To illustrate this, take $b = (b(t))_{t \in [0, T]}$ as the underlying Brownian motion of one trajectory and $\tilde{b} = -b$ as the underlying Brownian motion of the other. Then for any $t \in [0, T]$ $|b(t) - \tilde{b}(t)| = 2|b(t)|$ can become very large in general. In the next section we will introduce a way on how to handle this problem.

In the upcoming sections we only consider the case of the one-dimensional semilinear stochastic heat equation on $H = L^2(0, 1)$ with space-time white noise and Dirichlet boundary conditions known from Example 3.13.

6.1. Strong error approximation

This section is based on the approach of [LPS14] to numerically approximate the strong error. Let X be the solution of (84) and let $X_{N,M}$ be a corresponding approximation. Further, as usual, let $(t_k)_{k=0, \dots, M}$ be a set of equidistant points which satisfy $0 = t_0 < t_1 < \dots < t_M = T$ with $t_{k+1} - t_k = T/M = h$ for all $k = 0, \dots, M-1$. Then for any t_k the strong error of $X_{N,M}(t_k)$ is given by

$$\|X(t_k) - X_{N,M}(t_k)\|_{L^2(0,1)} = \mathbb{E}_\Omega \left[\|X(t_k, \omega) - X_{N,M}(t_k, \omega)\|_{L^2(0,1)}^2 \right]^{1/2} \quad (97)$$

for $\omega \in \Omega$, as defined in Definition 4.1. In the following discussion, we only consider $k = M$ such that $t_k = T$.

As mentioned above, in many cases we cannot sample directly from X due to the lack of the existence of an explicit solution. Hence, we will now introduce three reference solutions. For this consider $N_{ref}, M_{ref} \in \mathbb{N}$ with $N_{ref} \gg N$ and $M_{ref} \gg M$. Then consider the three approximations $X_{N_{ref}, M}$, $X_{N, M_{ref}}$ and $X_{N_{ref}, M_{ref}}$, where the first one has a large number of spatial dimensions and thus can be seen as a reference solution for the spatial dimension, the second one has very tight discrete grid of $[0, T]$ with grid width $h_{ref} = T/M_{ref}$ and thus can be seen as a reference solution for the temporal dimension and the third one combines both conditions and thus can be seen as a total reference solution. Note that it is reasonable to consider separate reference solutions for spatial and temporal dimensions as soon as one wants to simulate only the spatial or the temporal error. Consequently we get three strong error representations depending on what the given task is:

- 1) $\|X_{N_{ref}, M}(T) - X_{N, M}(T)\|_{L^2(\Omega)}$ for the strong error of the spatial approximation,
- 2) $\|X_{N, M_{ref}}(T) - X_{N, M}(T)\|_{L^2(\Omega)}$ for the strong error of the temporal approximation,
- 3) $\|X_{N_{ref}, M_{ref}}(T) - X_{N, M}(T)\|_{L^2(\Omega)}$ for the strong error of the fully discrete approximation.

By definition the strong error is the mean of the squared pathwise error

$$\|X(T, \omega) - X_{N,M}(T, \omega)\|_{L^2(0,1)}$$

of all trajectories $\omega \in \Omega$. Note that $X(T, \omega)$ can be replaced by $X_{N_{ref},M}(T, \omega)$, $X_{N,M_{ref}}(T, \omega)$ or $X_{N_{ref},M_{ref}}(T, \omega)$ if a reference solution is needed. As described in the third problem above, we need to make sure that for any trajectory ω the underlying Brownian motions of X and $X_{N,M}$ are the same. When using a reference solution we can explicitly state the dependence. For this we first need to consider the structure of the approximation of the noise. There are two main types:

Type 1) When the stochastic part is approximated by increments of the underlying Brownian motion:

$$\Delta W_k^N = W_N(t_{k+1}) - W_N(t_k),$$

like in the semi-implicite Euler-Maruyama method (see Section 4.3.1), or

Type 2) when the stochastic part is simulated by considering the whole stochastic convolution:

$$\int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s) \stackrel{D}{=} \sum_{l=1}^N \phi_l \left(\frac{1}{2\lambda_l} (1 - e^{-2\lambda_l h}) \right)^{1/2} R_k^l,$$

like in the exponential Euler method and the semilinear replacement method.

Note that by the structure of the approximation schemes, we consider the noise separately for every spatial dimension l , $1 \leq l \leq N$. Thus, we must only consider the temporal dimensions here. In the following, let $h_{ref} = T/M_{ref}$ denote the grid width of the reference method, such that $0 = t_1^{ref} < \dots < t_{M_{ref}}^{ref} = T$ is an equidistant partition of $[0, T]$. Further let $h = \kappa h_{ref}$ for $\kappa \in \mathbb{N}$. The idea behind the following is to restructure the approximation of the noise so that we can use the noise of the reference solution to generate the noise of the approximation.

Type 1) The following is based on [LPS14, p.439]. From before we know that

$$W_N(t) = \sum_{l=1}^N \phi_l \beta_l(t),$$

where $\beta_l(t)$ is a real-valued one-dimensional Brownian motion for every $l = 1, \dots, N$. Thus, knowing that $\beta_l(t_{k+1}^{ref}) - \beta_l(t_k^{ref}) \sim \mathcal{N}(0, h_{ref})$ for all $l = 1, \dots, N$ and $k = 0, \dots, M_{ref} - 1$ (due to stationary and independent increments), we can express the increment $W_N(t_{k+1}^{ref}) - W_N(t_k^{ref})$ by

$$W_N(t_{k+1}^{ref}) - W_N(t_k^{ref}) = \sum_{l=1}^N \phi_l (\beta_l(t_{k+1}^{ref}) - \beta_l(t_k^{ref})) = \sqrt{h_{ref}} \sum_{l=1}^N \phi_l R_{k_{ref}}^l$$

with

$$R_{k_{ref}}^l = \frac{\beta_l(t_{k+1}^{ref}) - \beta_l(t_k^{ref})}{\sqrt{h_{ref}}} \stackrel{iid}{\sim} \mathcal{N}(0, 1).$$

Then with $h = \kappa h_{ref}$ we can express the increment $W_N(t_{k+1}) - W_N(t_k)$ by

$$W_N(t_{k+1}) - W_N(t_k) = \sum_{n=0}^{\kappa-1} \left(W_N(t_k + t_{n+1}^{ref}) - W_N(t_k + t_n^{ref}) \right).$$

Type 2) Consider

$$W_{A_N}^{ref}(t_{k+1}) := \int_{t_k^{ref}}^{t_{k+1}^{ref}} e^{A_N(t_{k+1}-s)} dW_N(s).$$

By the Scheme (87) we know that we can simulate this stochastic convolution by

$$\sum_{l=1}^N \phi_l \left(\frac{1}{2\lambda_l} (1 - e^{-2\lambda_l h_{ref}}) \right)^{1/2} R_{k_{ref}}^l$$

with $R_k^l \stackrel{iid}{\sim} \mathcal{N}(0, 1)$ for $l = 1, \dots, N$ $k = 0, \dots, M_{ref} - 1$. Thus, by setting $h = \kappa h_{ref}$, we get

$$\begin{aligned} W_{A_N}(t_{k+1}) &= \int_{t_k}^{t_{k+1}} e^{A_N(t_{k+1}-s)} dW_N(s) \\ &= \sum_{n=0}^{\kappa-1} e^{A_N(\kappa-(n+1))h_{ref}} \int_{t_k+t_n^{ref}}^{t_k+t_{n+1}^{ref}} e^{A_N(t_k+t_{n+1}^{ref}-s)} dW_N(s) \\ &= \sum_{n=0}^{\kappa-1} \sum_{l=1}^N \phi_l e^{-\lambda_l(\kappa-(n+1))h_{ref}} \left(\frac{1}{2\lambda_l} (1 - e^{-2\lambda_l h_{ref}}) \right)^{1/2} R_{k+n_{ref}}^l, \end{aligned}$$

with $R_{k+n_{ref}}^l := R_{\kappa k_{ref}+n_{ref}}^l$. In particular, for $\langle \phi_l, W_{A_N}(t_{k+1}) \rangle_{L^2(0,1)}$ we get

$$\langle \phi_l, W_{A_N}(t_{k+1}) \rangle_{L^2(0,1)} = \sum_{n=0}^{\kappa-1} e^{-\lambda_l(\kappa-(n+1))h_{ref}} \left(\frac{1}{2\lambda_l} (1 - e^{-2\lambda_l h_{ref}}) \right)^{1/2} R_{k+n_{ref}}^l. \quad (98)$$

To better understand the idea behind the above calculations let us consider a simple example with $h = 3h_{ref}$ and $t_k = t_0$. Then we have $t_k = t_0 = t_0^{ref}$ and $t_{k+1} = t_1 = t_3^{ref}$. Hence,

$$\begin{aligned} \int_{t_0}^{t_1} e^{A_N(t_1-s)} dW_N(s) &= \int_{t_0^{ref}}^{t_3^{ref}} e^{A_N(t_3^{ref}-s)} dW_N(s) \\ &\stackrel{(*)}{=} e^{A_N(t_3^{ref}-t_1^{ref})} \int_{t_0^{ref}}^{t_1^{ref}} e^{A_N(t_1^{ref}-s)} dW_N(s) \\ &\quad + e^{A_N(t_2^{ref}-t_1^{ref})} \int_{t_1^{ref}}^{t_2^{ref}} e^{A_N(t_2^{ref}-s)} dW_N(s) \\ &\quad + \int_{t_2^{ref}}^{t_3^{ref}} e^{A_N(t_3^{ref}-s)} dW_N(s) \\ &= \sum_{n=0}^2 e^{A_N(2-n)h_{ref}} \int_{t_n^{ref}}^{t_{n+1}^{ref}} e^{A_N(t_{n+1}^{ref}-s)} dW_N(s), \end{aligned}$$

where we used $\int_{t_0^{ref}}^{t_1^{ref}} e^{A_N(t_3-s)} dW(s) = \int_{t_0^{ref}}^{t_1^{ref}} e^{A_N(t_3-t_1^{ref}+t_1^{ref}-s)} dW(s)$ in $(*)$.

While the stochastic part of the exponential Euler method can be purely represented by type 2, it is more complicated for the semilinear replacement method. The following remark will briefly introduce the problem.

Remark 6.1

Let $0 = y_0 < y_1 < \dots < y_J = 1$ be a partition of $[0, 1]$ with equidistant points $(y_j)_{j=0,\dots,J}$ and grid width $y_{j+1} - y_j = 1/J =: \delta$ and let $N = LJ$. Recall, that the approximation of the semilinear replacement method $\tilde{X}_{J,L,M}(t_k)(y_j)$ is given by

$$\tilde{X}_{J,L,M}(t_k)(y_j) = \sum_{l=1}^{J-1} \left(\sum_{m \in \mathcal{I}_l, m < JL} X_{N,M}^l(t_k) + \tilde{R}_k^{l,L} \right) \phi_l(y_j).$$

Compared to the solution generated by the exponential Euler method, $\tilde{X}_{J,L,M}(t_k)$ has an additional stochastic term, given by $\tilde{R}_k^{l,L}$ for every l . Moreover, $\tilde{R}_k^{l,L} \sim \mathcal{N}(0, s_l^2)$ with $s_l^2 = \sum_{m \in \mathcal{I}_l, m \geq LJ} \frac{1}{2\lambda_l}$ depends on $N = LJ$. Luckily, the normal distribution is stable with respect to summation. Hence, in all cases, (with a spatial, temporal or full reference solution) it is quite easy to express $\tilde{R}_k^{l,L}$ of the approximation by its part from the reference solution with an additional centered gaussian random variable to match the missing variance term. Since we do not simulate this method in terms of strong convergence, we will not go into further details here.

Knowing how to handle the noise, we now want to approximate the strong error. As (97) indicates we must first estimate the expected value and then approximate the $L^2(0, 1)$ -Norm. In the following we use $X_{N_{ref}, M_{ref}}(T)$ as our reference solution at time T but below also holds for the other two types of reference solutions. First, we will use Monte-Carlo simulations to estimate the expectation. This leads to

$$\|X_{N_{ref}, M_{ref}}(T) - X_{N,M}(T)\|_{L^2(\Omega)}^2 \approx \frac{1}{D} \sum_{d=1}^D \|X_{N_{ref}, M_{ref}}^d(T) - X_{N,M}^d(T)\|_{L^2(0,1)}^2,$$

with D independent copies $X_{N_{ref}, M_{ref}}^d(T)$ and $X_{N,M}^d(T)$, $1 \leq d \leq D$ of $X_{N_{ref}, M_{ref}}(T)$ and $X_{N,M}(T)$. Note that one can understand $X_{N_{ref}, M_{ref}}^d(T)$ and $X_{N,M}^d(T)$ as the d 'th sample path at time T , such that it is reasonable to write

$$\|X_{N_{ref}, M_{ref}}^d(T) - X_{N,M}^d(T)\|_{L^2(0,1)}^2 := \|X_{N_{ref}, M_{ref}}(T, \omega_d) - X_{N,M}(T, \omega_d)\|_{L^2(0,1)}^2.$$

To approximate the norm let us consider the partition of $[0, 1]$ defined in Remark 6.1. We can then use the trapezium rule (See A.4) to estimate the norm. We get

$$\|X_{N_{ref}, M_{ref}}^d(T) - X_{N,M}^d(T)\|_{L^2(0,1)}^2 \approx \delta \sum_{j=1}^J \left(X_{N_{ref}, M_{ref}}^d(T, y_j) - X_{N,M}^d(T, y_j) \right)^2.$$

And thus overall

$$\begin{aligned} \|X_{N_{ref}, M_{ref}}(T) - X_{N,M}(T)\|_{L^2(\Omega)}^2 &\approx \frac{1}{D} \sum_{d=1}^D \|X_{N_{ref}, M_{ref}}^d(T) - X_{N,M}^d(T)\|_{L^2(0,1)}^2, \\ &\approx \frac{1}{D} \sum_{d=1}^D \delta \sum_{j=1}^J \left(X_{N_{ref}, M_{ref}}^d(T, y_j) - X_{N,M}^d(T, y_j) \right)^2. \end{aligned}$$

In order for this approximation to be good, D and J must be large. Hence, simulating the strong error needs high computational effort as described in Problem 3 above.

It should be noted that by approximating the expected value and the L^2 -norm, we get an additional sampling error. While we will not discuss the error arising by the trapezium rule, the Monte-Carlo error will be discussed in the next section.

The strong error approximation can be used for two tasks. First, one can show the strong convergence rate numerically by simulating the error for different values N and M . Second, one can use it to compare two different numerical schemes. Note that theoretical convergence results are always asymptotic. Therefore, it is possible that the constants appearing there vary strongly depending on the method. This can lead to problems in a numerical comparison between two methods. Therefore, in this case a precision parameter ϵ should be chosen and then it should be looked how large the dimensions must be in each case so that the strong error of an approximation falls below this precision value [see discussion in JKW11].

One additional Flawback of this method is that one cannot mix the types defined above in the sense that one can't use a reference method of type 1 when one wants to calculate the error of a numerical scheme of type 2. This is because the approximations of the noises are so different that a noise of one type cannot be calculated by the noise of the other type. So, for example, you cannot use the exponential Euler method as a reference method and the semi-implicite Euler-Maryurama method as the approximation to compare with.

6.2. Weak error approximation

As seen in section 4.1 the weak convergence rate is at least as large as the strong convergence rate. Moreover, as a rule of thumb the weak error is twice as small as the strong error [see LP17, Chapter 1]. Thus, the sampling error generated by approximating the expectation using Monte-Carlo simulations plays a more important role. In the following we will talk about the influence of the sampling error in the context of weak error analysis. The following is based on [LP17].

As known from Section 4.1 for a sufficiently smooth functional $\varphi : H \rightarrow \mathbb{R}$ the weak error is given by

$$|\mathbb{E}[\varphi(X(T))] - \mathbb{E}[\varphi(X_{N,M}(T))]| = |\mathbb{E}[\varphi(X(T)) - \varphi(X_{N,M}(T))]|. \quad (99)$$

Note that we need to approximate the expectation of $\varphi(X_{N,M}(T))$ and thus, we need the definition of the empirical expected value.

Definition 6.2

Let H be a Hilbert space and let $Y \in H$ be a random element. The *empirical expected value* is given by

$$\mathbb{E}_D[Y] := \frac{1}{D} \sum_{d=1}^D Y_d,$$

where $(Y_d)_{d=1,\dots,D}$ is a set of independent copies of Y .

Note, that we already used this definition in the previous section. If the solution process does not have an explicit solution, such as in the semilinear stochastic heat equation, then we need a reference solution. In this case, the expected value of this reference solution must also be estimated by the empirical expected value. We will use the fully discrete reference solution $X_{N_{ref}, M_{ref}}(T)$ but the following will also hold for the spatial or temporal reference solution introduced before. We get the following weak error approximation:

$$\left| \mathbb{E}_D[\varphi(X_{N_{ref}, M_{ref}}(T))] - \mathbb{E}_D[\varphi(X_{N, M}(T))] \right| = \left| \mathbb{E}_D[\varphi(X_{N_{ref}, M_{ref}}(T)) - \varphi(X_{N, M}(T))] \right|.$$

Note that we have

$$\begin{aligned} & \underbrace{\left| \mathbb{E}_D[\varphi(X_{N_{ref}, M_{ref}}(T)) - \varphi(X_{N, M}(T))] \right|}_{\text{approx. weak error}} \\ &= \underbrace{\left| \mathbb{E}[\varphi(X_{N_{ref}, M_{ref}}(T)) - \varphi(X_{N, M}(T))] \right|}_{\text{exact weak error}} \\ & \quad + \underbrace{\left| \mathbb{E}[\varphi(X_{N_{ref}, M_{ref}}(T)) - \varphi(X_{N, M}(T))] - \mathbb{E}_D[\varphi(X_{N_{ref}, M_{ref}}(T)) - \varphi(X_{N, M}(T))] \right|}_{\text{Monte-Carlo error}}. \end{aligned}$$

So, as stated in [LP17] if the exact weak error is very small the Monte-Carlo error will be the driving factor of the approximated weak error. Hence, one needs a large number of samples to get a low Monte-Carlo error and thus a good approximation of the real weak error. As mentioned in Problem 3 this leads to large computational costs.

Remark 6.3

Note that the influence of the sampling error can be reduced by using the multi-level Monte Carlo method instead of the classical Monte Carlo method. See [LP17] for more details.

The left hand side of (99) indicates, the weak error approximation is not based on pathwise comparisons and thus allows to use different schemes for the reference solution $X_{N_{ref}, M_{ref}}(T)$ and the approximation $X_{N, M}(T)$. While this makes it easier to compare different methods, it also brings the disadvantage that we cannot reuse already calculated terms. In particular, this has a big impact on the computational effort. A first analysis of comparing computational times can be found in [LP17]. As a consequence, it depends on the given task and on the computational possibilities, which weak error estimation one chooses.

6.3. Quadratic Variation Approximation

While both, the simulation of the strong error and the simulation of the weak error need a reference method as soon as no explicit solution is available, we will now introduce a method which does not rely on reference solutions but on central limit theorems. In the following we will consider power variations, in particular the quadratic variation. Even though this leads to functionals which are not smooth enough for the (weak) convergence results to hold, they are still an important topic

in the statistical theory of SPDEs and moreover the total variation distance is connected to the Distribution of a stochastic element and thus controlling the total variation distance also controls the weak convergence [see Hil20]. For the following we consider $[b, 1 - b] \subset [0, 1]$ for $b \in [0, 1/2)$ and for $J \in \mathbb{N}$ let $(y_j)_{j=0, \dots, J}$ be a set of equidistant points of $[b, 1 - b]$ with $b = y_0 < y_1 < \dots < y_J = (1 - b)$ satisfying $y_{j+1} - y_j = \frac{1-2b}{J} =: \delta$. Further, let $X_{N,M}$ be a fully discrete approximation of the semilinear stochastic heat equation. The realized temporal and spatial quadratic variations are then given by

$$V_t = \frac{1}{JM\sqrt{h}} \sum_{k=0}^{M-1} \sum_{j=0}^{J-1} (X_{N,M}(t_{k+1})(y_j) - X_{N,M}(t_k)(y_j))^2 \quad (100)$$

and

$$V_{sp} = \frac{1}{JM\delta} \sum_{k=0}^{M-1} \sum_{j=0}^{J-1} (X_{N,M}(t_k)(y_{j+1}) - X_{N,M}(t_k)(y_j))^2 \quad (101)$$

[e.g. see HT21]. Then, it can be shown [see HT21] that under sufficient assumptions, there exist two central limit theorems, one for the spatial case and one for the temporal case, which are valid for approximations of semilinear parabolic SPDEs. In the following, we state the required assumptions and then briefly introduce the theorems.

Let F be a Nemytskii operator induced by $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $F(\nu)(x) = f(\nu(x))$ for all $\nu \in L^2(0, 1)$, $x \in [0, 1]$. Then, next to all previous assumptions needed for the existence of a solution (see Section 3.1), f and its derivative f' also need to be of at least polynomial growth [see HT21, p.6, Equation (7)]. This assumption is satisfied for two important classes of functions. First, for $f \in C_0^\infty(\mathbb{R})$ since f and its derivative f' are both globally Lipschitz continuous, and second, for f being a polynomial of odd degree with negative leading coefficient.

The central limit theorem for the temporal quadratic variation is given by the following theorem.

Theorem 6.4 ([HT21], Theorem 3.1)

Let $J = o((h)^{-p})$ for some $p < 1/2$. For V_t as defined in (100)

$$\sqrt{JM} \left(V_t - \frac{1}{\sqrt{\pi}} \right) \xrightarrow{\mathcal{D}} \mathcal{N} \left(0, \frac{B}{\pi} \right), \quad J, M \rightarrow \infty \quad (102)$$

holds with $B := 2 + \sum_{G=1}^{\infty} (2\sqrt{G} - \sqrt{G+1} - \sqrt{G-1})^2$.

For the spatial quadratic variation V_{sp} , the central limit theorem is given by the next theorem.

Theorem 6.5 ([HT21], Theorem 3.2)

Let $M = o(J)$. For V_{sp} as defined in (101)

$$\sqrt{JM} \left(V_{sp} - \frac{1}{2} \right) \xrightarrow{\mathcal{D}} \mathcal{N} \left(0, \frac{1}{2} \right), \quad J, M \rightarrow \infty \quad (103)$$

holds.

Remark 6.6

The conditions $J = o(\sqrt{M})$ for the realized temporal quadratic variation and $M = o(J)$ for the realized spatial quadratic variation are necessary for the previous two theorems to be satisfied. We refer to the discussion of [HT19, p.7] for more details.

Note that this method has advantages when comparing different numerical schemes. The most important one is that it does not rely on a reference solution but rather on central limit results. Thus, one can easily use it also for problems with no explicit solution without the need to worry about an approximation error of a reference solution. In particular this bypasses the problem of the strong error approximation when having two different methods with different types of approximating the noise and thus needing two different reference solutions, as described in the discussion of Section 6.1.

On the other side, if one wants to compare two different schemes using this method, one needs to sample many Monte-Carlo iterations to get a reliable marginal distribution which is close to the asymptotic gaussian distribution. Hence, one needs a large amount of simulations and thus runs into high computational costs.

6.4. Approximating the nonlinearity

In addition to the noise, one must also take care of the nonlinearity F when approximating semilinear SPDEs. As seen in Scheme (87), F often appears in the context of an inner product

$$F_N^l(X_{N,M}(t_k)) = \langle F_N(X_{N,M}(t_k), \phi_l \rangle_H.$$

Is $H = L^2(0, 1)$ then this inner product is an integral and also needs to be approximated. In the following, this approximation of the integral representing the inner product will be examined in more detail.

Consider the semilinear stochastic heat equation on $L^2(0, 1)$ with additive space-time white noise and Dirichlet boundary conditions as introduced in Example 3.1. Further, let F be a Nemytskii operator induced by $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $F(\nu)(x) = f(\nu(x))$ for all $\nu \in L^2(0, 1)$, $x \in [0, 1]$. For the following, let $\nu \in L^2(0, 1)$ be arbitrary but fixed. In what follows we will concentrate on the case when the spatial discretization is done via the spektral Galerkin method such that we deal with $P_N F = F_N$ where P_N is the orthogonal projection operator projecting on the space spanned by the first N eigenfunctions of A . From previous examples we know that the Laplace operator A has eigenfunctions $\phi_l(x) = \sqrt{2} \sin(\pi l x)$, $l \geq 1$ and these eigenfunctions create an orthonormal basis of $L^2(0, 1)$ such that we can write $F(\nu) : (0, 1) \rightarrow \mathbb{R}$, $x \mapsto F(\nu)(x)$ by

$$F(\nu)(x) = \sqrt{2} \sum_{l=1}^{\infty} F^l(\nu) \sin(\pi l x)$$

with coefficients

$$F^l(\nu) = \langle \phi_l, F(\nu) \rangle_{L^2(0,1)} = \sqrt{2} \int_0^1 f(\nu(x)) \sin(\pi l x) dx. \quad (104)$$

Similarly, for $P_N F(\nu) = F_N(\nu)$ we have

$$F_N(\nu)(x) = \sqrt{2} \sum_{l=1}^N F_N^l(\nu) \sin(\pi l x).$$

Note that for $1 \leq l \leq N$ the coefficients $F_N^l(\nu)$ coincide with $F^l(\nu)$ from (104), since by construction of F_N

$$\langle \phi_l, F(\nu) \rangle_{L^2(0,1)} = \langle \phi_l, F_N(\nu) \rangle_{L^2(0,1)}, \quad 1 \leq l \leq N$$

holds. To avoid dealing with the integral in (104) we will approximate $F_N^l(\nu)$ in a way that it is easy to translate into a numerical scheme. The following is based on the explanations of [LPS14, p.31ff and p.115].

Consider $\hat{F}(\nu)$, defined by

$$\hat{F}_N(\nu)(y) := \sqrt{2} \sum_{l=1}^N \hat{F}_N^l \sin(\pi l y).$$

where $\hat{F}_N^l$ is the approximation of F_N^l using the trapezium rule (see Section A.4). Then

$$\begin{aligned} \hat{F}_N^l &= \sqrt{2} \delta \left(\frac{f(\nu(0)) \sin(\pi l 0)}{2} + \sum_{j=1}^{J-1} f(\nu(y_j)) \sin(\pi l y_j) + \frac{f(\nu(1)) \sin(\pi l 1)}{2} \right) \\ &= \sqrt{2} \delta \sum_{j=1}^{J-1} f(\nu(y_j)) \sin(\pi l y_j), \end{aligned} \tag{105}$$

where $(y_j)_{j=1, \dots, J}$ are defined as usual with $\delta = 1/J$. Therefore, we get

$$\hat{F}_N(\nu)(y) = \sqrt{2} \sum_{l=1}^N \left(\sqrt{2} \delta \sum_{j=1}^{J-1} f(\nu(y_j)) \sin(\pi l y_j) \right) \sin(\pi l y).$$

Note that $\hat{F}$ is also notated by $I_N F$ and is called trigonometric interpolant [e.g. see LPS14, p.32].

7. Simulations

In this chapter we want to show numerically the theoretical results from Chapter 5 and 6 by means of simulations. For this purpose, we first consider the strong convergence of the exponential Euler method and use the theory from Section 6.1 to simulate its rates. Then we look at the marginal distribution of the realized temporal and spatial quadratic variation and compare it with the results of the respective central limit theorem known from Section 6.3. Throughout this Chapter we will consider the one-dimensional stochastic heat equation on $L^2(0, 1)$ with additive space-time white noise and Dirichlet boundary conditions of the form

$$\begin{cases} dX(t) = \left[\vartheta \frac{d^2}{dx^2} X(t) + f(X(t)) \right] dt + dW(t), & x \in [0, 1], \quad t \in [0, T] \\ X(t)(0) = X(t)(1) = 0 \\ X(0) = 0, \end{cases} \quad (106)$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ generates the Nemytskii operator F and will be chosen appropriately. Moreover, as usual for the temporal dimension let $(t_k)_{k=0, \dots, M}$ be a partition of $[0, T]$ with equidistant values $0 = t_0 < \dots < t_M = T$ with $t_{k+1} - t_k = T/M = h$ and for the spatial dimension let $(y_j)_{j=0, \dots, J}$ be a partition of $[0, 1]$ with equidistant values $0 = y_0 < \dots < y_J = 1$ with $y_{j+1} - y_j = 1/J = \delta$. Let $T = 1$.

7.1. Strong error simulations

For this section let $f \in C_0^\infty(\mathbb{R})$ be $(F2)$ from Example 5.12 and thus given by

$$f(x) = \begin{cases} 10e^{1/(x^2-1)}, & x \in (-1, 1) \\ 0, & \text{otherwise.} \end{cases}$$

Note that $A = \vartheta \frac{d^2}{dx^2}$ is the weighted Laplace operator and, has negative eigenvalues given by $-\lambda_l = \vartheta \pi^2 l^2$ with corresponding eigenfunction $\phi_l(x) = \sqrt{2} \sin(\pi l x)$ for $l \geq 1$ and $x \in [0, 1]$, as we saw before. Let $\vartheta = 0.1$. Further, for given $N, M \in \mathbb{N}$, let $X_{N,M}$ be the approximation of the solution process X generated by the exponential Euler method. Note that the solution of this problem does not have an explicit form and therefore we need reference solutions for the following simulations. For this we use the in Section 6.1 introduced $X_{N_{ref}, M}$, $X_{N, M_{ref}}$ and $X_{N_{ref}, M_{ref}}$. The strong error then will be approximated by the in Section 6.1 introduced formula

$$\|X_{N_{ref}, M}(T) - X_{N, M}(T)\|_{L^2(\Omega)}^2 \approx \frac{1}{D} \sum_{d=1}^D \delta \sum_{j=1}^J \left(X_{N_{ref}, M}^d(T, y_j) - X_{N, M}^d(T, y_j) \right)^2,$$

based on $D = 50$ Monte-Carlo iterations and $J = 100$ (similarly for $X_{N, M_{ref}}$ or $X_{N_{ref}, M_{ref}}$).

7.1.1. Strong error simulation for spatial and temporal discretizations

We first simulate the spatial and temporal convergence rates.

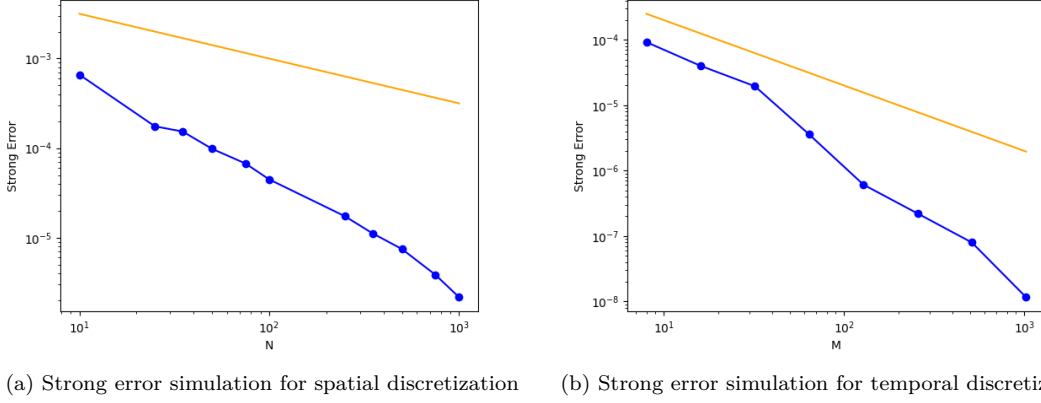


Figure 5: Strong error simulation of spatial and temporal Discretization using the exponential Euler method with reference solutions generated by $N_{ref} = 2000$ and $M_{ref} = 2048$. The orange line represents the respective theoretical rates of $1/2$ (left) and 1 (right).

For the spatial convergence rate, we use $X_{N_{ref},M}$ as the reference method and choose $N_{ref} = 2000$ and $M = 100$. We then generate approximations $X_{N,M}$ for $N \in \{10, 25, 35, 50, 75, 100, 250, 350, 500, 750, 1000\}$ and plot their strong error with respect to the reference method. The results are shown as blue dots in the log-log plot represented in Figure 5a. The orange line serves as a reference and corresponds to the theoretical spatial strong convergence rate of $1/2$ known from Corollary 5.10. We see that the simulations of the strong error in this case converge even faster than $1/2$.

For the temporal convergence rate we use $X_{N,M_{ref}}$ as the reference method and choose $M_{ref} = 2048$ and $N = 100$. we then generate approximations $X_{N,M}$ for $M \in \{8, 16, 32, 64, 128, 256, 512, 1024\}$. and plot their strong error with respect to the reference method. Note that every M is chosen so that $M_{ref} = \kappa M$ for some $\kappa \in \mathbb{N}$, so that the same noise can be used - see Section 6.1 for more information. The results are shown as blue dots in the log-log plot represented in Figure 5b. Again, the orange line serves as reference and corresponds to the theoretical temporal convergence rate of 1 as seen in the discussion of Corollary 5.10. Again, we see that the simulations of the strong error in this case converge even faster than 1 . Note, however, that the reason the simulations yield better rates may be that N_{ref} and M_{ref} from the respective reference solution are not large enough and that 50 Monte-Carlo iterations might be not enough.

7.1.2. Strong error simulation with respect to computational costs

Next, we simulate the strong error with respect to computational effort. As explained in the discussion of Corollary 5.10, the computational costs of the exponential Euler method correspond to the arithmetic operations $N \cdot M$. Thus, based on the convergence rates of the spatial and temporal discretization, it makes sense to choose $N = M^2$. The theoretical total convergence rate is then $1/3$ in this case.

To simulate the strong error with respect to the computational effort, we use the

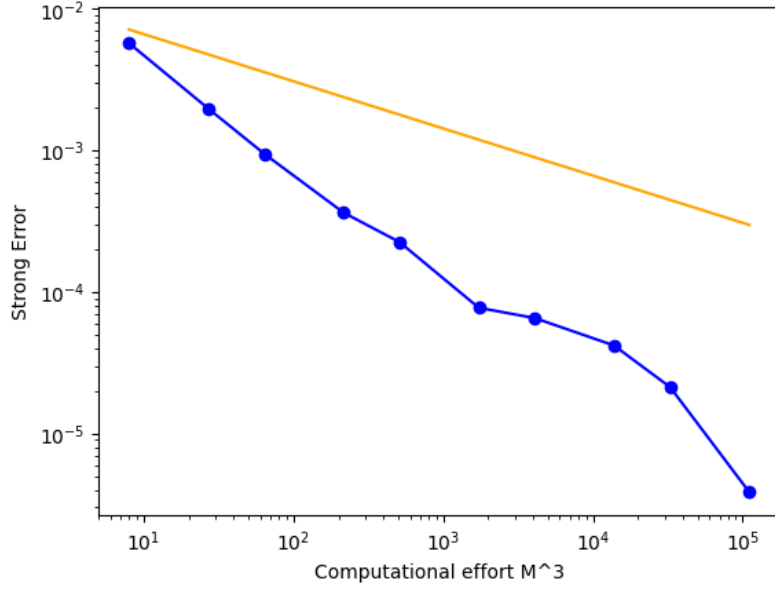


Figure 6: Strong error simulation of the exponential Euler method with respect to computational effort $N \cdot M$ with $N = M^2$ and reference solutions generated by $M_{ref} = 96$ and $N_{ref} = 96^2$. The orange line represents the theoretical rate $1/3$.

reference solution $X_{N_{ref}, M_{ref}}$ with $M_{ref} = 96$ and $N_{ref} = 96^2$. For the approximations $X_{N, M}$ we use $M \in \{2, 3, 4, 6, 8, 12, 16, 24, 32, 48\}$ and $N = M^2$. As before, we made sure that for each M , M_{ref} is a multiple of M so that we can use the same noise. Figure 6 shows a log-log plot of the simulated strong errors with respect to the computational costs of the approximation, represented by the blue dots. Moreover the orange line represents the theoretical convergence rate $1/3$ with respect to the computational costs. We see that the theoretical result can be supported numerically.

7.2. Realized quadratic variation simulations

The goal of this section is to simulate the marginal distribution of the realized temporal and spatial quadratic variations presented in Section 6.3 for the exponential Euler method and the semilinear replacement method for fixed parameters M , N , L and J . Based on the limit theorems presented in Section 6.3, we can use a centered gaussian distribution as a reference and thus compare the goodness of each approximation by looking at the difference between the histogram of Monte-Carlo iterations of realized quadratic variations and the density of the reference distribution. This idea is based on [Hil20, Section 4].

The central limit theorems hold if F is sufficiently regular. In Section 6.3 we mentioned, that this is the case for F being the Nemytskii operator induced by f , where f is either in $C_0^\infty(\mathbb{R})$ or f is an odd polynomial with negative leading coefficients. Hence, in the following we will consider Both cases. Similarly as before, $f \in C_0^\infty(\mathbb{R})$

we be given by

$$f(x) = \begin{cases} e^{1/(x^2-1)}, & x \in (-1, 1) \\ 0, & \text{otherwise,} \end{cases}$$

and for f being a polynom of odd degree, we choose

$$f(x) = -x^3, \quad x \in \mathbb{R}.$$

We will first look at the realized temporal quadratic variation and then at the spatial ones. Let $\vartheta = 1$.

7.2.1. Realized temporal quadratic variation simulations

In this section we consider the realized temporal quadratic variations given by (100). To avoid boundary effects, we use a partition $(\tilde{y}_j)_{j=1,\dots,J}$ of $[b, 1-b] \subset [0, 1]$ with $\tilde{y}_{j+1} - \tilde{y}_j = \frac{1-2b}{J} = \tilde{\delta}$ and with $b > 0$. Moreover, we choose $M = 1000$ and $J = 10$ such that the condition $J = o(\sqrt{M})$ is satisfied.

The goal is to simulate, for fixed M, N, J and L , the marginal distribution of the realized temporal quadratic variation for both the exponential Euler method and the semilinear replacement method, and then to compare it with the gaussian distribution given by the central limit Theorem 6.4. This will be done first for $f \in C_0^\infty(\mathbb{R})$ and then for f being a polynom of odd degree.

$\mathbf{f} \in C_0^\infty(\mathbb{R})$:

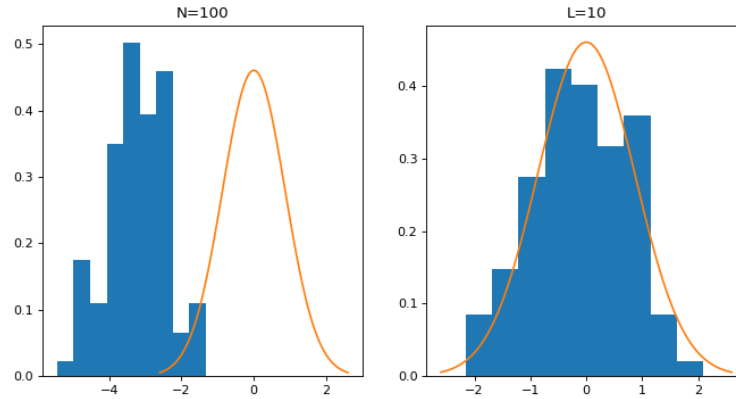


Figure 7: Histogram of 100 Monte-Carlo iterations for realized temporal quadratic variation based on the exponential euler method (left) and the semilinear replacement method (right) - with $f \in C_0^\infty(R)$. The orange line corresponds to the centered gaussian distribution.

First consider $N = 100$ and $L = 10$, then $LJ = 100$ and the computational costs of simulating the exponential Euler method and the semilinear replacement method are roughly the same. Then, Figure 7 shows the marginal distribution of $D = 100$ Monte-Carlo iterations of the realized temporal quadratic variation. The

left hand side shows the simulations of the exponential Euler method and the right hand side the one of the semilinear replacement method. the orange line represents the gaussian distribution given by the central limit theorem. It is clearly to see that even though both methods have roughly the same computational effort, the semilinear replacement methods provides significant better results.

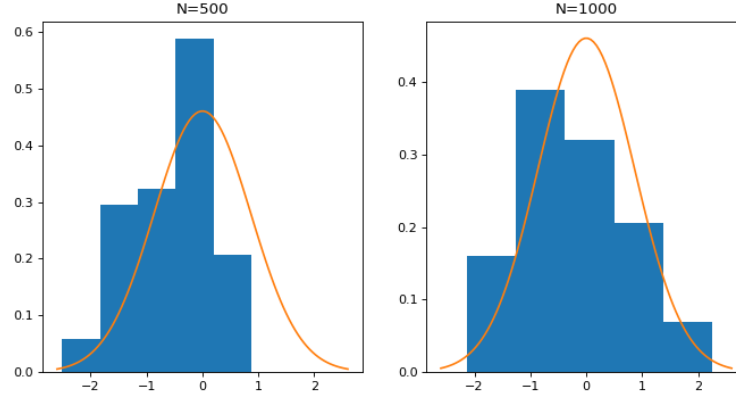


Figure 8: Histogram of 50 Monte-Carlo iterations for realized temporal quadratic variation based on the exponential euler method with different spatial dimensions - with $f \in C_0^\infty(R)$. The orange line corresponds to the centered gaussian distribution.

Even if one increases N significantly in the exponential Euler method, the marginal distribution is still skewed, and only for $N = 1000$ it provides good results as one can see in Figure 8. There we used $N = 500$ for the left simulation and $N = 1000$ for the right simulation and only $D = 50$ Monte-Carlo iterations.

f being a polynom of odd degree with negative leading coefficient:

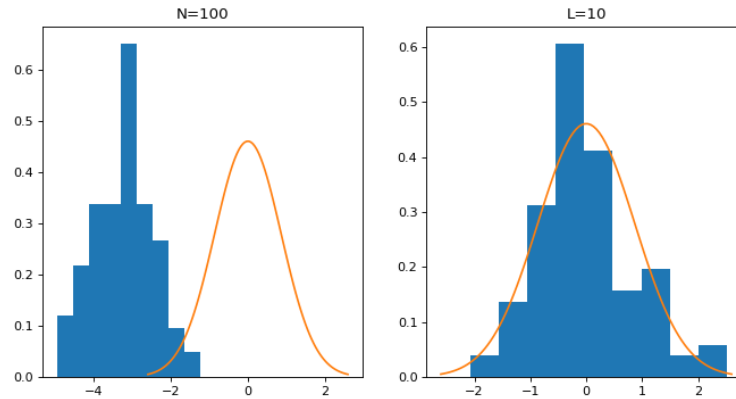


Figure 9: Histogram of 100 Monte-Carlo iterations for realized temporal quadratic variation based on the exponential euler method (left) and the semilinear replacement method (right) - with f being a polynom of odd degree with negative leading coefficient. The orange line corresponds to the centered gaussian distribution.

For f given by $f(x) \mapsto -x^3$ we ran the same simulations with the same parameters. As expected, the results are similar. The semilinear replacement method gives much better results than the exponential Euler method when the computational effort is about the same, as Figure 9 shows. Only with significantly more spatial dimensions does the exponential Euler method produce similarly good results, as can be seen in Figure 10.

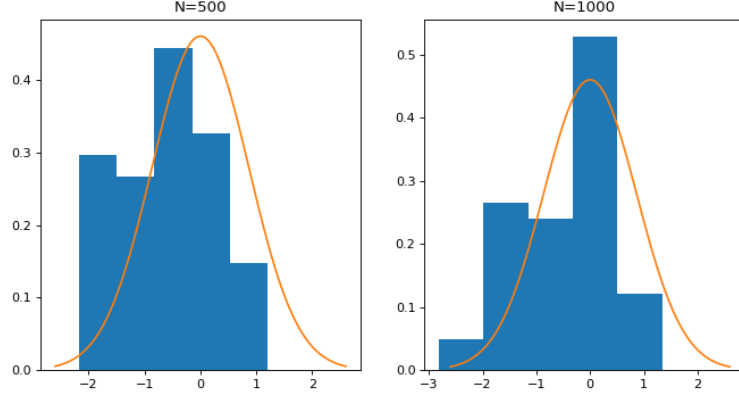


Figure 10: Histogram of 50 Monte-Carlo iterations for realized temporal quadratic variation based on the exponential Euler method with different spatial dimensions - with f being a polynomial of odd degree with negative leading coefficient. The orange line corresponds to the centered gaussian distribution.

7.2.2. Spatial quadratic variation simulations

In this section we consider the realized spatial quadratic variations given by (6.5). Other than for the realized temporal quadratic variations, we now consider again the classical partition $(y_j)_{j=0,\dots,J}$ of $[0, 1]$ with $y_{j+1} - y_j = 1/J = \delta$ known from before. However, since we chose $X(0) = 0$, we will sum from 1 to M rather than from 0 to $M - 1$. Thus the realized spatial quadratic variation becomes

$$V_{sp} = \frac{1}{JM\delta} \sum_{k=1}^M \sum_{j=0}^{J-1} (X_{N,M}(t_k)(y_{j+1}) - X_{N,M}(t_k)(y_j))^2.$$

Moreover, we choose $M = 50$ and $J = 500$ and thus the condition $M = o(J)$ is satisfied and we will use $D = 20$ Monte-Carlo iterations for every simulation in this Section. Again, the goal is to simulate, for fixed M , N , J and L , the marginal distribution of the realized spatial quadratic variation for both the exponential Euler method and the semilinear replacement method, and then to compare it with the gaussian distribution given by the central limit Theorem 6.5. This will be done first for $f \in C_0^\infty(\mathbb{R})$ and then for f being a polynomial of odd degree.

$\mathbf{f} \in \mathbf{C}_0^\infty(\mathbb{R})$:

As for the realized temporal quadratic variation we first choose N and L so that the computational effort of both methods is about the same. Therefore, choose

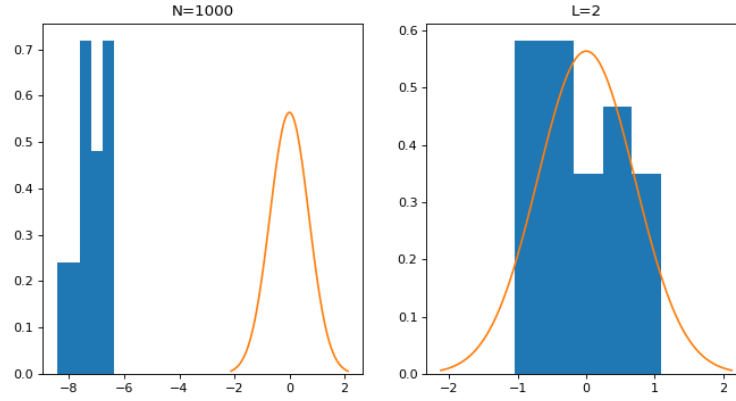


Figure 11: Histogram of 20 Monte-Carlo iterations for realized spatial quadratic variation based on the exponential euler method (left) and the semilinear replacement method (right) - with $f \in C_0^\infty(R)$. The orange line corresponds to the centered gaussian distribution.

$N = 1000$ and $L = 2$. Then $JL = 1000$ holds. Figure 11 shows the marginal distribution of $D = 20$ Monte-Carlo iterations of the realized spatial quadratic variation for the exponential Euler method (left) and the semilinear replacement method (right) compared to the gaussian distribution given by the central limit Theorem 6.5.

Figure 11 shows that, as for the temporal version, the semilinear replacement method provides significant better results. And even for much larger N the exponential Euler method still produces a skewed marginal distribution, as one can see in Figure 12 for $N = 5000$ (left) and $N = 10000$ (right).

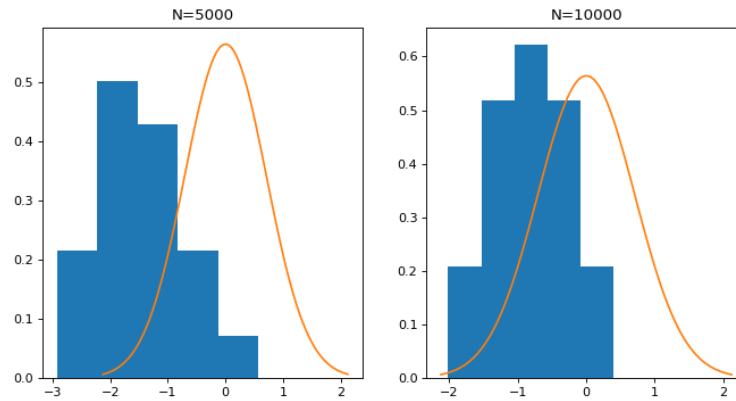


Figure 12: Histogram of 20 Monte-Carlo iterations for realized spatial quadratic variation based on the exponential euler method with different spatial dimensions - with $f \in C_0^\infty(R)$. The orange line corresponds to the centered gaussian distribution.

f being a polynomial of odd degree with negative leading coefficient:

In the case where f a polynomial of odd degree with negative leading coefficient, we choose $N = 1500$ and $L = 3$ such that $JL = 1500$ and the computational

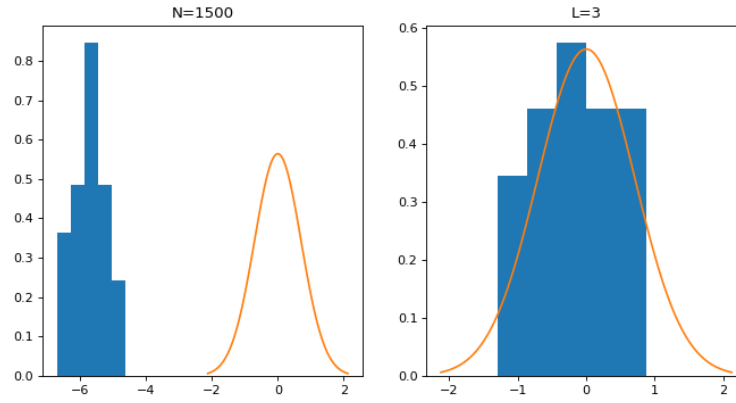


Figure 13: Histogram of 20 Monte-Carlo iterations for realized spatial quadratic variation based on the exponential euler method (left) and the semilinear replacement method (right) - with f being a polynomial of odd degree with negative leading coefficient. The orange line corresponds to the centered gaussian distribution.

effort again is about the same. Then, as for $f \in C_0^\infty(\mathbb{R})$, Figure 13 shows that the marginal distribution of 20 Monte-Carlo iterations of the semilinear replacement method (right) behaves much more like the centered gaussian distribution know from Theorem 6.5 than the marginal distribution of 20 Monte-Carlo iterations of the exponential Euler method (left). And even with much larger N this skew is only partly fixed, as Figure 14 shows for $N = 5000$ (left) and $N = 10000$ (right).

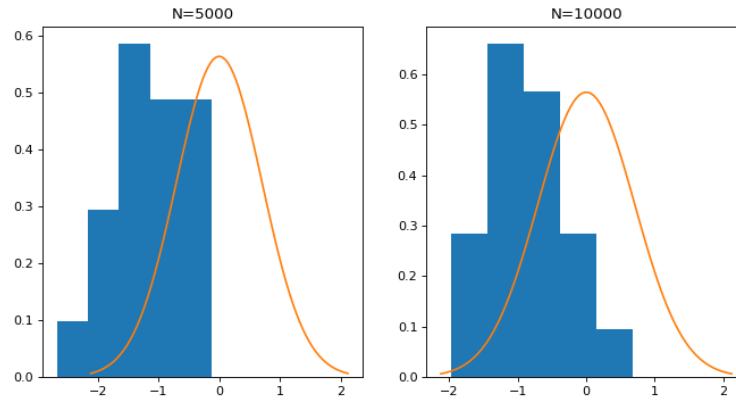


Figure 14: Histogram of 20 Monte-Carlo iterations for realized spatial quadratic variation based on the exponential euler method with different spatial dimensions - with f being a polynomial of odd degree with negative leading coefficient. The orange line corresponds to the centered gaussian distribution.

Remark 7.1

Note that in this analysis we have not considered the additional computational effort required by the semilinear replacement method to generate the additional random variables. While it is not computationally expensive to generate a gaussian distributed random variable, as the parameters J and L increase, this can add up and have a significant impact.

7.3. A note on computational effort

As we discussed in Chapter 6, the computational effort to simulate SPDEs is very high. In particular, this is true for the simulations in this thesis, since all simulations performed here are based on the Monte Carlo method and therefore require a lot of independent copies. Therefore, a tradeoff between informativeness of the graphs and computational effort was taken for the simulations produced in this work. For more accurate statements, on the one hand the number of Monte Carlo simulations would have to be increased significantly, and on the other hand the parameters M , and N of the reference solution, which are needed in Section 7.1, would have to be much larger.

8. Conclusion

The consideration of semilinear parabolic SPDEs as a construct on an infinite dimensional Hilbert space allows, under sufficient conditions, to prove the existence of a solution of such an equation and to present it as a stochastic process with infinite dimensional values. However, this solution often cannot be represented explicitly, so one needs methods to discretize the spatial and temporal dimensions to simulate such a stochastic process. Here, special attention must be paid to the nonlinearity and the irregularity of the noise. The influence of the nonlinearity is usually limited by assuming a sufficient smoothness of the function. The noise can be represented as an infinite sum of independent and identically distributed real-valued Brownian motions. Its irregularity generally causes a discrete approximation in the time dimension to converge only slowly towards the infinite-dimensional continuous solution.

With the exponential Euler method, which was first introduced in [JK09a], we have described a numerical method for time discretization, which, with the help of a smoothing factor for the noise, provides a fast strong convergence in the time dimension. We have proved that under sufficient, with under very restrictive, assumptions a temporal convergence rate of $1 - \epsilon$ can be achieved, where ϵ can be chosen arbitrarily small. Furthermore, we have shown that the fully discretized model, consisting of the spectral Galerkin method for the spatial discretization and the exponential Euler method for the temporal discretization, converges strongly and its convergence rate is composed of the individual convergence rates of the spatial and temporal discretizations. The individual spatial and temporal strong convergence rates proven for this complete approximation scheme in theory were then also shown numerically using simulations. Even more, in the special case of the semilinear stochastic heat equation, the individual spatial and temporal convergence rates were combined to show a total convergence rate with respect to computational costs. This theoretical result was then also shown numerically by means of simulations.

Next, we showed that the exponential Euler method can be represented by Ornstein-Uhlenbeck processes and then introduced a modified method, the semilinear replacement method, where we did not truncate high dimensions of the process but replaced them by centered gaussian processes. Note that this is a generalization of the idea that can be found in [Hil20] to semilinear SPDEs. Using central limit theorems for realized temporal and spatial quadratic variations, we then compared the exponential Euler method with the semilinear replacement method and were able to numerically demonstrate the superiority of the semilinear replacement method for this application using simulations.

It remains open to also theoretically validate the semilinear replacement method. Here, especially the strong convergence is interesting, since this was proved for the similar exponential Euler method in this work. Special attention has to be paid to the additional error term of the nonlinearity, which results from replacing the higher dimensions of the process by a centered gaussian distributed random variable. This is the task for future research.

Also, it should be noted that in the simulations the expected value was replaced by Monte Carlo iterations. This leads to an additional sampling error, which can sometimes account for a large part of the approximated strong (or weak) error. Therefore, in such cases it may be useful to use a multi-level Monte-Carlo approach [see LP17].

Finally, it should be mentioned that due to the high computational effort, simulations were performed only on a small scale. Therefore, neither for the simulation of the strong convergence rate nor for the simulation of the marginal distribution of the realized temporal and spatial quadratic variations was it possible to determine exactly the parameter sizes needed to achieve a predefined precision of the approximation. This task also remains open for future research.

Appendices

A. Basic concepts and results of probability theory

A.1. Useful equalities and inequalities

In this subsection we will state some helpful and well-known inequations and tools. We will not provide proofs for the most statements but rather refer to the exact.

The first two inequalities are classical results and we need them in the proof in Section 5.2.2 to handle integrals and the $L^2(\Omega)$ -norm.

Theorem A.1 (Hölder inequality, [LPS14] p.9)

Suppose that $1 \leq p, q \leq \infty$ with $1/p + 1/q = 1$. For $u \in L^p(X)$ and $\nu \in L^q(X)$,

$$\|u\nu\|_{L^1(X)} \leq \|u\|_{L^p(X)} \|\nu\|_{L^q(X)}$$

In particular, for $p = q = 2$, this is known as *Cauchy–Schwarz inequality* and is most common in its integral notation

$$\int |f(x)g(x)| dx \leq \left(\int |f(x)|^2 dx \right)^{1/2} \left(\int |g(x)|^2 dx \right)^{1/2}.$$

Theorem A.2 (Minkovsky inequality, [LPS14] p.9)

For $u, \nu \in L^p(X, Y)$, $p > 0$,

$$\|u + \nu\|_{L^p(X, Y)} \leq \|u\|_{L^p(X, Y)} + \|\nu\|_{L^p(X, Y)}.$$

Next we state a common tool to bound nonlinear functions, namely the Taylor expansion.

Theorem A.3 (Taylor expansion [LPS14] (Theorem A.1))

If $f \in C^{n+1}(\mathbb{R}^d, \mathbb{R}^m)$, then

$$f(x + h) = f(x) + Df(x)(h) + \dots + \frac{1}{n!} D^n f(x)(h)^n + R_n$$

where $x, h \in \mathbb{R}^d$ and the remainder R_n is given by

$$R_n = \frac{1}{n!} \int_0^1 (1-s)^n D^{n+1} u(x + sh)(h)^{n+1} ds.$$

Remark A.4

Note that we consider a more general nonlinearity $F : H \rightarrow H$. But since we only consider nonlinearities of Nemytskii type (e.g. $F(u)(x) = f(u(x))$ for an unique $f : \mathbb{R} \rightarrow \mathbb{R}$) the Taylor expansion is still applicable. Moreover one can also state a more general version of the Taylor expansion but it needs the definition of a Fréchet derivative [see Wer04, Theorem III.5.5].

The next lemma is known as Gronwall's inequality. While there are different versions of this inequality, we will state the integral case as well as the discrete case which fit our purposes best. The following lemma and its proof is based on [Kal80, Proposition 5.1.1].

Lemma A.5 (Gronwall's inequality (integral case))

Let $\alpha(t)$ and $\beta(t)$ be Lebesgue integrable functions on $[a, b]$ and suppose H is a constant such that

$$\alpha(t) \leq \beta(t) + H \int_a^t \alpha(s) ds, \quad t \in [a, b]. \quad (107)$$

Then

$$\alpha(t) \leq \beta(t) + H \int_a^t e^{H(t-s)} \beta(s) ds. \quad (108)$$

Further, if $\beta(t) \equiv B$ is a constant, then

$$\alpha(t) \leq B e^{H(t-a)}. \quad (109)$$

Proof. Let $A(t) = \int_a^t \alpha(s) ds$. The function $g(t) = A(t)e^{-Ht}$ is absolutely continuous on $[a, b]$ and

$$\begin{aligned} \frac{d}{dt} g(t) &= \alpha(t)e^{-Ht} - HA(t)e^{-Ht} = e^{-Ht}(\alpha(t) - HA(t)) \\ &\leq e^{-Ht}(\beta(t) + HA(t) - HA(t)) = e^{-Ht}\beta(t) \end{aligned}$$

a.e. on $[a, b]$. Thus

$$g(t) - g(a) = \int_a^t g'(s) ds \leq \int_a^t \beta(s)e^{-Hs} ds.$$

Further

$$g(t) - g(a) = A(t)e^{-Ht} - A(a)e^{-Ha} = A(t)e^{-Ht}$$

and therefore

$$A(t) \leq e^{Ht} \int_a^t \beta(s)e^{-Hs} ds.$$

Now, distinguish between two cases: The first case, where $\beta(t) \equiv B$ is a constant and the second one, where it is a Lebesgue integrable function.

Let $\beta(t) \equiv B$ be a constant, then we get

$$\alpha(t) = \frac{d}{dt} A(t) \leq \frac{d}{dt} \left(B e^{Ht} \int_a^t e^{-Hs} ds \right) = \frac{d}{dt} \left(\frac{B}{H} (e^{H(t-a)} - 1) \right) = B e^{H(t-a)}.$$

Now, let $\beta(t)$ be a Lebesgue integrable function. By the Condition (107) we get

$$\alpha(t) \leq \beta(t) + HA(t) \leq \beta(t) + H e^{Ht} \int_a^t \beta(s)e^{-Hs} ds = \beta(t) + H \int_a^t \beta(s)e^{H(t-s)} ds.$$

□

One can also state a discrete version of Gronwall's lemma. The following is one version that suits our purposes. It can be found in [Kru14, Lemma A.3].

Lemma A.6 (Gronwall's inequality (discrete Case))

Let $c \geq 0$ and $(x_k)_{k \geq 1}$ and $(b_k)_{k \geq 1}$ be nonnegative sequences. If

$$x_k \leq c + \sum_{j=1}^{k-1} b_j x_j, \quad \text{for } j \geq 1$$

then

$$x_k \leq c \exp\left(\sum_{j=1}^{k-1} b_j\right) \quad \text{for } j \geq 1.$$

The next theorem is very important in the theory of Hilbertspaces. It connects a space with its dual space. The following version can be found in [Wer04, Theorem V.3.6].

Theorem A.7 (Riesz-representation theorem)

Let H be a Hilbert space and H' its dual. Then, the map $\Phi : H \rightarrow H'$, $y \mapsto \langle \cdot, y \rangle$ is a bijection. Moreover it is isometric and conjugate linear (e.g. $\Phi(\lambda y) = \bar{\lambda} \Phi(y)$). Thus, for every $x' \in H'$ there exists exactly one $y \in H$ such that $x' = \langle x, y, \rangle$ for $x \in H$, and $\|x'\|_{H'} = \|y\|_H$ holds.

A.2. Bochner integral

This section is devoted to the Bochner integral, a generalization of the classical lebesgue integral, where we also consider Hilbert space valued functions (such that the integral is also Hilbert space valued). We will state the most important properties for this work and refer to [PR07, Appendix A] for more information.

As for the derivation of the classical lebesgue integral we define the Bochner integral via simple functions. Let $\mathcal{S}$ be the space of all simple functions $f : \Omega \rightarrow X$. Then for every $f \in \mathcal{S}$ the *Bochner integral* is defined as

$$\int f d\mu := \sum_{k=1}^n x_k \mu(A_k)$$

with $x_k \in X$ and $A_k := \{f = x_k\}$. Now let

$$L^1(\Omega, \mathcal{F}, \mu; X) = \{f : \mu \rightarrow X \mid f \text{ is strongly measurable with respect to } \mathcal{F}, \\ \text{and } \int \|f\|_X d\mu < \infty\}.$$

Then, it can be shown that $\mathcal{S}$ is dense in $L^1(\Omega, \mathcal{F}, \mu; X)$ such that $\bar{\mathcal{S}} = L^1(\Omega, \mathcal{F}, \mu; X)$ and thus we can define the Bochner integral $\int f d\mu$ for all $f \in L^1(\Omega, \mathcal{F}, \mu; X)$. Next we like to state the most important properties for this work:

Theorem A.8 (Bochner inequality [PR07], Prop. A.2.1)

Let $f \in L^1(\Omega, \mathcal{F}, \mu; X)$, then

$$\left\| \int f d\mu \right\| \leq \int \|f\| d\mu$$

holds.

Theorem A.9 ([PR07], Prop. A.2.2)

Let $f \in L^1(\Omega, \mathcal{F}, \mu; X)$, then

$$\int L \circ f d\mu = L \left(\int f d\mu \right)$$

holds for all $L \in L(X, Y)$, where Y is another Banach space.

A.3. Functional analysis basics

The following is a collection of useful definitions and properties from the world of Functional Analysis that is needed throughout this work. This collection is mainly based on [Wer04], [LR17] and [LPS14]. Let H be a separable Hilbert space.

Definition A.10 (Linear operator [LR17] p. 83)

Let X and Y be real Banach spaces. A mapping $A : X \rightarrow Y$ is called *linear operator* if $A(ax + by) = aA(x) + bA(y)$ for all $a, b \in \mathbb{R}$ and $x, y \in X$.

Definition A.11 (Continuous operator ([Wer04])(Definition II.1.1))

A continuous linear mapping between two normed spaces is called *continuous operator*.

A continuous linear operator $A : X \rightarrow Y$ is also bounded (e.g. there exists a constant $M > 0$ such that $\|Ax\|_Y \leq M \|x\|_X$ for all $x \in X$) [see Wer04, Theorem II.1.2]. Moreover, if $Y = \mathbb{R}$, then A is also called *functional*. As usual, [e.g. see Wer04, p.46], we denote the space of continuous (and thus bounded) linear operators $A : X \rightarrow Y$ by $L(X, Y)$. Hence, we will call $A : X \rightarrow Y$ a bounded linear operator if $A \in L(X, Y)$. We can also define a norm on $L(X, Y)$ - the so-called *operator norm* denoted by $\|\cdot\|_{Op}$.

Theorem A.12 (Operator norm [Wer04](Theorem II.1.4))

$\|A\|_{Op} = \sup_{\|x\|_X \leq 1} \|Ax\|_Y$ defines a norm on $L(X, Y)$.

A special case of a space of bounded linear operators is given by the dual space X' , which now will be introduced.

Definition A.13 (Dual space [Wer04](Definition II.2.1))

The space $L(X, \mathbb{R})$ of bounded linear functionals A on a normed space X is called *dual space of X* and will be denoted by X' .

Definition A.14 (Nuclear operator ([Wer04])(Definition VI.5.1))

An operator $A \in L(X, Y)$ is called *nuclear*, if there exist sequences (x'_n) in X' and (y_n) in Y with $\sum_{n=1}^{\infty} x'_n(x) y_n < \infty$ such that

$$Ax = \sum_{n=1}^{\infty} x'_n(x) y_n, \quad \forall x \in X$$

is satisfied.

Definition A.15 (Trace ([Wer04])(Definition VI.5.7))

Let H be an infinite dimensional Hilbert space and let $A : H \rightarrow H$ be a nuclear operator. Then the *trace* of A is defined as

$$\text{tr}(A) := \sum_{n=1}^{\infty} \langle A\phi_n, \phi_n \rangle_H,$$

for any orthonormal basis $(\phi_n)_{n \geq 1}$ of H .

For the next definition, let A^* be the adjoint of A [see LR17, Theorem 3.1.25]).

Definition A.16 (Trace-class operator [LR17] (Definition 3.1.39))

Let X, Y be two separable Hilbertspaces and let $A : X \rightarrow Y$ be a bounded linear operator. Then A is called *trace-class* if $\sum_{n \geq 1} \langle \sqrt{A^*A} \phi_n, \phi_n \rangle < \infty$, for every orthonormal collection $(\phi_n)_{n \geq 1}$.

If $A : H \rightarrow H$ is a self-adjoint nuclear operator on a separable Hilbert space H , such that $A^* = A$, then $\sqrt{A^*A} = A$ holds. Moreover, if A is self-adjoint and its eigenfunctions $(\phi_n)_{n \geq 1}$ with corresponding eigenvalues $(\lambda_n)_{n \geq 1}$ create an orthonormal basis of H , then the trace of A is given by

$$\text{tr}(A) = \sum_{n=1}^{\infty} \langle A\phi_n, \phi_n \rangle_H = \sum_{n=1}^{\infty} \lambda_n \langle \phi_n, \phi_n \rangle_H = \sum_{n=1}^{\infty} \lambda_n.$$

Moreover, if A is of trace-class, then $\sum_{n=1}^{\infty} \lambda_n < \infty$.

Definition A.17 (Hilbert-Schmidt operator [LR17] (Definition 3.1.39))

Let X, Y be two separable Hilbertspaces and let $A : X \rightarrow Y$ be a bounded linear operator. Then A is called *Hilbert-Schmidt*, if $\sum_{n \geq 1} \|A\phi_n\|_Y^2 < \infty$, for every orthonormal collection $(\phi_n)_{n \geq 1}$.

Remark A.18

Let X, Y be separable Hilbertspaces and let $A : X \rightarrow Y$ be a bounded linear operator and let A^* be its adjoint. Then following [LR17, Theorem 3.1.41] the following properties are equivalent:

- (i) A is nuclear,
- (ii) A is trace-class,
- (iii) A^* is trace-class,
- (iv) $A = BC$ where B and C are Hilbert-Schmidt operators.

Next we need the basics of orthogonal projection operators which are an important instrument in the spectral Galerkin approximation. The following statements are based on [Wer04, §IV.6 and §V.3] and [LPS14, §1.2]. For this, Let G be a closed subspace of H .

A.3.1. Orthogonal projections

Definition A.19 (Projection [Wer04] p.161)

Let U be a subspace of H . Then the mapping $P : H \rightarrow U$ which satisfies $P^2 = P$ is called *projection*.

Definition A.20 (Orthogonal complement ([LPS14], Definition 1.36))

The *orthogonal complement* of a closed subset $G \subset H$ is

$$G^\perp := \{u \in H : \langle u, \nu \rangle_H = 0 \text{ for all } \nu \in G\}.$$

Theorem A.21 ([Wer04] Theorem V.3.4)

Let $G \neq \{0\}$ be a closed subspace of H . Then there exists a linear projection $P_G : H \rightarrow G$ with $\|P_G\|_{Op} = 1$ and $\ker(P_G) = G^\perp$. $(I - P_G) : H \rightarrow G^\perp$ is also a projection with $\|I - P_G\|_{Op} = 1$. Furthermore $H = G \oplus G^\perp$.

Definition A.22 (Orthogonal projection)

Let P_G be the linear projection given by Theorem A.21. Then P_G is called *orthogonal projection*.

Note that, using the settings of the previous theorem, given any $\nu \in H$, $p \in G$ and $q \in G^\perp$ such that $\nu = p + q$ the orthogonal projections P_G and $(I - P_G)$ satisfies $P_G(\nu) = p$ and $(I - P_G)(\nu) = q$. Moreover, using the orthogonality of p and q (e.g. $\langle p, q \rangle_H = 0$) we have

$$\|\nu\|_H^2 = \|p\|_H^2 + \|q\|_H^2.$$

A.3.2. Differentiation on Hilbert spaces

Definition A.23 (Differentiation of nonlinear function [Wer04] p.112f.)

Let X and Y be normed spaces, $U \subset X$ open and $f : U \rightarrow Y$ a function.

- (a) f is called *Gâteaux-differentiable* at $x_0 \in U$ if there exists a continuous, linear operator $T \in L(X, Y)$ with

$$\lim_{h \rightarrow 0} \frac{f(x_0 + hv) - f(x_0)}{h} = Tv, \quad \forall v \in X. \quad (110)$$

- (b) f is called *Fréchet-differentiable* at $x_0 \in U$, if the convergence from (110) is uniform with respect to $v \in B_X$.

Obviously, if f is Fréchet-differentiable it is also Gâteaux-differentiable.

Theorem A.24 (Mean-value theorem for nonlinear operator [Wer04] p.120)

Let X and Y be normed spaces and let $U \subset X$ be an open subspace. Let $f : U \rightarrow Y$ be Gâteaux-differentiable. Further let $I = \{x_0 + \lambda u | \lambda \in [0, 1]\}$ an interval in U . Then the following holds:

$$\|f(x_0 + u) - f(x_0)\| \leq \sup_{\xi \in I} \|Df(\xi)\| \|u\|. \quad (111)$$

Proof. View part (b) of the proof of [Wer04, Theorem III.5.4]. □

A.4. The Trapezium rule

In the following, we mainly represent [LPS14, Appendix A.2]. The *trapezium rule* is given by

$$\int_a^b u(x)dx \approx \frac{1}{2(b-a)}(u(a) + u(b)), \quad (112)$$

which has an error of order $\mathcal{O}(|b-a|^3)$. To get a more accurate approximation we can divide the interval $[a, b]$ into subintervals $[x_j, x_{j+1}]$, e.g. choose $0 = x_0 < x_1 < \dots < x_N = b$ with $x_{j+1} - x_j = \frac{b-a}{N} = h$ for all $j \in \{1, \dots, N-1\}$. Then the *composite trapezium rule* is given by

$$\int_a^b u(x)dx \approx h \left(\frac{u(a)}{2} + \sum_{j=1}^{N-1} u(x_j) + \frac{u(b)}{2} \right) = T_N(u) \quad (113)$$

which has an error order of $\mathcal{O}(h^2)$. It can be shown [see LPS14, Theorem A.6] that if $u \in C^2(a, b)$ with $u(a) = u(b)$, then there exists $\xi \in [a, b]$ such that

$$\int_a^b u(x)dx - T_N(u) = \frac{b-a}{12} h^2 u''(\xi).$$

Therefore the smoother u is, the more accurate is the composite trapezium rule (and also the original trapezium rule, since it is a special case with $h = (b-a)$).

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